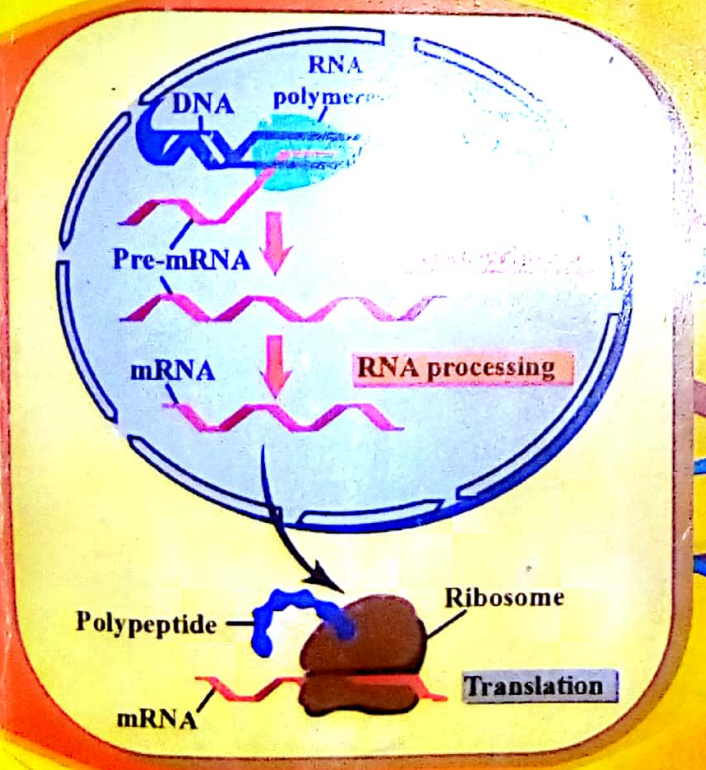
Extreme Series

BIOLOGY

GRADE 11 - 12



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Mossisa Wakshum (Msc)
Yosef Yamba (Msc)

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Extreme

BIOLOGY

For Grade 11-12

Based on Common Currently Used

National and International Curriculum

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Table of content

Unit – 1 The Science of Biology
- The method of science 1
- Tools of a biologist 12
- The relevance and promise of biological science 23
- Biology and HIV/AIDS 28

Unit – 2 Biochemical Molecules
- Classification of biological molecules 36
- Organic molecules 44

Unit – 3 Enzymes
- Nature of enzymes 101
- Function of enzymes 119
- Factors affecting the function of enzymes 128

Unit – 4 Cell Biology
- Cell theory 157
- Types of cells 170
- Parts of the cell and their function 177

Unit – 5 Energy Transformation
- Respiration 231
- Photosynthesis 265

Unit – 6 Microorganisms
- Microorganisms 323
- Shapes and arrangements of bacterial cells 326
- Gram – positive and Gram – negative bacteria 326
- Types of Cells 327

6.2 The Ecology & uses of bacteria 331
- How microorganisms cause disease 332
- Role of bacteria in the environment 334
- Role of bacteria in industry 336
- Genetic modification engineering 336

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- Steps of genetic engineering 337

6.3 viruses 342

- Types of viruses 342
- Reproductive cycle of viruses 343
- HIV & AIDS 344
- Stages of HIV infection 345
- Illustrative examples 346
- Review Question 347
- Answer to Review Question 351

Unit – 7 Ecology 367

7.1 Recycling of Matter in Ecosystems 368

- The Nitrogen cycle 369
- Dangers of overuse of Fertilizers 372
- The carbon cycle 372
- The phosphorus cycle 373
- The water cycle 374
- Illustrative Examples 376

7.2 Ecological Succession 382

- Illustrative Examples 384

7.3 Biomes 387

7.3.2 Aquatic Biomes 392

7.3.3 Ecosystem of Ethiopia 393

- Recommended Conservation measures 396
- Illustrative Examples 397

7.4 Biodiversity 402

- Illustrative Examples 407

7.5 Population 410

- Factors controlling population size 410
- How do populations grow? 411
- Human Populations 413
- Illustrative Examples 415
- Review Question 430
- Answers to Review Questions 441

Unit – 8 Genetics 442

8.1 Genetic crosses 448

- Illustrative Examples 480

8.2 Molecular Genetics 481

- Structure of DNA 484
- DNA replication 485
- Gene & Organism cloning 486
- Genetic Engineering 486
- Dolly , an example of in vivo cloning 486
- Main steps of genetic engineering 487
- Potential benefits of genetic engineering 488
- Genetic finger printing 488
- Steps of genetic finger – printing 489
- Is using gene technology ethical 489
- Illustrative Examples 494

8.3 Protein synthesis 495

- The genetic code 496
- Properties of the genetic code 497
- Transcription in Eukaryotes 498
- Translation 499
- Steps of translation 500
- Introns & Exons 501
- Examples of protein 501
- Amino acids 504
- Control of gene expression 506
- Illustrative Examples 512

8.4 Mutations 518

- Illustrative Examples 536
- Review Questions 544
- Answers to review Questions 546

Unit – 9 Evolution

9.1 Theories of origin of life 546

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- Illustrative Example 551

9.2 Theories of Evolution 554
- Lamarckism 554
- Darwinism 554
- Neo-Darwinism 555
- Illustrative Examples 556

9.3 The Evidence for Evolution 557
- Illustrative Examples 561

9.4 The process of Evolution 568
- Types of natural selection 572
- Polyploidy & Its Importance 573
- Divergent & Convergent Evolution 575
- Illustrative Examples 575

9.5 The Evolution of humans 576
- Lucy & Ardi 578
- Human evolution & brain size change 579
- Are humans still evolving? 579
- Hominid evolution 580
- Illustrative Examples 581
- Review Questions 583
- Answers to review Questions 596

Unit – 10 Behaviour

10.1 Introduction 601
- Biological definition of behavior 606
- Plant responses 606
- Photoperiodism 606
- Tropisms 606
- Nastic movements 607
- Response in simple animals 609
- Impact on environment & resources 610
- Impact on animal welfare 611
- Impact on science education 611
- Review Examples 611

10.2 Innate behavior 609
- Types of innate behavior 609
- Reflex actions 609
- Orientation behavior 610
- Instinctive behavior 610
- Structure of the nervous system 610
- Biological clocks 611
- Imprinting 612

10.3 Learned behavior 613
- Types of learned behavior 614
- Habitation 614
- Sensitization 615
- Associative learning 615
- Classical conditioning 615
- Pavlov 's experiment on dogs 616
- Operant conditioning 616
- Latent learning 616
- Insight learning 617
- Review Examples 617

10.4 Examples of behavior patterns 618
- Courtship behavior 620
- Methods of communication 620
- Types of pheromones 621
- Territorial behavior 621
- Defending a territory 622
- Social behavior 623
- Benefits of social behavior 623
- Social behavior in bees 623
- Illustrative Examples 624
- Review Questions 627
- Answers to review Questions 637
- Basic biology terms (A – Z) 644

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UNIT ONE

The Science of Biology

1.1 The Methods of Science

The Science of Biology

- Biology is a branch of natural science that is concerned with study of life or living things.
- The word biology comes from two Greek words, *Bios* means life, and *logy* means study.
- Therefore, biology is concerned in studying all living things from single celled organisms like bacterias, protozoans and unicellular algae's to multicellular organisms including plants and animals.
- Biologists are those who are involved in the study of living organisms under different categories. Some of the areas of study in which biologists get involved include;
  - **Taxonomists** are involved in the methods and study of classification of organisms.
  - **Microbiologists** studies about microorganisms.
  - **Astrobiologist** –engaged in researched to find evidence of life forms in other planets.
  - **Pale biologist** studies about how life is originated on the earth and how it is evolved.
    - It is mainly concerned in the study of fossils.
  - **Ecologist** – engaged in the study of relationship between organisms and their environment.

  - **Genetist** – involved in the study of heredity variation, physiological basis of DNA and the genetic material.
  - **Neurologist** - engaged in the study of the complex structure of neurons and nervions coordination.
  - **Parasitologist** – involved in the study of the parasites.
    - Note: Parasite is an organism that derives its food from living organism of another species (host).
  - **Zoologist** – involved in the study of animals.
  - **Botanist** – involved in the study of plants.
  - **Entomologist** – involved in the study of insects.

Science

Science is a systematic body of knowledge developed as a result of observations and practical experience of human beings in their attempt to understand and control nature.

- In science knowledge is acquired on the basis of scientific method.
- Scientific method involves a series of steps followed by scientists to solve problems. Thus, scientific method is important:
  - To solve problems
  - To discover facts and principles
  - To formulate laws and theories
- Science is also referred to as experimental study because it involves the use of experiments in obtaining and proving information's.

Scientific Method

- Scientific method is important for scientists including biologists to approach their works in order to
  i) Look in to the nature
  ii) Investigate solutions for problems
  iii) Gain knowledge

Scientific Methods follow the following steps:

1. Ask a question or identify the problem
   - Problem identification or questioning is based on:
     i) Observations that can be made directly or indirectly.

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- Directly through seeing, hearing, tasting, smelling and teaching.
- Indirectly through systematic observations which require knowledge and skills.
  Eg. Using special equipments like microscopes.

ii) Facts
iii) Imaginations

2. Back ground research – Involves carrying further observations about the problem from:
   - Media broad casts
   - Magazines
   - Internets
   - People who are knowledgeable about the area of the problem
   - Anyone who looked into the problem, etc.
3. Hypothesizing means putting together a tentative explanation to account for an observation.
   - If the hypothesis is scientific, it is testable through experiments.
   - A hypothesis is an educated guess about what a biologist thinks the explanation of an observation made.
   - Therefore a hypothesis
     - Should consider all the observed facts relevant to the problem
     - Should lead to a prediction which is suitable for testing.
   - There are two forms of a hypothesis:
     i) **Inductive hypothesis** – *is a generalization that process from specific to general to generate a hypothesis to discover new scientific generalization.*
     ii) **Deductive hypothesis** formulated generalizations from general to specific conditions.
4. Testing the hypothesis through experiments
   - In testing the hypothesis, the hypothesis should be subjected to planned and controlled experimental testing.
   - The goal is to control all the variables except the one under the study.

Note: **Variables** are events or conditions subject to change.

Examples are – Amount of light
- Temperature
- Moisture
- Concentration of substances
- Number of organisms

- There are three categories of variables:

i) **Independent variable** – are variables over which the experimenter can impose control during experimentation.
   - They are factors that the scientist changes or manipulates.

ii) **Dependent variable** – *are variables over which the experimenter has no control.*
   - They are factors that the scientist measures to see if it changes when the independent variable changes.

iii) **Controlled Variables** – are conditions that could affect the outcome of the experiment.
   - They are factors other than the independent variable such as temperature, lighting conditions, number of seeds, volume of liquid added to avoid influencing results.

5. Analyze and interpret the test results
6. **Conclusions** – experimental results and analysis that help to reject or accept the hypothesis.
   - If the hypothesis is supported through repeated experimental results it will be accepted.

Accuracy, reliability and validity in scientific experiments

- **Accuracy** – refers to how precisely something has been measured or counted.
- **Reliability** – a measure of how dependable and consistent the results of an experiment are.
  - This means, after repeated investigation the results show very close or the same results.
- **Validity**: *related to whether an experiment is measuring what is intended to be measured or not. Thus, all the variables should be controlled for an experiment to be valid.*

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Writing Reports in Scientific Experiments

- Scientific investigations are published in different print and online webs.
- The steps to be followed in writing scientific reports include;
  - Title – what is to be investigated?
  - Hypothesis – a perdition of a particular experiment.
  - Procedure – clear description of experimental procedure.
    - Apparatus to be used
    - Chemicals used
    - Controls etc.
    - Result
    - Conclusions
    - Evaluation of the procedure
    - Acknowledgement

Illustrative Examples

1. Which one of the following corresponds to the first step of the scientific Method? (UEE 2002),
   A. Analyze the data collected.
   B. Identify topic to be studied
   C. Undertake the research
   D. State the results

Explanation: The Correct sequential order in scientific method is;

- Identification of the topic to be studied
- Undertaking the research
- Analyzing the data collected
- Stating the result

Answer: B

2. Which area of biology studies how organisms undergo growth, differentiation and morphological changes? (UEE,2001)
   A. Physiology C. Taxonomy
   B. Development D. Cytology

Explanation:

- **Physiology** deals with functions of the body.
- **Development** is a broad term that includes growth, differentiation and changes (morphological).
- **Morphology** is the study of the external structures of organisms.
- **Taxonomy** – is the study of classification of organisms.
- **Cytology** – is the study of structures and functions of cells.

Answer: B

3. Based on the five steps listed below, choose the alternative that shows the correct sequence of application in actual research in biology.
   1) Performing an experiment
   2) Conducting preliminary observation about a particular problem in biology
   3) Statement of a biological theory
   4) Collecting of experimental data
   5) Hypothesis formulation

   A. 1,2,3,4,5 C. 2,5,1,4,3,
   B. 5,4,3,2,1, D. None of the above

Explanation: The correct sequence in application of biological research:

1) Conducting prelimary observation about a particular problem.
2) Formulating a hypothesis
3) Performing an experiment
4) Collecting experimental data
5) Making a statement of biological theory

Answer: C

4. What does ornithologist study:
   A. The biology of mammals
   B. The biology of frogs and reptile'
   C. The biology of fishes
   D. The biology of birs

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Explanation:

- Ornithology is th study of birds.
- Ornithologist is a biologist who studies about birds.

Answer: D

5. A field biologist who studies the behavior of primates in the forest is most likely expected to frequently use which method of data collection?
   A. Experiment
   B. Animal dissection
   C. Observation
   D. Comparative biochemistry

Explanation:

- Animal dissection is important in looking and studying the internal structure of the body (heart, lung, kindly, etc).
- Biochemistry is involved in the study of the chemical substances and structure that makes up the body.
- In order to study the behavior (courtship, feeding, adaptation, etc) of animals in the forest, the best way of gaining sufficient information is observation.

Answer: C

6. Controlled experiment is used to increase which of the following quality of the experiment.
   A. Accuracy
   B. Validity
   C. Reliability
   D. All are possible answers

Explanation:

- **Accuracy** – is how precisely something has been measured or counted.
- **Reliability** is a measure of how dependable and consistent the results of the experiment are.
- **Validity** – is about weather the experiment measures what is intended to measure or not.
  - Unless there is a contoll over all the variables, it cannot be valid.

Answer: B

7. Which of the following is not true about scientific method?
   A. It is the way of collecting information for certain problems.
   B. The steps are the same in all scientific researches.
   C. It follows certain steps to solve scientific problem.
   D. It helps to increase scientific knowledge and understanding.

Explanation: Steps in scientific methods may vary according to type of problem to be investigated.

Answer: B

8. Variables if they do not kept constant they change the outcome of the experiment:
   A. Independent variable
   B. Dependent variable
   C. Controlled variable
   D. Confounding variable

Explanation:

- **Independent variable** – are variables over which the experimenter can impose control during experimentation.
- **Dependent variables** are variables over which the experiment has no control.
  - These are factors other than independent variable.
  - It includes; Temperature, light, etc.

9. Science solve the natural problems through
   A. Testing the observed problems.
   B. Spraying to the divine power.
   C. Collecting ideas and facts through reading
   D. Discussion with elders.

Explanation: Science is always experimental. This implies all the information including the observed problems should be tested in order to teach some results or mechanisms that indicate appropriate solutions and add body of knowledge in to the science.

Answer: A

10. Hypothesis that is formulated should be:
    A. Never been disproved
    B. Tested through experimentation

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C. Proved to be a theory necessarily
D. None of the above

Explanation: Hypothesis is a tentative explanation or idea about observed problems.

- A hypothesis must be testable through experimentation. If it is proved it will be accepted, while if it is disproved it will be rejected.
- Thus, a hypothesis:
  - Can be proved or disproved.
  - It tested through experimentation.
  - If it is proved and accepted can be added as a body of knowledge

Answer: B

11. In scientific methods, which one of the following is used to test a hypothesis? (UEE, 2003)
A. Observation
B. Theory
C. Experiment
D. Prediction

Explanation:

- **Observation** – is used to collect information and identify problems.
- Theory is a hypothesis that stands the test by different people at different places through time.
- **Experiment** is a method that is used to test a hypothesis.
- **Prediction** is an educated guess as to how a biologist thinks the experiment turn out.

Answer: C

12. After dividing plant materials in to two groups, an experiment has applied a fertilizer to 'group A' plants and not to group B plants. What group is 'Plant B' called?
A. Control group
B. Experimental group
C. Test group
D. Variable group

Explanation:

- Control groups are groups to which no treatments are applied.
- Experimental groups are groups over which treatments are applied.

Answer: A

13. Which one of the following best defines modern biology?
A. The study of animals
B. The study of plants and animals
C. The study of all living things
D. The study of transfer, of energy

Explanation: Biology is a branch of natural science that is concerned in the study of living things.

Answer: C

14. Which of the following corresponds to the beginning step of a scientific work?
A. Testing hypothesis
B. Making observation
C. Conducting experiments.
D. Drawing conclusions

Explanation: The correct sequence is a scientific work is;

- Making initial observations.
- Formulating a hypothesis
- Testing a hypothesis by conducting experiments
- Drawing conclusions

Answer: B

15. In an experiment that is testing the effect of temperature on the germination rate of certain seeds, which one of the following should the experimental and control group differ?
A. The amount of water they receive.
B. The age of the seeds assigned to them.
C. The number of seeds assigned to them.
D. The temperature at which they are kept

Explanation: See example number 12.

Answer: D

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16. What is the reason that Louis Pasteur used a swan –necked flask instead of straight necked one in his experiment designed to disprove the theory of spontaneous generation?
   A. To allow free passage of air to the broth inside the flask
   B. To prevent the escape of any microorganism from the flask.
   C. To keep the broth in the flask not to kill microorganisms.
   D. To trap particles from the air that might enter the flask before reaching the broth.

Explanation:

- Louis Pasteur is a biologist who disproved the theory of spontaneous generation for once and last.
- A swan necked (S-shaped) flask helps to trap particles from the air that might enter the flask before reaching the broth.
- Theory of spontaneous generation was based on the idea that certain non-living things can be directly changed in to living things.

Answer: D

17. A biologist applied the scientific method repeatedly, gathered a large amount of supporting experimental data and finally described a pattern or relationship between different factors. What is the best term that refers to facts established in this way?
   A. Theory B. Hypothesis C. Prediction D. Law

Explanation:

- **Theory** is a hypothesis that stands the test by different people at different places through time.
- **Hypothesis** is tentative statement or idea about observed facts.
- **Prediction** – expectation that give as how the biologist thinks an experiment will turn out.
- **Law** – if variations are observed and requires further experiments.

Answer: A

18. Which of the following steps in the scientific method comes following observation?

Explanation: The Steps in scientific method are:
   i) Initial observation
   ii) Formulation of hypothesis
   iii) Performing an experiment to test the hypothesis
   iv) Collecting experimental data
   v) Making statement of biological data

19. What is a theory in biology ?
   A. The outcome of the hypothesis is true.
   B. A proof that shows hypothesis is true.
   C. A hypothesis supported by experiments and / or observations.
   D. An opinion of educated gues resulting from observations.

Explanation:
See example 17 **Answer: C**

20. Describe the importance of scientific method for scientists

Explanation: Scientific method helps scientists:
   - To discover facts and principles
   - To solve problems
   - To formulate laws and theories.

21. Distinguish between inductive hypothesis and deductive hypothesis.

Explanation:

Deductive hypothesis	Inductive hypothesis
• Ideas formulated from general to specific conditions.	• Generalization that proceeds from specific to general in order to discover scientific generalizations

22. Clearly discuss the meaning of
   - Independent variable
   - Dependent variable
   - Controlled variables

Explanation:

- **Independent variable** – is a condition or an event under the study.
- **Dependent variable** – Ones that can possibly change because of the presence of or change in an independent variable.
- **Controlled variables** – conditions that could affect the outcome of an experiment but that do not, because are held constant.

1.2. Tools of a biologist

- Biologist need various types of equipments or apparatus to be used in order to carry out biological studies.
- Some of the major equipments that biologists use include
   - Microscope
   - Bunsen burner
   - Disecting kits

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- Centrifuges
- Sterilizers
- Measuring equipments like balance, cyclinders,
- Petridishes
- Pipettes and syringe
- Thermometer, etc

Measuring tools

i) Measuring Volumes are used to measure the volume of liquids and volume of gases.
Fig. Measuring cylinder

ii) Balance used for measuring mass

iii) Thermometer – used for detecting the hotness or coldness of an object.

iv) Gas syrings is used to measure the volume of gases like oxygen

2) Dissecting kits

- Dissection is the process of cutting a part separate tissues of an organism for studying the anatomical structure of an organism.
- Dissection is carried to study internal organs like heart, kidney, lung, etc.
- Biologists can even dissect small specimens by using a microscope.
- Dissecting kits contains:
  - Scissors
  - Forceps
  - Magnifying glass
  - Scalpels
  - Mounting needles

3) Petridish

Petri dish – is a shallow circular glass or plastic dish used for growing microorganisms or cultured cells in a suitable medium.

- They are filled with a solid nutrient medium called agar.
  Note: Nutrient medium is a collection of nutrients to grow microorganisms.
- They are used to:
  - Show the effectiveness of antibiotics against certain bacteria.
  - Show how well different concentration of enzymes digest a substance

- **Micro propagation** is a technique of growing plants from just few cells on special agars.
- **Agar** is a solid nutrient medium that is mainly extracted from red algae.

4) Microscope

- A microscope is considered to be a basic tool of a biologist
- Microscopes can be classified as :

ii) **Optical (light) microscopes** – are microscopes that use a beam of light.
- They have relatively less magnification and resolution capacity.
- They can be used in the laboratory or can be used in fields.

iii) **Electron microscopes** – are microscopes that use beam of electrons.

Remember: magnification is multiplying the size of an object.

- The magnification capacity of light (optical) microscope is 2000 times while the maximum magnification of election microscope is 2000000 times.
- **Resolution** is the ability to distinguish between two points that are close together.

Electron Microscope

Transmission electron microscope	Scanning electron microscope
• Image is focused on to a fluorescent screen. • Electron beam passes through the specimen and then falls on a phographic plates or a fluorescent screen. • Shows the fine detail of cell structure	• Image is viewed on a type of "television" screen. • The specimen is coated with in gold film. When the electron beam strikes various points on the surface of a specimen, secondary electrons are eruitted whose intensity varies with the countour of the surface. • Views only the surface of an object.

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4) Centrifuge

Centri fuge is a machine that spins at high speed to separate components of different molecular weights.

- For instance, it is used to separate solid particles from liquids
- It is used in biological laboratories, pharmaceutical industries, hospitals, etc
- **Ultracentrifuge** – is used to spin the tubes at faster speed to separate extremely light particles in to the bottom of the tube. Thus , it is used to separate plant and animal cells.

What do biologists use in the field?

- For studying an ecosystem a biologist need to go out of the laboratories and study an interaction between organisms and their environment.
- To perform the above activities a biologist is required to:
  - Take measurements in relation to abundance or distribution of organisms.
  - Take samples from the environment (soil, water, rock, trees, etc) to be analyzed in the laboratories.
  - To collect specimens
  - Identify and analyze the collected specimens.
- In measuring the abundance or distribution of organisms in an environment biologists are required to use:

i) **Quadrate** – is a sample area enclosed with in frame, usually square within which a plant community or sometimes animal community is analyzed.

a. It is placed randomly on the ground in order to count organism.
b. The number and type of organisms in each quadrate is recorded.
c. Finally, the anundance of organisms is estimated based on the recordings from the sample quadrates.
d If you have 5 quadrates in which each quadrate has an area of $1m^2$ (1mx1m) over an area of $5om^2$, calculate the total estimate of a given species of a

plant, if the number of individuals in each quadrate is as follows.

Quadrat 1=4
Quadrat 2=5
Quadrat 3=3
Quadrat 4=6
Quadrat 5=4

Quadrats can be confined to land as well as water habitat.

ii) **Transect line** – is a line that shows the type and change in distribution of vegetation which exists on a line across a habitat.

- Record the types plants which touch the rope (transect line) at certain intervals.
- For example, you can record type of plant that touches the rope every 1 meter.

Collecting and preserving specimens

- Collecting plant specimens – plants can be easily collected if they are smaller in size and parts of a plant (leaf, flower) can also be collected.
- Preserving plant specimens – can be preserved temporarily in plastic bags or plastic jars.
  - Their parts can also be preserved in a plant press

Remember: plant press is made up of news papers to separate individual specimens and cupboards that provides support in order to preserve the specimen in shape and form.

Collecting animals

- Collecting animals is relatively difficult because of their locomotory nature.
- Some of the equipments that are used in collecting animal include:

i) **Pit fall traps** – is a container that is sunk in to the ground in order to collect insects that crawl on the ground.

ii) **Sweep nets** – are nets with canvas bags that are used to collect insects and other invertebrates from long grasses

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iii) **Funnel** - for collecting insects from the soil.
iv) **Potter (aspirator)** - to collect or suck small organisms on tree trunks or leaf.

- Some other equipments used by biologists on the field include:
  - pH kit - to measure the pH
  - Data logger - for recording informaitons
  - Flow meter – to measure the rate of flow of water
  - Field microscope – to investigate the structure of specimens in the field while they are fresh.
  - Theodolite - to measure the height of trees or slopes in the area.
  - **Global positioning system receiver (Gps)** is used for recording positions quickly and accurately.

Illustrative Examples

23. Which of the following field equipment is used to determine an exact location of a place?
A. Theodolite
B Global positioning system
C. Field microscope
D. Field pH kit

Explanation:

- **Theodolite** is used to measure the height of trees or slopes in the area.
- **Global positioning system (Gps)** – used for recording positions quickly and accurately.
- **Field microscope** for investigating the structure of specimens in the field while they are fresh.
- **pH kit** - for measuring the pH of soil and water.

Answer: B

24. For which of the following is a theodolite used in biology education?
A Measuring the height of trees.
B Recording position where a species is found.

C. Measuring the rate of flow of water in a cell.
D. Measuring the pH of water or soil

Explanation: (See example 23)

Answer: A

25. A machine that spins at high speed to separate components of different molecular weight is:
A. Incubator B. Centrifuge C. Balance D. cylinder

Explanation:

- **Centrifuge** – is a machine that spins at high speed to separate compenent of different molecular weight.
- **Balance** – is used for measuring mass.
- **Measuring cylinder** is closed to measure the volume of liquids and volume of gases.

Answer: B

26. Which of the following activities is not required by biologists in a field study?
A. Taking measurements in relation to the distribution of organisms.
B. Collecting specimens from the field.
C. Indentifying and analyzing the collected specimens.
D. None of the above.

Explanation:

In a field study biologists are required to perform the following activities:

1) Take measurements in relation to abundance or distribution of organisms.
2) Take samples from the environment to be analyzed in the laboratories.
3) To collect specimens.
4) To identify and analyze the collected specimens.

Answer: D

27. A technique of growing plants from few cells on special agar is:
A. Nutrient Medium C. Genetic engineering
B. Micro propagation D. Breeding

Explanation:

- **Nutrient Medium** is a collection of nutrients, to grow microorganisms under controlled condition in the laboratories.

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- **Micro propagation** - is a technique of growing plants from a few cells on special agars.
- **Genetic engineering** is a technique of altering an organisms genotype by inserting another organism characteristics in to its DNA.

Answer: B

28. A microscope helps to distinguish between two points that are close together. This is related to;
    A. Magnification
    B. Staining
    C. Resolution
    D. Mounting

Explanation:
A microscope has two major abilities:

i) **Magnification** is multiplying the size of an object.
ii) **Resolution** is the ability to distinguish between two points that are close together.

Answer: C

29. Which of the following is TRUE in relation to scanning electron microscope?
    A. Shows fine detail of cell structure
    B. Image is focused on to a fluorescent screen.
    C. Image is viewed on a television screen
    D. It has less magnification and resolution power.

Explanation: In scanning electron microscope:

- Image is viewed on a television screen.
- The specimen is coated with gold film
- Views only the surface of an object.

Answer: C

30. Which of the following is not a measuring tools?
    A. Measuring cylinder
    B. Balance
    C. Dissection
    D. Thermometer

Explanation:

- Measuring tools used by biologists include:
  - Measuring cylinder
  - Balance
  - Thermometer
- **Dissection** – is method of studying internal organs and structures of an organism.

Answer: C

31. Cutting parts of separate tissues of an organism for studying the anatomical structure of an organism
    A. Test tube
    B. Petridish
    C. Scarpels
    D. Forceps

Explanation:

- Petridish is a shalow circular glass or plastic dish used for growing microorganisms or cultured cells in a suitable medium.
- Pertidish consists of a solid nutrient medium called **agar**.

Answer: A

34. What would be the approximate image size of a cell with the size of 0.03 microns-viewed under a compound microscope with magnification powers marked as 10 x on eye piece and 100 x on the objective lenses?
    A. 10 micros
    B. 20 Micros
    C. 30 microns
    D. 40 Micros

Explanation:
Total magnification = 0.03 microns x 10 x 100 = 30 microns

Answer: C

35. Which tool of the biologist is more stable for cultivating bacteria in the laboratory? (UEE 2004)
    A. Test tubes
    B. Microscopes
    C. Petri dishes
    D. Beakers

Explanation:
To grow bacteria's in laboratories under controlled conditions they are cultured in to a nutrient medium on a petridish.

Answer: C

36. Which of the following types of microscopes is the best to show the details of the surface of an object?
    A. Optical microscope
    B. Compound microscope
    C. Scanning electron microscope
    D. Transmission electron microscope

Answer: D

37. Which of the following laboratory equipment is used to separate the organelles of a cell according to their density?
    A. Incubator
    B. Centrifuge
    C. Measuring cylinder
    D. Filter paper with fine pores

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Explanation:

- **Centrifuge** – is a machine that is closed to separate components of different molecular weights.

Answer: B

38. Type and change in distribution of vegetation which exists on a line across a habitat is:
A. Quadrat B. Transect C.Balance

Explanation:

- **Quadrat** – is a sample area enclosed with in fame which is placed randomly on the ground for counting organisms rounded inside it.
- **Transect line** is used to know the change in distribution of vegetation on a line across a habitat.

Answer: B

39. Which of the following is not required in preservation of plant specimens?
A. Plastic bags C. Plant press
B. Plastic years D. Agar

Explanation:

After collecting plant specimens, it can be preserved:

- Temporary in plastic bags organs.
- Plant press to preserve the specimen is shape and form

Answer: C

40. Insects and invertebrates on long grasses are collected by ?
A.Pitfall traps C. Tulgren funnel
B. Sweep nets D. Potter

Explanation:

- **Pitfall traps** – to collect insects crawling on the ground.
- **Sweep nets** – to collect insects and other invertebrates from long grasses.
- **Tulgen funnel** – for collecting insects from the soil.

- **Potter** – to collect (suck) small organisms from tree trunks or leaf.

Answer: B

41. Trace the two major types of microscope with their distinguishing features

Explanation:

Optical Microscope	Electron Micro scope
• It uses beam of light • It has relatively less magnification and resolution power. • The highest magnification capacity of light microscope is 2000x and resolution of up to 500times than human eye.	• It uses beam of electrons • It has highest magnification and resolution power. • It magnifies up to 2 million times and resolution of 200,000 times human eye.

42. What makes collecting animal specimens difficult when comp aired with plants?

Explanation:

It is difficult to collect animal specimens since animals are capable of moving from one place to another place while, plant specimens are easy to collect.

43. List and describe the use of some equipments that are used by biologists in the field.

Explanation:

- Among various equipments used by biologists in the field some of them include:
  - **pH kit** – for measuring the pH of soil and water.
  - **Data logger** – for recording information's.
  - **Flow meter** – to measure the rate of flow of water.
  - **Field microscope** – For investigating the structure of specimens in the field while they are fresh.
  - **Theodolite** – to measure the height of trees or slopes in the area.

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- **Global positioning system (GPS)** – is used for recording positions quickly and accurately.

44. What is the term used to express the ability of a microscope to show details clearly?
    A. Enlargement
    B. Reduction
    C. Magnification
    D. Resolution

Explanation:

- Microscope has two abilities. These are:

(1) **Magnification** – increasing the size of a object and show the multiplied size of an object from the original one.

(2) **Resolution** – is the ability of microscope to show the detailed structure of an object. This enables us to identify between two points which are close.

Answer: D

1.3 The relevance and promise of biological science

- Biologists are striving to increase areas and advancements in which biological science benefits people more.
- Some of the field areas in which the promise and relevance of biological science are involved include:
  - Agriculture
  - Medicine
  - Environment
  - Energy, etc

i) Biology and agriculture

- One of the current issues on the world specially in developing countries is a demand of food supply. This is also related to an increasing number of human population.
- Biological science devices ways to increase food supply through agricultural productivity.
- Some of the relevant and promising areas in which biologists are striving and researching include, developing plant varities that:
  - Can adapt to new conditions.
  - Are capable of producing crops in short period of time.

  - Are resistant to drought and disease.
  - Are highly productive
  - Are capable of giving high quality products.

ii) Biology and Medicine

- Biological science has a huge relationship with medicine.
- Some of the relevance of biological knowledge in medicine are:
  - To device effective methods of contraception
  - To educate people about the need of controlling population growth.
  - For advising people on ways of keeping themselves and their family healthy.
- Medical professionals have specialized biological knowledge and expertise. Some of these are:
  - **Medical laboratory technicians** – test samples required (blood, urine, sputum, feaces) and provide reports for doctors.
  - **Radiographers** – produce X-rays to help in diagnosing a disease
  - **Drug development researches (pharmacologists)** – work in development of new drugs.
  - **Neurologist** – studies about the morphology, physiology and pathology of the nervious system.

iii) Biology and Environment

- Ecologists are those biologists who are involved in environmental studies.
- Ecologists work in:
  - Monitoring the impact of global warming on the environment.
  - Issues related to abundance and distribution of species in the environment.
  - Give advice to governments and the society how best to conserve environments.
  - Introduction of new species that will maintain the balance of species with in an environment.

Biotechnology

- Biotechnology is the use of microorganisms and enzymes to make things that people want.

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- Biotechnological applications can be seen from two perspectives, traditional and new applications.
  - Yeasts are used in production of:
    - i) Alcoholic drinks such as Tej, Tella, Katikala, Beer, whisky, etc. The process is called **alcoholic fermentation**.

$$C_6H_{12}0_6 \longrightarrow 2C_2H_5OH + 2CO_2 + 2ATP$$

    - ii) Backing of bread, injera, cake, etc
- Bacteria ferment the milk ((lactic fermentation) to produce cheese and yoghurt.
- **New applications of biotechnology** – involves technological application in a broader scale industrial production process.
  - The biotechnological applications brought the following relevancies and promises to our life:
    - Cures for genetic diseases.
    - Industrial process like production of insulin.
    - Genetic modification of plants to meet specific needs.
    - Cloning
    - Production of antibiotics, etc.
  - *Genetic engineering is a process which involves altering an organisms genotype by inserting another organisms gene in to its DNA.*
    - It is a new area in modern biotechnological applications.

Illustrative Examples

45. Which of the following is recently active areas of research in biology today?
    A. Taxonomic study
    B. Study about the cell theory.
    C. Ecological research
    D. Stem cell research.

Explanation:

Stem cells are cells from an early embryo that consists of unspecialized (un differentiated) cells that have a potential to form any type of specialized cells.

- Currently biologists are working on mechanisms of developing new organs and tissues in order for transplantation of diseased or damaged tissues or organs of the body.

Answer: D

46. To which major area of relevance and applications of biology is the production of transgenic organisms related? (UEE 2004)
    A. Agriculture
    B. Environment
    C. Biotechnology
    D. Medicine

Explanation:

- **Transgenic organisms** are animals and plants in to which genes from another species have been deliberately introduced by genetic engineering.
- **Genetic engineering** is one of the most recent issue in biotechnology.

Answer: C

47. Which biotechnological process has practical application in criminal investigation?
    A. DNA finger print.
    B. Bio-fuel production.
    C. Single cell protein technology.
    D. Use of biosensor to test blood glucose level.

Explanation:

In criminal investigations various evidences can be used as an evidence. Some of these include blood group, fingerprint, eye witness, etc. However, currently DNA finger prints are highly acceptable in giving accurate evidence in criminal investigations.

- DNA is a life code in which all persons have variation in their DNA make up.

Answer: A

48. Which of the following is not a specified role of biologists who are involved in environmental studies?
    A. Monitoring issues related to global warming.
    B. Trial of HIV vaccines.
    C. Advising the society on to best conserve environment
    D. Working on abundance of species in an environment.

Explanation:

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The roles of biologists (ecologists) who are involved in environmental studies are:

i) Monitoring the impact of global warming on environment.
ii) Issues related to abundance and distribution of species in the environment.
iii) Give advice to governments and the society on how best to conserve environments.
iv) Introduction of new species that will maintain the balance of species within an environment.

Answer: A

49. The types of microorganisms that are used traditionally to make cheese and yoghurt are:
A. Bacterias B. Yeast C. Pencillium D. Fungus

Explanation:

- Traditional applications of biotechnology is important in:
  - Making of cheese and yoghurt by bacterial actions.
  - Production of alcoholic drinks by the action of yeast.
  - Backing of bread and injera by the action of yeast.

Answer: A

50. List out some relevance of biology in agriculture.

Explanation: Biology is very important in production of plants that:

- Are capable of adapting to changes.
- Produce crops in a short period of time.
- Resistant to drought and disease.
- Highly productive
- Capable of giving high quality products.

51. Describe the role of the following medical professionals.

- Medical researches
- Pharmacologists.
- Neurosurgeons
- Medical laboratory technicians
- Radiologists

Explanation:

Medical professionals have various areas of specialties or work areas these are:

- **Medical researches** – research into ways of controlling and functioning of disease causing organisms.
- **Pharmacologists** work in development of drugs.
- **Neurologists** studies about the morphology, physiology and pathology of the nervous system.
- **Medical laboratory technicians** test samples required and provide reports for the doctor.
- **Radiologists** produce X-rays to help in diagnosing a disease.

52. Clearly discuss some of the areas in which new biotechnological applications are important.

Explanation:

The major areas in which new/modern biotechnological applications are important include.

- Agriculture
- Energy production
- Medicine
- Food, etc

53. Clearly distinguish between lactic fermentation and alcoholic fermentation.

Explanation:

- Lactic fermentation is the process in which bacterial (*lactobacillus bulgaricus* and *strephtococus thermophillus* acts on lactose (milk sugar) to produce lactic acid.
  - The lactic acid ferments the milk and makes the milk to solidify in order to form a curd.
- **Alcoholic fermentation** – is the process in which yeast acts on sugar anaerobically and produce alcohol.

$$C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2CO_2 + 2ATP$$

1.4 Biology and HIV/AIDS

- **AIDS** – is disease that reduces the body's immune functioning.

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- It is an abbreviation which stands for acquired immuno deficiency syndrome.
- It is caused by the human immune Virus (HIV)
- Through time it reduces the T-helper cells which help in fighting diseases.
- When it reduces the immune capacity of the body the body will be easily affected by opportunistic diseases.
- It is a sexually transmitted disease. The routes of HIV transmission includes:
    - Unprotectected sexual contact with an infected person.
    - During blood transfusion
    - Sharing sharp objects which are infected.
    - From mother to child during pregnancy, birth or breast feeding.

Importance of biological knowledge in fighting AIDS

- Biological knowledge is quite important in controlling the spread of HIV/AIDS.
- Biology helps in fighting HIV/AIDS through:
    - Producing vaccines to control the spread of HIV in the body (Antivitroviral drugs)
    - Devicing ways of reducing or stopping the transmission of HIV from an infected person to a healthy person.
    - Trail of making vaccines for HIV with the aim of breaking the transmission path way by making people immune.
    - Creating awareness in the society about the issues related to responsible sexual behavior.
    - Creating consciousness about the ABC rules specially among the young men and women.
        A. Abstain from sexual contract
        B. Be faithful to one sexual partner
        C. Use condoms.

The life cycle of HIV

- HIV is a retrovirus.
  Retrovirus are viruses that copy their RNA in to DNA by an enzyme called **reverse transcriptase**. The DNA enters the host cell chromosome, where it may continue to the direct production of the virus particles.
- The steps in life cycle of HIV are:
    - Entry phase
    - Viral genetic material is converted to DNA.
    - The new DNA enters the host cell DNA.
    - The new DNA 'instructs' the cell to make more HIV.

Biologist and the battle against HIV

- Biologists have been working on ways of breaking the life cycle of HIV. Thus, one of the most important contribution is antiretroviral drugs (ART's)
- ARTs vary in combination and called **highly active-retro viral therapy** or HAART.
- In other ways biologists are working to produce long lasting vaccines against HIV. However, the nature of the virus to mutate rapidly made their works difficult.

Fighting HIV/AIDS

- The major control and preventation mechanisms that people should use in fighting HIV/AIDS include:
    - Avoid having many sexual partners. Thus, having a sexual inter course with one sexual partner.
    - Circumcision of males reduces the risk of HIV transmission to the men.
    - Avoid sharing sharp objects like needle, ra zor, etc.
    - Check yourself for HIV every interval of time through voluntary HIV counseling and testing.
    - Educating people. Specially those who are highly involved in sexual inter course with many partners. These include prostitutes and travelers mostly.

Illustrative Examples

54. Which of the following is true about the HIV/AIDS?
    A. HIV refers to the symptom that characterize the condition.
    B. AIDS refers to the causative agent
    C. AIDS refers to the symptom.
    D. HIV refers to its possible origin from chimpanzees

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Explanation:

- HIV is an abbreviation that stands for the human immunovirus.
- It reduces the immune capacity of the body.
- It is a virus that causes AIDS
- AIDS
  - It is an abbreviation for Acquired Immuno Deficiency Syndrome
  - It is a disease that reduces the body immune functioning.

55. The major route of HIV transmission is:
A. Risky sexual contact C. Sharing sharp objects
B. Blood transfusion D. Mother to child

Explanation. HIV can be transmitted by:

- Unprotected sexual contact with an infected person
- During blood transfusion
- Sharing sharp objects which are infected
- From mother to child during pregnancy, birth or breast feeding.

Answer: A

56. In life cycle of HIV, which step comes last?
A. Entry phase
B. Enterance of the new DNA in to the host cell.
C. Conversion of viral genetic material in to DNA
D. Instruction by the new DNA to make more HIV

Explanation:
The life cycle of HIV follows:

- Entery phase
- Viral genetic materials is converted to DNA
- The new DNA enters the host cell DNA
- The new DNA instructs the cell to make more HIV.

Answer: D

57 List out some of the methods that are important for fighting HIV/AIDS.

Explanation:

- Avoid having many sexual partners
- Circumcision of males.
- Avoid sharing sharp objects.
- Voluntary HIV counseling and testing

- Educating people

58. Clearly describe the what are opportunistic diseases.

Explanation:

- **Opportunistic diseases** are diseases that easily attack the body when it is in a weak immune state.
- For instance, Tuberculosis is an ideal example.

59. How can biological knowledge help is fighting HIV/AIDS?

Explanation:
Biology important infighting HIV/AIDS through:

- Production of vaccines (trial vaccines)
- Creating consciousness about ways of protecting the spread of HIV. This includes ABC rules etc.

60. Explain what are:

- Rotaviruses
- Antiretroviral drugs
- Highly active – retroviral therapy.

Explanation:

- Rotaviruses are viruses that copy their RNA into DNA
  - by an enzyme reverse transcriptase
  - HIV is a Rota virus.
- Antiretroviral drugs are drugs that break stage in life cycle of HIV.
- Highly active – retroviral therapy are ARTs that are used in combination.

Supplementary Exercise on Unit I

I. Matching

Column A	Column b
1. Taxonomist	a) Studies about micro organisms
2. Microbiologist	b) Studies the complex structure of neurons and nervious coordination
3. Astrobiologist	c) Studies about parasites
4. Paleabiologist	d) Studies how life is originated and evolved
5. Ecologist	e) Studies methods and study of classification
6. Genetist	f) Involved in the enrionmental studies
7. Neurologist	g) Studies about heredity
8. Parasitologist	h) Studies about insects
9. Entomologist	

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	i) Engaged in research in to final evidences of life forms in other plants

II. Fill the blank spaces

10. _____ is a tentative explanation based on observed facts.
11. Events or conditions that are subjected to changes are _____.
12. _____ is used to measure the volume of liquids and volume or gases.
13. An instrument that is used for detecting the hotness or coldness of an object is _____.
14. _____ is the process of cutting apart separate tissues of an organism for studying anatomical structure of an organism.
15. A solid nutrient medium that is extracted from red algae is _____.
16. _____ is a collection of nutrients to grow microorganisms.
17. The ability of microscope the increase the size of an object in _____.
18. _____ is used to spin the tubes at faster speed to separate extremely light particles into the bottom of the tube.
19. A sample area enclosed usually square with in which a plant community or animal community is analyzed is called _____.
20. _____ are nets with canvas bags that are used to collect insects and other invertebrates from long grasses.

III. Multiple Choice

21. Which of the following is not the role of biological science?
    A. Looking into the nature
    B. Investigating solutions for problems.
    C. Disproving religious beliefs
    D. Gaining knowledge
22. Variables over which the experimenter has no control is called:
    A. Independent variable
    B. Dependent variable
    C. Controlled variable
    D.None of the above
23. A measuring tool that used to measuring mass is:
    A. Measuring volume
    B. Balance
    C. Thermometer
    D. Gas syringe
24. Which of the following is not a dissecting tool ?
    A Scissor
    B. Forceps
    C. Scalpols
    D. Thermometer
25. Which of the following is incorrect about an optical microscope?
    A It uses a beam of light

    B. It has high resolving power
    C. It used beam of electrons
    D. It can be used in the laboratories
26. A quadrat can be;
    A. Used for sampling plant population
    B. Placed randomly on the ground
    C. Number and type of organisms be recorded
    D. All are possible answers.
27. Which of the following is not necessary for collecting animals?
    A. Pit fall trap
    B. Animal press
    C. Sweep net
    D. Full gren funnel
28. Afield tool that is used to measure the height of trees or slopes in the area is:
    A. pH kit
    B. Flow meter
    C. Theodolite
    D. Data logger
29. The use of microorganisms to make things that people want is:
    A. Biophysics
    B. Biotechnology
    C. Biochemistry
    D. Biometry
30. In which areas is biological science least important?
    A. Agriculture
    B. Medicine
    C. Energy
    D. None of the above
31. Biological knowledge is important in the areas of medicine for:
    A. Devising alternative energy forms
    B. Monitoring the impact of blobalwarming
    C. Devicing effective method of contraception
    D. Increasing crop productivity
32. Which of the following biotechnological application cannot be traditional?
    A. Lactic Fermentation
    B. Genetic engineering
    C. Alcoholic fermentation
    D. Selective breeding
33. A medical professional who works in testing samples required and proved results for doctors is:
    A. Ridiographers
    B. Neurologist
    C. Medical laboratory technician
    D. Pharmacologist
34. Which of the following is not important in fighting AIDs?
    A. Avoid hugging people living with HIV
    B. Avoid having many sexual partners
    C. Avoid sharing sharp objects
    D. Male circumcision

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Answer to supplementary exercise on unit 1

1.	E	2.	A	3.	I	4.	D	5.	F
6.	G	7.	B	8.	C	9.	H		

10. Hypothesis, 11. Variables, 12. Measuring cylinder
13. Thermometer, 14. Dissection, 15. Agar
16. Nutrient medium, 17. Magnification, 18. Ultracentrifuge
19. Quadrat 20. Swep net

21.	C	22.	B	23.	B	24.	D	25.	C
26.	D	27.	B	28.	C	29.	B	30.	D
31.	C	32.	B	33.	C	34.	A		

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UNIT TWO

Biochemical Molecules

2.1. Classification of biological molecules

Biological molecules can be classified in to two major Categories. These are **organic molecules** and **inorganic molecules.**

Organic molecules are molecules that contain carbon and hydrogen. They are the main structural components of living cells and tissues. The four major classes of organic molecules are **carbohydrates, lipids, proteins** and **nucleic acid.**

Inorganic molecules are biological molecules that contain either carbon or hydrogen or neither of them. They are relatively small and simple compounds such as **water, simple acids, bases** and **salts.**

Some important biological Molecules

Before proceeding to the detailed discussion of some elements that are frequently found in living organisms, let us discuss some of the most important terminologies used in biochemical studies.

- **Atoms** – are the smallest particles of chemical element. They are minute particles and cannot be seen. For instance carbon atoms make up element carbon. Atoms are composed of three basic particles; **protons, electrons** and **neutrons.**

I. **Protons** – are positively charged particles that are grouped in the nucleus (center).

II. **Neutrons** are also grouped in the nucleus but they have no electrical charge.

III. **Electrons** are negatively charged particles that are dispersed around the nucleus in layers or orbitals.

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- **Elements** are substances that compose all living and non-living things. Some of the elements that have specified role in our body are:
  - **Calcium (Ca)** is used for maintaining the strength of bones, teeth and muscles.
  - **Iron (Fe)** – is a major component of hemoglobin which is responsible for carrying oxygen around the body.
  - **Iodine (I)** – used for making the hormone thyroxine, which plays a vital role in regulation of the body's metabolic activities.
  - **Fluorine (F)** is used for making the tooth enamel.

N.B. **Enamel** is the outer protective layer of the teeth.

- **Molecules** are groups of two or more atoms held together by chemical bonds. One of the most important biological molecule is **water.**

Water: - is the most abundant inorganic molecule in living things. Accordingly, it covers around 75% of the body of living things. However, it varies based on the type and living style of organisms.

Water is formed by a hydrogen bonding which is formed when two hydrogen atoms are bonded to one oxygen atom.

Water is a polar substance. Thus, it has positive and negative charges on either end.

The hydrogen bond between hydrogen and oxygen that makes up water molecule is weak and short lived. However, the cumulative effect gives it its important properties. The attractive force between water molecules is known as **cohesion.** This cohesive property keeps water molecules to stay interlinked.

Characteristics and importance of water

Some of the importances of water include:

1. It serves as habitat or a place to live for variety of organisms.
2. It is a transport medium. Thus, water plays a vital role in transporting important substances in the body of organisms.

3. It is transparent. This property of water allows light to pass deep in to the water bodies in order to assist the photosynthetic process which is carried by aquatic plants and algaes.
4. It serves as a reactant in many chemical reactions. For example, water serves as a reactant in hydrolysis processes such as digestion.
5. It serves as a place for other reactions to take place.

- The most commonly known characteristics of water are:

a) Water has high specific heat

Specific heat is the amount of heat that must be absorbed in order for 1 gram of a substance to change its temperature in 1^0 Celsius. This means that large bodies of water like oceans, absorb a lot of heat and resist changes in temperature. This provides suitable environment for the organisms that live in them.

Because of its ability of having high specific heat water cools and looses heat quite slowly. Therefore, water is important for the proper functioning of enzymes in the body by maintaining body's temperature.

Note: Enzymes are sensitive to temperature

b) Water has a high latent heat of vaporization

Latent heat of vaporization is the amount of heat energy required to evaporate water. A lot of energy is required to return liquid water in to water vapor. As a result, evaporation of sweat significantly cools the body surface.

c) Water has high surface tension

surface tension is the tension at the surface of liquid resulting from unbalanced forces acting on the molecules at the surface. Thus, it is a measure of how difficult it is to break the surface of liquid. This property of water allows small organisms such as insects to walk on the water without breaking its surface.

High surface tension of water is brought due to strong attractive force between water molecules. This phenomena is called **cohesion.**

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d) Water is the universal solvent

solvent is a substance which dissolves the solute in order to form a solution. Thus, since water is highly polar molecule, it dissolves all polar and ionic substances. Accordingly, water is important for dissolving most of the substances in the body and transportation of various substances in and out of the body.

e) Water has high viscosity

Viscosity is the measure of how a fluid flows easily. Thus, water has high viscosity which makes it a transport medium.

Accordingly, viscosity of water helps for effective transportation of substances in the body.

f) Water has high adhesion property

Adhesion is the clinging of one substance to another. This property of water plays an important role in the survival of the plant by allowing capillarity action, which helps water flow up from the roots of a plant to the leaves.

Note: Capillarity is the ability of water to rise in fine tubes.

Illustrative Examples

1. All are organic compounds except:

A. H_2O B.CH_3COOH C. $C_5H_{10}O_5$ D. $C_6H_{12}O_6$

Explanation: Organic molecules necessarily contain carbon and hydrogen. But water molecule (H_2O) has only hydrogen atom. Thus, it is inorganic compound. **Answer: A**

2. Which of the following elements is the most abundant in our body?

A. carbon B. Hydrogen C. Nitrogen D. phosphorus

Explanation

Element	Percentage composition in the cell
Hydrogen	59%
Carbon	11%
Nitrogen	24%
Phosphorous and sulphur	2%

Answer: B

3. Sea lion can easily float on water. This is because:

A. The oil in the body of the animal is less denser than water and help it to float.

B. The fur under the skin give it large surface area to float on water.

C. The oil under the skin gives it larger surface tension to push against the water.

D. None

Explanation: - Surface tension on the surface of water results from unbalanced forces acting on the water molecules at surface. This enables small organisms to float on the surface of water. **Answer: A**

4. Two atoms of the same element are joined to produce:

A. compound B. Element C. Molecule D. Valance

Explanation: - Atomos of the same element combine and make up a molecule. Examples are N_2, H_2 and O_2.

- N_2 - - - Nitrogen molecule
- H_2 - - - Hydrogen molecule
- O_2 - - - Oxygen molecule.

Answer: C

5. Which of the following is true about atoms and molecules?

A. Atoms are the smallest particles of an element, while molecules are made up of more than one type of atoms.

B. Atoms combine to form a molecule, while a molecule contain just one type of atoms.

C. Atoms are the simplest particles of an element, while molecules always contain more than one atom.

D. Atoms combine to form particles, while molecules always contain more than one type of atom. **Answer: C**

6. The type of bond that exists between the hydrogen atom and oxygen atom to form water molecules is:

A. Hydrogen bond C. Covalent bond

B. Ionic bond D. Coordinate bond

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Explanation: - The bond that exists between hydrogen and oxygen atoms to form water molecule is **hydrogen bond.** Water's ability to form hydrogen bonds is responsible for much of the organization of living chemistry. Hydrogen bond is weak; however, its commulative effect is responsible for many important properties of water.

Answer: A

7. Which of the given spectra of light penetrates deep in to the ocean?

A. green B. Yellow C. Violet D. Blue

Explanation: -Blue light spectra penetrates deep in the ocean up to 300 meter.

Answer D

8. Which property of water enables plants in the water to photosynthesize?

A. solvent B. surface tension C. Transparent D. colorless

Explanation: - The transparency of water allows light to penetrate deep in to the water bodies. This allows plants (producers) in the water to get light for the photosynthesis.

Answer: C

9 Which of the following statement best describes the hydrolysis property of water?

A. Addition of hydrogen.

B. Addition of water to break polymer.

C. Removal of water to join monomer units.

D. strong attraction between water molecules.

Explanation: - Hydrolysis reaction is the reverse of condensation reaction which involves the addition of water in order to break polymers in to their monomer units.

Answer: B

10. What would happen, if water were more viscus?

A. Heart would unable to pump the blood.

B. Blood could not suspend organelle.

C. press and rub against delicate organs.

D. A and C are possible answers.

Explanation: - Viscosity is the measure of how fluid (liquid) easily flows. If the water were more viscous, the heart would not be able to move blood

through the blood vessels and if the water was less viscous, it would flow too easily and it would not carry or move the blood cells and other substances around the system efficiently.

Answer: A

11. What would happen if the water were viscus?

A. Blood can not carry blood cells.

B. The heart is unable to pump the blood.

C. It could not support the organelles.

D. A and C

Answer: D

12. Water has high specific heat because:

A. Dipole ends of water are joined by bond and this requires much amount of heat to break.

B. Hydrogen bond is strong and requires high energy to break apart water molecule.

C. The hydrogen bond insulates the heat to get in to water molecule.

D. Mass of water contains more hydrogen bond and most of the energy used to break the bond.

Answer: D

13. Which one is the biological advantage of water in having high specific heat capacity?

A. Enzymes work at constant temperature.

B. Water climbs up through xylem vessel.

C. Water walks on the surface of vessels.

D. to dissolve polar compound.

Explanation: -Because of its property of having high specific heat capacity water it gains and looses heat very slowly. This allows organisms to keep their body temperate constant which is very important for enzymatic activity.

Answer: A

14. Ice floats on top of water, this property is important:

A. To insulate the bottom part of the water from freezing

B. To insulate the heat produced by underwater organism from escaping out.

C. To facilitate the nutrient cycle.

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D. All are possible answers

Explanation: - Ice is less dense than liquid water. Thus, ice on the surface of water acts as an insulator and slows down the heat loss from the liquid underneath. This enables life under water to continue at warm condition. It also plays a role in facilitation of nutrient cycle in the water. **Answer: D**

15. Which property of water enables smaller insects to walk on the water?
    A. High heat of vaporization.
    B. High surface tension of water.
    C. High heat of fusion.
    D. High specific heat of water.

Explanation: - Look at example number 3 **Answer: B**

16. Which of the following is incorrect about the property of water?
    A. The only to exist in three states.
    B. It is vital constituent of a living cell.
    C. It is more viscous than other fluids.
    D. It is a medium for metabolic reactions.

Explanation: - Water is not more and less viscous than other fluids because of the reasons mentioned on example number 10. **Answer: C**

17. Of the constituents of protoplasm, which one is inorganic in nature?
    A. starsch B. protein C. water D. None

Explanation: -Water is inorganic by nature. This is because all organic substances necessarily contain carbon and hydrogen. **Answer: C**

18. Which of the following groups of substances are all in organic?
    A. Water, sugar, calcium carbonate
    B. Water, calcium carbonate, carbondioxide.
    C. Sugar, Fatty acid, amino acid
    C. Carbondioxide, aminoacid, fatty acid

Explanation: -

In organic	Organic
• Water • Calcium carbonate • Carbondioxide	• Sugar • Fatty acid • Aminoacid

Answer: B

19. The colomn of water continuously climb up through xylem vessel, because:
    A. Water has ability to absorb more heat.
    B. Water molecule bind together by hydrogen bond.
    C. Water expands through xylem vessel when it cools.
    D. Water has high surface tension.

Explanation: -Water mulecules climb up though xylem vessels for various reasons. These include:

- Adhesive property of water.
- Cohesive property of water.
- hydrogen bond between water molecules.
- Root pressure.
- Transpiration pull, etc.

Answer: B

20. Which is NOT the characterstics of water?
    A. Water has high specific heat.
    B. Water has high heat of vaporization.
    C. Water exhibits strong cohesion tension.
    D. Water is less dense than ice.
    E. Water is known as universal solvent

Explanation: -All properties of water mentioned are correct except 'D' Ice is less dense than water. **Answer: D**

2.2. Organic Molecules

Organic molecules are molecules that contain skeletons of covalently bonded carbon atoms. They necessarily contain atoms of carbon and hydrogen. These molecules are known as **organic molecules,** because once it was believed that they were thought to be produced by living things. Organic molecules are relatively large and complex as they are compared to inorganic molecules.

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Biologically important groups of organic molecules are:

- Carbohydrates
- Lipids
- Proteins
- Nucleic acids

1. Carbohydrates

Carbohydrates are organic molecules that are composed of only three elements. They are **carbon, hydrogen** and **oxygen**. The term carbohydrate meaning "hydrate (water) of carbon" which stems from the 2:1 ration of hydrogen to oxygen. Therefore, you have to always remember that the ratio of hydrogen and oxygen atoms in a carbohydrate molecule is always two to one. The general formula of carbohydrates can be indicated as:

$(CH_2O)_n$ .... For $n \geq 3$

Generally all carbohydrates have the following properties:

- All are Ketones and aldehydes.
- All contain several hydroxyl groups.
- Their chemistry is determined by their hydroxyl groups.

Carbohydrates have various importance and functions for living things. These are:

- They are the quickest and most readily available form of energy for the body.
- They serve as the storage forms of energy in plants, animals and other organisms.
  - **Starch** is the storage form of carbohydrate in plants.
  - **Glycogen** is the storage form of carbohydrate in animals.
- They are used as the structural components in different types of living organisms.
  - **Cellulose** – is the major component of cell wall in plants.
  - **Chitin** is the structural component of fungial cell wall.
  - **Peptidoglycan** is the structural component of bacterial cell wall.

Based on the number of sugar units which they contain carbohydrates can be classified in the three types. These are:

- Monosaccharides (simple sugars)

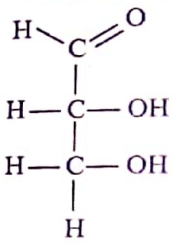
- Disaccharides (double sugars)
- Polysaccharides (complex sugars)

a) Monosaccharides

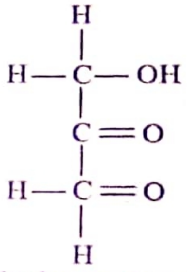
Monosaccharides are simple sugars that contain three to seven carbon atoms. They are the simplest and soluble forms of carbohydrates.

- They contain one sugar units. As it is mentioned above one sugar unit contains three to seven carbon atoms. Thus, some groups of simple sugars can be discussed as:

i) **Trioses** – contain three carbon atoms. They include glyceraldehydes, dihydroxy acetone which are intermediate components of in the process of photosynthesis.



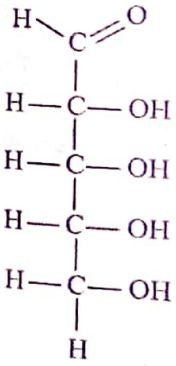
Glyceraldehydes
$C_3H_6O_6$



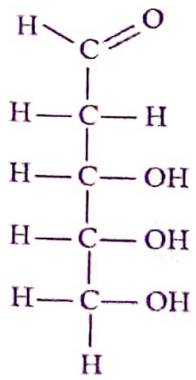
Dihydroxy acetone
($C_3H_6O_3$)

ii) **Pentose** – are five carbon sugars which serves as components of nucleic acids (RNA and DNA) and coenzymes (NAD, NADP).

∴ We will see more about coenzymes (NAD and NADP) under unit five.



Ribose
($C_5H_{10}O_5$)



Deoxyribose
($C_5H_{10}O_4$)

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- The straight structure of monosaccharides appear in rings when they are in solution. The straight form of glucose produces two ring forms. Namely, **alpha (α) glucose** and **beta (β) glucose.**

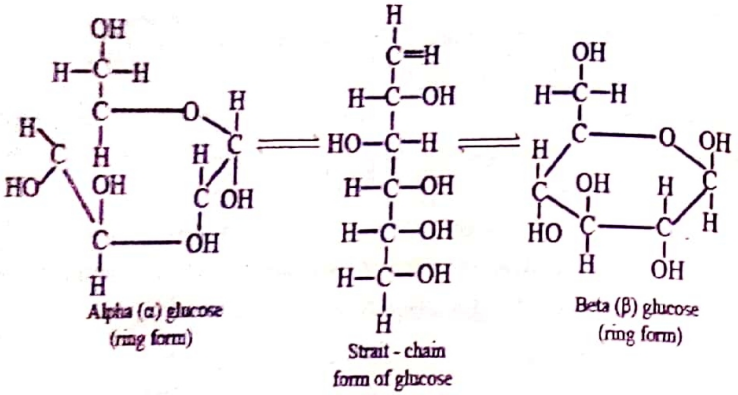


Fig. Lenear and ring forms of glucose

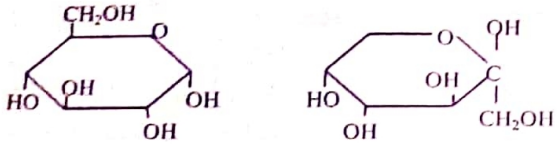
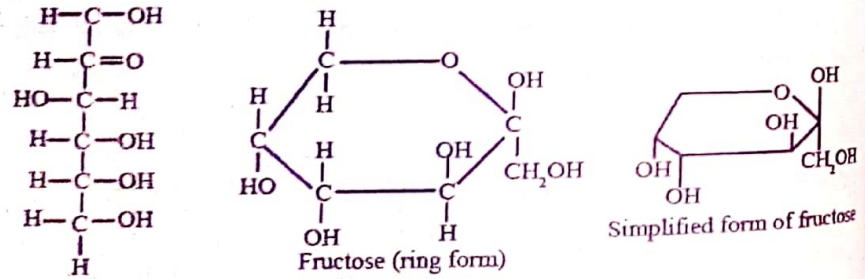


Fig. Simplified ring structure of glucose

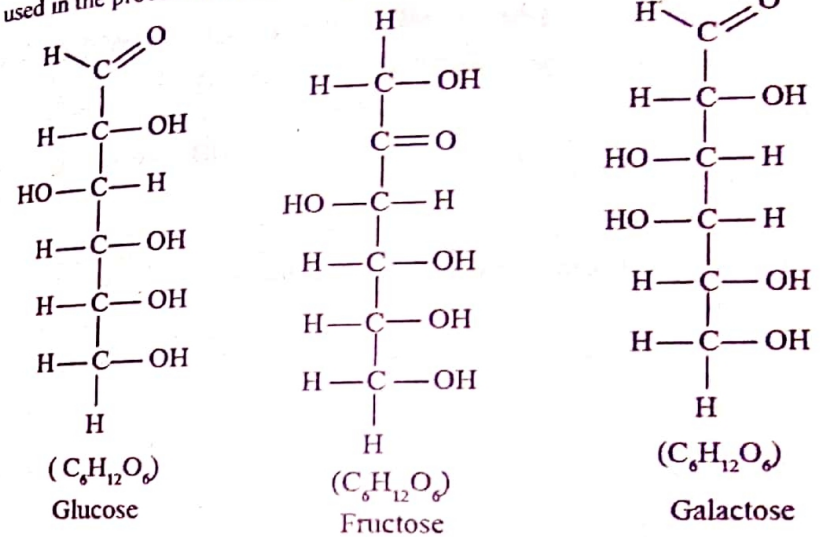
- Fructose has only one ring form



iii) Hexoses - are six carbon sugars. They include glucose, galactose and fructose which are all **isomers** of each other.

- Glucose ($C_6H_{12}O_6$) is the most abundant monosaccharide which is extreamly important in life processes. It is synthesized in the process of photosynthesis by green plants and algaes.

Glucose is used mainly as the raw material for the main cellular fuels used in the process of cellular respiration.



Note: Molecules that have the same chemical composition but different in structural arrangement are called isomers. Accordingly Trioses, Pentoses and hexoses have various isomeric molecules. For example hexoses such as glucose, galactose and fructose are isomers.

Based on the functional group which they posses monosaccharides can be classified as:

i) Aldoses – contain aldehyde group with the formula CHO.

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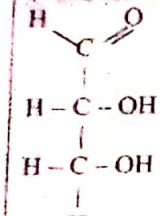
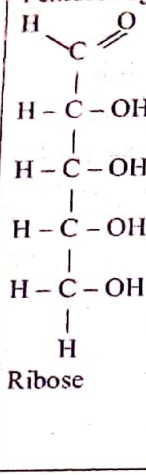
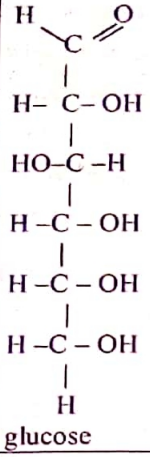
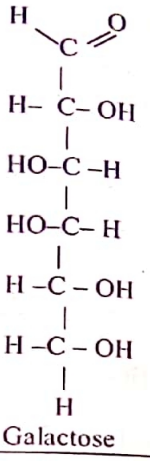
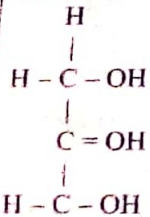
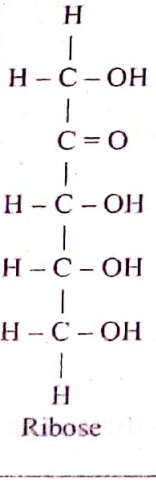
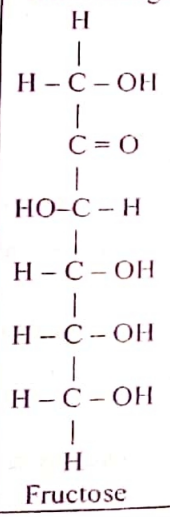
Examples:

Triose sugars	Pentose sugars	Hexose sugars	
($C_3H_6O_3$)			
			
Glyceraldehyde	Ribose	glucose	Galactose

ii) Ketoses – Contain Ketone groups with the Formula C = O.

Examples:

Triose sugars	Pentose sugars	Hexose sugar
($C_3H_6O_3$)	($C_5H_{10}O_3$)	
		
Bihydroxy acetone	Ribose	Fructose

c) Polysaccharides

Polysaccharides are complex and stable forms of carbohydrate in the body. They are composed of repeating units of simple sugars, usually glucose.

Thus, polysaccharides are macromolecules. Polysaccharides are the most abundant groups of all carbohydrates.

The monomer units (sugars) that make a polymer (complex sugar) are joined by condensation reaction.

The number of sugar that present in polysaccharide molecule varies from one molecule to the other.

Polysaccharides may appear in a single long chain or branched chain.

Polysaccharide molecules have very different properties because:

- They are composed of different stereoisomers.
- The glucose units are arranged differently.

Polysaccharide molecules have various importances. Among these some of them are:

- They are storage forms of carbohydrate in plants and animals.
- They are structural components of plant, Fungial and bacterial cell walls. As mentioned earlier:
  - **Cellulose** is the structural component of plant cell wall.

Some of biologically important polysaccharide molecules include:

i) Starch

Starch is the storage form of carbohydrate in plants. It is a polymer consisting of repeating units of glucose molecules. The monomers (glucose sub units) are joined by 1-4 linkages. Starch molecule occurs in two forms, namely **amylase** and **amylopectin.**

- **Amylose** is the simpler form which exists in a lenear form (unbranched). It winds in to helix when in contact with water which allows the reaction between starch and iodine solution. The amylase iodine solution which appears blue.
  - Glucose molecules are joined by 1, 4-glycosidic bond.
- **Amylopectin** is complex and branched. It is a more common form that usually consists of about 1000 units in unbranched chain.

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- Branching takes place at about every 20 to 25 units and involves a 1-6, glycosidic linkage. However, on the lenear part of amylopecting glucose molecules are joined by 1-4, glycosidic linkage.
- The branched nature of amylopectin makes it to have many ends which allows it to be quickly hydrolyzed by enzymes in order to release glucose molecules.

Fig. structure of amylopectin

Starch molecule is suited to its function because:

- ✓ Both amylase and amylopectin are compact molecules, so they can be stored in small space and do not affect the metabolic process.
- ✓ Both amylase and amylopectin are insoluble in water.
- ✓ Since amylopectin have many ends they can be quickly hydrolyzed to release glucose. This process requires enzymatic action.

ii) Glycogen

Glycogen is sometimes referred to as **animal starch**, since is the storage form of carbohydrate in animals. It is more soluble than starch. This is because glycogen is more branchned than amylopectin or it has more α - 1, 6 glycosidic linkages, it can be easily hydrolyzed. This ability of glycogen is very important since animals have higher metabolic rate.

Glycogen is mainly stored in the liver and muscle cells of animals.

iii) Cellulose

Cellulose is the structural component and most abundant carbohydrate in plants which accounts for 50% or more of all the carbon in plants.

Cellulose is an important structural component of plant cell walls and it is compose of repeating units of β-glucose molecules joined by β-1, 4-glycosidic linkage.

The glycosidic bonds in cellulose molecules are splitted by enzymes. These enzymes are lacking in humans, so we are unable to use cellulose as a nutrient.

However, ruminant animals have symbiotic bacterias that produce enzymes for the digestion of cellulose.

Note: Cellulose is unbranched polysaccharide that is compased of approximately 10,000 glucose units joined by glycosidic bonds.

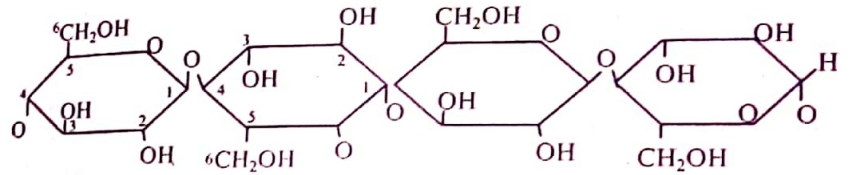

Fig: cellulose (formed by 1-4 linkage β - glycose monomers)

Illustrative Examples

21. Which of the following is the function of carbohydrate?
    A. substrate for respiration.
    B. structural components of cells.
    C. store information.
    D. store energy.

Explanation:-Carbohydrates have wide range of importances.

Among these some of them are:

- They are substrates for the process of respiration.
- They are convenient to store chemical energy.
  * Examples: Starch and glycogen.
- They are structural components of a cell.
  Examples: cellulose, chitin

22. Carbohydrates are classified based on

Answer: C

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A. The number of carbon they contain
B. The number of sugar unit
C. Their role
D. Their structure

Explanation: - Based on the number of sugar (saccharide) unit they contain, carbohydrates can be classified as:

- Monosaccharides (simple sugars).
- Disaccharides (double sugars).
- Poly saccharides (complex sugars).

∴ One sugar unit consists of 3-8 carbon atoms. **Answer B**

23. The functional groups of crbohydrate are:

A. SH and COOH
B. NH_2 and COOH
C. C=O and COH
D. OH and C

Explanation: The functional groups of carbohydrate are aldoses (CHO) and ketoses (C = O) **Answer: C**

24. Which of the following monosaccharide is ketose?

A. Glucose B. Fructose C. Galactose D. All are possible

Explanation:- Glucose and galactose are **aldoses** because they have C = O group on the first carbon.

- Fructose is **ketose** which have carbonyl group bonded to two other carbons. **Answer: B**

25. The sugars that are used to make nucleic acids are:

A. Glucose and Fructose
B. Sucrose and ribose
C. Deoxyribose and glucose
D. Deoxyribose and ribose

Explanation:- Among the nucleic acids:

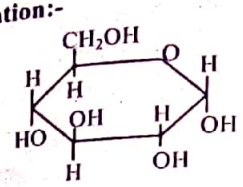
- DNA consists of deoxyribose sugar
- RNA consists of ribose sugar **Answer: D**

26. Alpha (α) glucose differs From Beta (β-glucose) in:

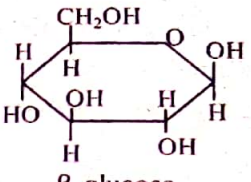
A. The -OH group is absent in the 1st carbon of α-glucose.
B. The -OH group on carbon number one of α-glucose is mirror image of β-glucose.

C. There is no difference in the arrangement of their OH- groups.
D. α-glucose has double bonds, while β-glucose does not posses double bonds.

Explanation:-



α - glucose



β-glucose

Answer: B

27. The process by which disaccharides are produced is:

A. Condensation reaction between two monosaccharides
B. Hydrolysis of monosaccharides
C. Peptide bond between monosaccharides
D. All are possible

Explanation:- Condensation reaction involves joining of two molecules to form larger molecule by producing water as a biproduct. Thus, disaccharides are formed by the condensation reaction between two monosaccharides. $C_6H_{12}O_6 + C_6H_{12}O_6 \longrightarrow C_{12}H_{22}O_{11} + H_2O$

Answer: A

28. The reaction in which two molecules combine to form a larger molecule and water is produced as a bi-product is:

A. Hydrogenation
B. hydrolysis
C. condensation
D. Dehydrogenation **Answer: C**

29. Which of the following describes hydrolysis?

A. Taking up excess hydroxide ions.
B. Making a polymer by removing water.
C. Making water by combining on acid and base.
D. Adding water to break a polymer in to monomer units.

Explanation: The process of adding water to a polymer inorder to break it in to its monomer units is called **hydrolysis reaction.**

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Answer: D

30. All are structural components of living organisms except:
    A. Glycogen C. chitin
    B. cellulose D. All are possible

Explanation:- Cellulose is the structural component of plants cell wall and chitin is the structural component of fungial cell wall. However, glycogen is the storage form of carbohydrate in animals.

Answer: A

31. Amylopectin hydrolyzes faster than amylase, because:
    A. it winds itself into helix when contracts.
    B. It has branches.
    C. It is simpler.
    D. It is unbranched.

Explanation:- Amylopectin is branched, thus it creases its surface area to Be easily hydrolyzed by water.

Answer: B

32. Which of the following is an important component of the exoskeleton of grasshopper?
    A. cuticle B. chitin C. protein D. oil

Explanation:- Chitin which is a complex sugar is a structural component of exoskeleton of insects and cell wall of Fungi.

Answer: B

33. How many carbon atoms are contained in a single molecule of sucrose?
    A. 6 B. 12 C. 24 D. 48

Explanation: Sucrose has a molecular formula of $C_{12}H_{22}O_{11}$. Thus, the number of carbon atoms contained in sucrose molecule are 12 carbon atoms.

Answer: B

34. Which of the following sugars do babies get from the milk of their mothers?
    A. Maltose B. Sucrose C. Lactose D. Fructose

Explanation:-

Sugar	Found in
Maltose	Found in germinating seeds like barley.
Sucrose	Found in sugar cane and table sugar
Lactose	Found in milk. _____

Answer: C

35. 1, 4-glycosidic bond is absent in
    A. Maltose B. Lactose C. Starch D. Sucrose

Explanation:- Sucrose which is formed by condensation reaction of glucose and Fructose is joined by α-1, 2 glycosidic bond.

Answer: D

36. Hexoses are:
    A. Disaccharide molecule that contain six carbon atoms.
    B. Monosaccharides that contain six hydrogen atoms.
    C. Monosaccharides that contain six carbon atoms.
    D. Disaccharides that contain six oxygen atoms.

Explanation: Hexoses are monosaccharides that contain six carbon atoms ($C_6H_{12}O_6$).

Answer: C

37. Which of the following is a polysaccharide?
    A. Glucose B. Glycogen C. Maltose D. Lactose

Explanation:

Type of Sugar	Example
Monosaccharide	Glucose, galactose, fructose
Disaccharide	Sucrose, Maltose, Lactose
Polysaccharide	Starch, Chitin, Cellulose, Glycogen

Answer: B

38. How many water molecules can be produced from 100 monosaccharides joined to form a polysaccharide?
    A. 100 B. 90 C. 99 D. 50

Answer: C

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39. The molecular formula of glucose is $C_6H_{12}O_6$. What is the polymer of 12 units of glucose that have been linked by condensation reaction?

A. $C_{12}H_{144}O_{72}$ B. $C_{72}H_{122}O_{61}$ C. $C_{61}H_{144}O_{72}$ D. $C_{24}H_{44}O_{24}$

Explanation: When 12 units of glucose molecule are joined by condensation reaction 11 molecules of water (H_2O) will be removed. Thus,

$= 12\ (C_6H_{12}O_6) - 11\ (H_2O)$

$= C_{72}H_{144}O_{72} - H_{22}O_{11}$

$= C_{72}H_{122}O_{61}$

Answer: B

40. Which of the following can produce more glucose by hydrolysis reaction?

A. Maltose B. Sucrose C. Disaccharide D. Starch

Explanation:- Starch is a complex sugar which consists of long chain of glucose units which are joined by glycosidic bonds. Thus, if starch is hydrolyzed many glucose molecules are produced.

Answer: D

41. The glucose branching from the strait chain of α-glucose is joined by which o the following bond?

A. α-1, 6 glycoside C. α-1, 4 glycoside

B. α-1, 2 glycoside D. None

Explanation:- The glucose that branches from straight chain of α-glucose molecule is joined by α-1, 6 glycosidic linkage. For example, amylopectin branches on α-1, 6 glylosidic linkage. **Answer: A**

42. Amylopectin differs from amylase by:

A. It is made up of α-glucose monomer

B. It forms helix in water.

C. It is branched.

D. It is a polysaccharide

Amylopectin	Amylase
• It is complex and branched • It consists of about thousand units which are joined by glycosidic bond. • Branching occurs at every 20 to 25 carbon. • Easily hydrolyzed	• It is linear (straight) chain of molecule which is made up of hundreds of α-glucose units joined by glycosidic bond. • It gets hydrolyzed slowly in comparision.

Answer: C

43. An enzyme that breaks amylase in to maltose is:

A. Maltose B. Amylase C. Cullulose D. Invertase

Answer: B

44. Iodine gives blue color in starch, because:

A. staining changes white starch in to blue black.

B. Iodine molecule is embedded in the helix and changes the light absorbing property.

C. The iodine dissolves the starch and make it blue black.

D. None

Answer: B

45. The polysaccharide which is most abundant in nature is:

A. starch B. Glucose C. cellulose D. Glycogen

Explanation:- Cellulose is the most abundant carbohydrate that accounts for 50% or more of all carbons in plants and it is the most abundant organic molecule on the eart. **Answer: C**

46. Which of the following glycosidic linkage forms glycogen?

A. α-1, 4 linkage

B. α-1, 6 linkage

C. α-1, 4 – straight chain and α-1, 6 branch

D. α-1, 4 straght chain and β-1, 4 branch

Explanation:- The glycogen is formed by straight chains of α-1, 4 glycosidic linkange and branchs by α-1, 6 linkage.

Answer: C

47. Which of the following is produced by unbranched chain of glucose joined by β-1, 4 glycosidic bond?

A. Amylase B. Amylopectin C. cellulose D. Glycogen

Explanation:- In cellulose β- glucose is the monomer and the linkages are

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β -1, 4 bond.

Answer: C

48. Which of the following cannot be digested by human gut?

A. cellulose B. Starch C. Glycogen D. All are possible

Explanation:- Human gut has no enzymes that can digest cellulose. While, ruminant animals have symbiotic bacteria that produce cellulose digesting enzymes in their gut.

Answer: A

49. The main difference between cellulose and starch molecule is:

A. Starch dissolves in water, while cellulose is insoluble.

B. They are made up of different types of glucose.

C. cellulose cannot be broken in to monomer units, while starch can.

D. None of the above.

Explanation:- Cellulose is made up of β-glucose, while starch is made up of α-glucose.

✓ Cellulose and starch share the following common properties:

- They do not dissolve in water.
- They can be broken down in to their monomer units.

Answer: B

50. Which of the following pairs of elements are found in all carbohydrates in addition to the element carbon?

A. Nitrogen and oxygen C. Hydrogen and nitrogen

B. Hydrogen and oxygen D. phosphorous and nitrogen

Explanation:- All carbohydrates are composed of three types of elements. Namely **carbon, hydrogen** and **oxygen.**

Answer: B

51. If one mixes a sample of fruit juice and some drops of benedicts solution and obtains a brick-red precipitate up on warming the mixture, what does the juice contain?

A. starch B. Reducing suga C. sucrose D. protein

Explanation:- Reducing sugars are monosaccharides and disaccharides which change a fruit juice in to a brick red precipitate up on the addition of Fheling or Benedicts solution.

Answer: B

52. All of the following carbohydrates are polysaccharides except:

A. Glucose B. cellulose C. starch D. Glycogen

Explanation:- Polysaccharides are complex sugars which are composed on a chain of sugars that are joined by glycosidic linkage. Some examples of polysaccharides are **cellulose, starch, glycogen** and **chitin.**

Answer: A

53. Which of the following is a polysaccharide?

A. sucrose B. Glucose C. cellulose D. Fructose

Answer: C

54. In which of the following compounds are both members of the pair are polymers of carbohydrates that naturally occur in plants? (UEE 2005)

A. starch and chitin C. Glycogen and cellulose

B. starch and cellulose D. Glycogen and chitin

Explanation:- Starch and cellulose are commonly found in plants because:

- **Starch** is the storage form of carbohydrate in plants.
- **Cellulose** is the structural component of plant cellwal.

Answer: B

55. Which one of the following molecules in living things is NOT an organic molecule? (UEE 2006)

A. sugar B. Nucleotide C. Amino acid D. Water

Explanation:- All except water are organic substances.

Answer: D

56. How many carbon and oxygen atoms are there in a molecule of maltose?

A. 6 carbon and 6 oxygen C. 11 carbon and 12 oxygen

B. 12 carbon and 11 oxygen D. 24 carbon and 12 oxygen

Answer: B

57. Choose the one that represents a chemical formula of an organic compound?

A. $C_{12}H_{22}O_{11}$ B. $Mg(OH)_2$ C. NH_3 D. NaCl

Explanation:- Organic substances necessarily contain carbon and hydrogen

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atoms. Thus, $C_{12}H_{22}O_{11}$ is organic substance.

Answer: A

2. Lipids

Lipids are diverse groups of organic substances that are composed of elements **carbon, hydrogen** and **oxygen.** However, the proportion of oxygen is much more less than that of carbohydrates. They have oily or greasy consistency and are relatively insoluble in water. However, they are soluble in non polar organic solvents like benzene and alcohols.

There are various groups of lipids, among these biologically important groups of lipids include.

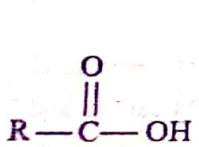
- Neutral fats
- steroids
- phoshplipids
- waxes
- carotenoids

Most of the lipid consists of **fatty acid** and **glycerol** molecules.

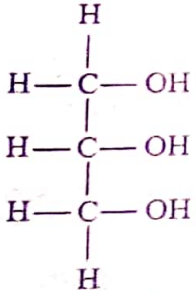
i) Neutral fats

Neutral fats are the most abundant groups of lipids in living things. They are rich in their energy content which is more than twice as much energy per gram as do carbohydrates. Neutral fats are composed of glycerol and fatty acid molecules. Glycerol is joined to one, two or three molecules of fatty acids. Accordingly:

- **Monoglyceride** – when glycerol combines chemically with one fatty acid.
- **Diglyceride** – when glycerol combines chemically with two fatty acids.
- **Triglyceride** – when glycerol combines chemically with three fatty acids.



Fatty acid



Glycerol

Glycerol is a three carbon alcohol that contains three - OH groups and **fatty acid** is a long, straight chain of carbon atoms with carboxyl group (-COOH) at one end.

There are about 30 different types of fatty acids which are commonly found in animal lipids.

The formation of neutral fats involves a condensation reaction between glycerol and fatty acids. Fatty acids are joined to glycerol by an **ester linkage** of hydroxyl groups (-OH) from glycerol and carbon (COOH) groups of the fatty acids.

Triglycerides

Triglycerides are the most commonly known groups of lipids that occur in nature. They can be classified as fats and oils. Fats are solids and oils are liquids at room temperature. The biological properties of triglycerides is determined by:

- The length of chain of carbon atoms.
- The degree of saturation.

Accordingly, they can be grouped as **saturated** and **unsaturated fatty acids.**

i) **Saturated fatty acids** contain maximum possible number of hydrogen atoms and all the carbon. Carbon bonds in the hydrocarbon chain are single bonds.

- Saturated fats tend to be solids at room temperature
- Examples for saturated fat are butter and animal fat.

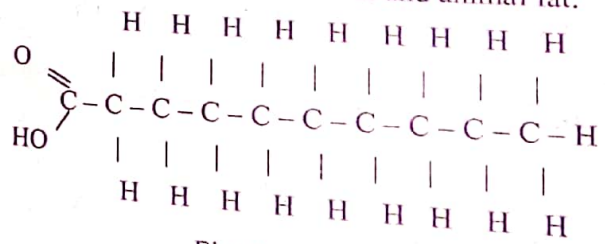


Fig. Saturated fatty acid

Illustrative Examples

58. All are the differences between lipid and carbohydrate except:

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A. All lipids are insoluble in water, while most carbohydrates dissolve in water.

B. Lipids produce twice as much energy than the same mass of carbohydrate.

C. Lipids contain much oxygen than the same mass of carbohydrate.

D. None

Answer: D

59. A tissue responsible for storage of fat is:

A. Muscle B. Nerve C. Connective D. Adipose

Explanation:- Fat is mainly stored in the inner layer of the skin called **hypodermis** which mainly consists of adipose tissues.

Answer: D

60. Hydrolysis of triglyceride yields:

A. One glycerol and three fatty acids.

B. Two glycerol and two fatty acids.

C. One fatty acid and three glycerols.

D. One glycerol and two fatty acids.

Explanation:- Triglyceride is composed of a reaction between one glycerol and three fatty acids.

Answer: A

61. The functional group of fatty acids is:

A. Hydroxyl C. sulphahydral

B. carboxylic D. NH_2

Explanation:- The functional groups of fatty acids are – COOH (Carboxylic groups)

Answer: B

62. Which of the following is true about saturated fatty acids? (UEE 2006 E.C)

A. They have single bonds between carbon atoms.

B. They are the same as polyunsaturated fatty acids.

C. They are the same as monounsaturated fatty acids.

D. They have double bonds between the carbon atoms.

Explanation:- See example number 67

Answer: A

63. How are unsaturated fats distinguished? (UEE, 2006 E.C)

A. They are made up of glucose and fructose.

B. They are made up of aminoacids and glycerol.

C. They have double bonds in their carbon chain.

D. They have only single bonds at the points of attachment.

Explanation:- See example number 67

Answer: C

64. Which of the following is produced when a lipid molecule is hydrolyzed? (UEE, 2006 E.C)

A. Aminoacid and water C. Fatty acids and glycerol

B. Aminoacid and glucose D. Glucose and glycerol

Explanation:- Hydrolysis of lipid molecule forms **fatty acid** and **glycerol** molecules.

Answer: C

65. Which of the following is generally expected to give better health benefit when present in human diet? (UEE 2005E.C)

A. Animal fat C. Monounsaturated fatty acid

B. saturated fatty acid D. polyunsaturated fatty acid

Explanation:- Polyunsaturated fatty acids help to prevent cholesterol being laid down in the linings of arteries and prevent heart disease.

Answer: D

66. Distingush between fatty acid and glycerol

Explanation:

Glycerol	Fatty acid
• It is an alcohol • It is made up of 3 carbons having each end with 'OH'. • It is the back bone of triglyceride	• It is an acid • It is covalently bonded with hydrocarbon at each end with COOH. • It is a tail of triglyceride

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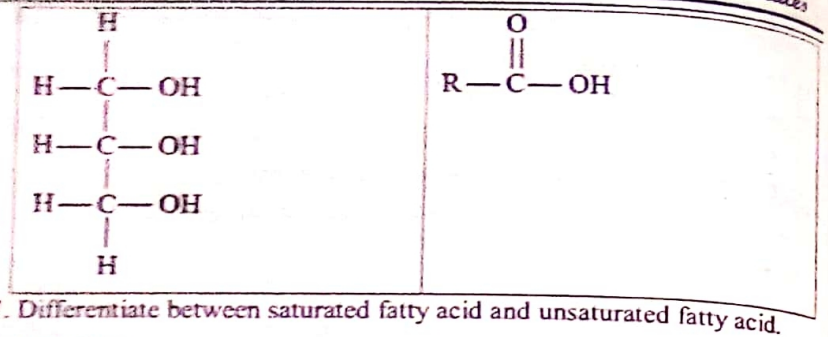

67. Differentiate between saturated fatty acid and unsaturated fatty acid.

Explanation:

Saturated fatty acid	Unsaturated fatty acid
• contain single bond between each carbon atom • Each carbon contains maximum number of hydrogen (saturated with hydrogen) • solid at room temperature since the fatty acid align together to resist heat. • More energetic since it contains larger number of hydrogen	• One or more multiple bonds between carbon atoms. • The double or triple bonds replace (reduce) the number of hydrogen. • Liquid at room temperature since the fatty acids separate at the point of double or triple bond. • Less energetic because of less number of hydrogen

68. All are lipids except:

A. chlophyll B. steriod C. cholesterol D. chitin

Explanation:- Chitin is a polysaccharide carbohydrate.

Answer: D

69. Compared to saturated fat, unsaturated fat contains less?

A. oxygen B. Fatty acid C. Glycerol D. Hydrogen

Explanation: See example number 67

Answer: D

70. Oils are liquids at room temperature. This is because

A. They are made up of unsaturated hydrocarbon.

B. They are made up of saturated hydrocarbon.

C. They are mainly composed of longchain of glycerol.

D. They have high proportion of oxygen.

Explanation:- Oils are unsaturated fatty acids. Thus, unsaturated fatty acids contain triglycerides that can easily separate at the double bond.

Answer: A

71. Lipids that are made up of unsaturated hydrocarbon have low melting point. This is because:

A. Double or triple bond is weaker than single bonds.

B. The strengths of bonds reduce because of less number of hydrogen.

C. The triglycerides separate at the point of double or triple bond.

D. The presence of low concentration of oxygen reduce the bond strength.

72. Which of the following is true about fatty acids?

A. As the degree of unsaturation increases, the melting point also increases.

B. As the degree of unsaturation increases, there will be more double bonds and the lipid becomes more liquid.

C. As the degree of unsaturation increases, the amount of energy contained in the lipid also increases.

D. As the chain length increases, the melting point decreases.

Explanation:- See example number 67

Answer: C

73. Which of the following statement is true about saturated fatty acids?

A. saturated fat contain one or more double bond

B. saturated fat contain maximum number of hydrogen

C. saturated fat make up the majority of plant oil.

D. food composed of saturated fat are good for our health

Explanation: Saturated fatty acids:

• contain maximum number of hydrogen

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- contain single bonds between each carbon atom
- They are mostly found in animal fat which is not good for our health.

Answer: B

74. The most abundant type of lipid in cell membrane is:

A. phospholipid C. channel protein
B. steroid D. carrier protein

Explanation:- Phospholipids are the most aboundant types of lipids in the cell membrane because they exist in a bi-layer form in which the hydrophobic tails are sandwiched between the hydrophilic tails.

Answer: A

75. The lipid that is used to make sex hormone is:

A. Triglyceride B. Wax C. Neutral fat D. Steroid

Explanation:- Steroids are molecules that contain four interlocked rings. The biological importance of steroids includes:

- Cholestrol
- Bile salts
- Male and female sex hormones.
- Hormones secreated by adrenal cortex.

Answer: D

76. Which of the following lipids are stored below the dermis of homeotherms?

A. Lipids made up of unsaturated fatty acids.
B. Lipids that are made up of unsaturated fatty acid.
C. Lipids that are made up of cholesterol.
D. None

Explanation:- Saturated fatty acids are solids at room temperature. Thus, they have a tendency to be stored under the skin.

Answer: B

77. Which of the following are NOT included in lipids?

A. Nitrogenous bases C. steroids
B. Waxes D. phospholipids

Explanation: Nitrogeneous bases are rather protein by their content. They are components of nuctleotides.

Answer: A

78. Which of the following helps to the biological membrane in a fluid state?

A. Waxes B. phospholipids C. cholesterol D. water

Answer: B

79. What makes phospholipid molecules arrange themselves in to a bi-layer in water medium?

A. The double bonds of the fatty acids.
B. The length of the fatty acid chains.
C. The presence of hydrophilic and hydrophobic groups.
D. Their high solubility in water.

Answer: C

80. The property of triglyceride depends on:

A. Double or triple bond on the fatty acid.
B. The length of glycerol.
C. The property of glycerol.
D. None of the above.

Explanation: The property of triglycerides is determined by:

- The length of the chains
- The degree of saturation.

Answer: A

Proteins

- Proteins are organic molecules that have central importance in chemistry of life. They contain elements carbon, hydrogen, oxygen, nitrogen and often sulphur and phosphorous.
- Proteins are polymers that consists of repeating units called **amino acids** which are joined by **peptide bonds.**
- Amino acids consist of a carboxyl group, an amino group, and R-group which is variable and attached to a central carbon atom. Amino acids are the building blocks of protein.

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- There are about 20 different types of aminoacids that are responsible for building thousands of different types of proteins. Amino acids differ from one another by their R-groups of side chains which are bonded to the alpha carbon.

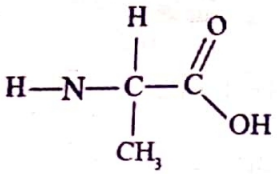
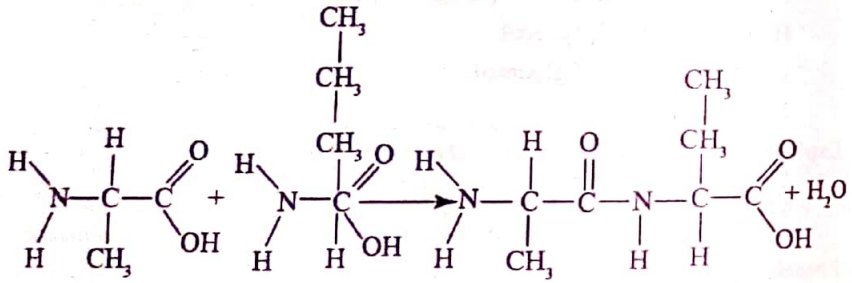


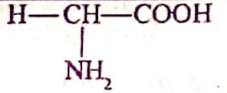
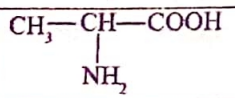
Fig. The general structure of aminoacids

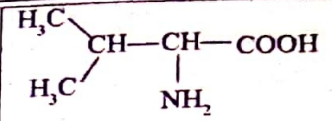
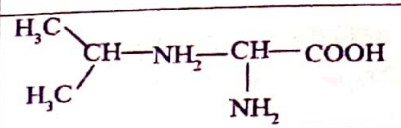
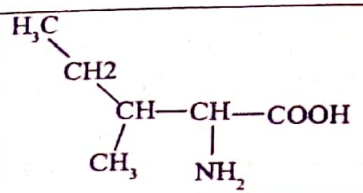
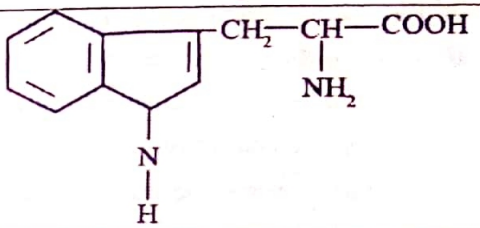
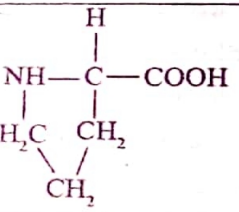
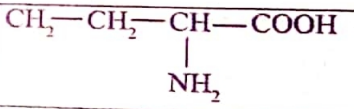
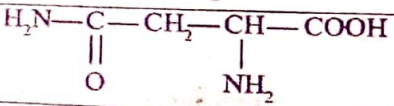
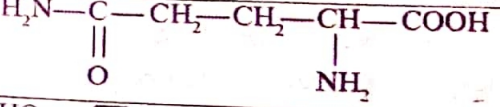
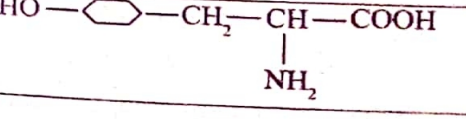
- The peptide bond between a chain of amino acids in making protein molecule is called **peptide chain.** When one molecule of amino acid joins with the other aminoacid molecule, water is released as a byproduct since they join by **condensation reaction.**



Amino acid Amino acid Depeptide

Some of the amino acids that are commonly found in proteins

Name	Structure
Glycine (Gly)	
Alanine (Ala)	

Valine (Val)	
Leucine (Lev)	
Isoleucine (Ile)	
Tryptophan (Trp)	
Proline (Pro)	
Methionine (Met)	
Asparagine (Asn)	
Glutamine (Gln)	
Tyrosine (Tyr)	

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Levels of organization in protein structure

As mentioned above, the function of proteins depend on their shapes. The shapes of proteins can be classified in to four levels of structure. These are **primary, secondary, tertiary** and **quaternary structures.**

i) **Primary structure** – is the sequence of amino acids in a polypeptide chain. Thus, it is the leaner sequence of aminoacids.

- The primary structure of each protein is coded by DNA which determines the type and position of each aminoacid in the polypeptide chain.

Example: - Insulin hormone which is secreated by the pancreas consists of 51 amino acid units in two linked chains.

ii) **Secondary structure** – involves coiling of peptide chain in to helix or some other irregular conformation. It is formed when a peptide chain folds or coils in to a helix or α-plated sheet.

- The common secondary structure is proteins molecule known as **alpha (α) helix.** The helical structure is maintained by the formation of hydrogen bonds between aminoacids in successive turns of spiral coil.
- In alpha helical structure the hydrogen bonding occurs between atoms within the same polypeptide chain.
- Alpha helix is the basic structural unit of fibrous proteins such as wool, hair, skin and nails.
- Beta plated sheet is a second type of secondary structure proteins.
- In β- plated sheet the hydrogen bonding takes place between polypeptide chains.

iii) **Tertiary structure** – involves further folding of the secondary structure and formation of new bonds to hold the tertiary structure in place.

- The tertiary structure of proteins:
  - ✓ has three dimensional structures

- The new bonds include:
  - ✓ Ionic attraction between R-groups of aminoacid subunits in adjacent loops of the same polypeptide chain.
  - ✓ Ionic attraction between R – groups with positive charges and those with negative charges.
  - ✓ Hydrophobic interactions resulting from non polar R-groups to associate in the anterior of the globular structure away from the surrounding water.
  - ✓ Disulphide bonds (-s-s-) are the covalent bonds that link the sulfur atoms of two crystalline subunits.

iv) **Quaternary structures** – are structures that are formed when two or more polypeptide chains (folded in to tertiary structure) become associated in the final structure of protein.

Hemoglobin, the protein in the red blood cells that is responsible for oxygen transport is an example of a globular protein with quaternary structure. It consists of 574 aminoacids arranged in four polypeptide chains (two identical alpha and two identical beta chains).

- The chemical formula of hemoglobin is:

  $C_{3032}\ H_{4816}\ O_{872}\ N_{780}\ S_8\ Fe_4$

- Proteins can also be classified as:
  - ✓ **Fibrous proteins** – are the polypeptide chains arranged in long sheets. Have tertiary structure that resembles along string or fiber.
    Example: collagen and keratin
  - ✓ **Globular proteins** – are polypeptide chains that are tightly folded in to a compact spherical shape.
    Example: Most enzymes, receptor proteins
- Functions of proteins include:
  - ✓ They are structural components of plasma membrane.

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- ✓ They are important for the immune system (antigen and antibody).
- ✓ Important to build the structure of chromosome (DNA wounds around the protein histone to form a chromosome)
- ✓ Making enzymes which play a vital role in metabolic process.

Illustrative Examples

81. Which of the following is NOT composed of protein?

A. sex hormones C. Hair

B. Muscle D. Enzymes

Explanation:- Muscles, hair and enzymes are composed of protein. While, sex hormones are composed of steroids which are lipids.

Answer: A

82. The element that is commonly found in protein but not in triglycerides is:

A. oxygen B. Nitrogen C. carbon D. iron

Explanation:-Triglycerides and proteins commonly contain elements carbon, hydrogen and oxygen. However, proteins differ from other organic compounds because they contain additional elements like **nitrogen.**

Answer: B

83. What is the role of protein in the body?

A. to transport oxygen.

B. To defend the body from disease.

C. To catalyze biological reactions.

D. All

Explanation:- Proteins have wide range of importance for our body. Among these some of them are:

- ✓ To repair damaged tissues.
- ✓ To build body's muscles
- ✓ To make enzymes that catalyze biological reactions.

- ✓ To maintain strength of the body and helping to fight disease.
- ✓ To make hemoglobin, etc.

Answer: D

84. Which polymer is made up of different classes of monomers?

A. carbohydrates B. protein C. Fat D. Nucleotides

Explanation:

Polymer	Monomer
Carbohydrates (complex sugar)	Sugar
Protein	Aminoacid
Fat	Fatty acid and glycerol
DNA	Nucleotides

Answer: C

85. What is the number of naturally occurring amino acids that make protein?

A. 5 B. 20 C. 4 D. 2

Explanation: -There are about 20 different types of aminoacids that occur in nature. This aminoacids combine in different number, sequence and arrangement inorder to make different types of protein molecules.

Answer: B

86. When an enzyme is denatured by heat or extreme PH, which one of the following does it lose?

A. peptide bond C. secondary structure

B. primary structure D. Tertiary structure

Explanation:- pH and temperature affect the bond holding the tertiary structure of enzymes in place such as disulphide bridge, hydrogen bond, ionic bond between R-groups but not peptide bond.

Answer: D

87. In the process of aminoacid condensation, which one of the following happens?

A. Oxygen is used up

B. Carbondioxide is released.

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C. Water is released as a bi-product

D. protein is broken down in to aminoacids

Explanation:- Condensation reactions release water as a bi-product.

Answer: C

88. All proteins contain carbon, hydrogen, oxygen and what other element?

A. chlorine B. Florine C. Nitrogen D. Sulphur

Explanation:- All proteins contain four major elements. These are **carbon, hydrogen, oxygen** and **nitrogen**. In addition some groups of protein may contain sulphur and phosphorous.

Answer: C

89. One of the following molecules is building units of protein molecule

A. Aminoacide B. Glucose C. Nucleic acid D. Fatty acid

Explanation:- The building (monomer units) blocks of all protein molecules are aminoacids.

Answer: A

90. Which of the following pairs of molecules are known to have catalytic activity?

A. Lipids and proteins

B. Carbohydrates and proteins

C. proteins and RNA

D. proteins and DNA

Explanation:- Catalyst is a substance that affects the rate at which the chemical reaction takes place. Accordingly enzymes which are globular proteins and RNAs have catalytic activity.

Answer: C

91. Which of the following is NOT part of aminoacid?

A. Hydrogen B. Carboxyl C. Hydroxyl D. Amino group

Explanation:- Aminoacid molecules are built around carbon atom which is attached to:

- ✓ Hydrogen atom
- ✓ Amino group (NH_2)
- ✓ R- group
- ✓ Carbonlyl group (COOH)

Answer: C

92. How many peptide bonds are formed from 10 aminoacids?

A. 10 B. 8 C. 5 D. 9

Explanation:- Ten aminoacids can be joined by nine peptide links (bonds) between aminoacids.

Answer: D

93. Which of the following do not take part in peptide bond formation?

A. NH_2 B. R-group C. COOH D. Hydrogen

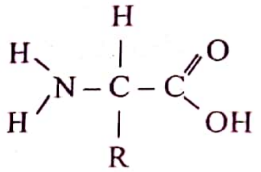
Answer: C

94. What are the functional groups of aminoacids?

A. R-group B. COOH C. NH_2 D. B and C

Explanation:- The general structure of aminoacids can be shown as:

Answer: D



95. The primary structure of protein is firmly maintained by:

A. Hydrogen bond C. peptide bond

B. Ionic bond D. sulphahydral bond

Explanation:- Primary structure of protein is formed when many amino acids are joined by peptide bond to form a polypeptide chain.

Answer: C

96. Which of the following is incorrect about the primary structure of proteins?

A. It is determined by the sequence of nucleotides on DNA molecule.

B. It is branched.

C. It is determined by the sequence of amino acid.

D. They are base for secondary and tertiary structure

Explanation:- The primary structure of proteins is determined by the type and position of each amino acid in the polypeptide chain folds.

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Answer: C

97. The primary structure of proteins is due to:
   A. Interaction among the side chain of R-group of the amino acid.
   B. The coiling due to hydrogen bonding between amino acids.
   C. The number and sequence of amino acids.
   D. All

Answer: C

98. Which of the following element is found in proteins and nucleic acids but not in carbohydrates?
   A. carbon B. Nitrogen C. oxygen D. Hydrogen

Explanation:- Carbohydrates contain elements carbon, hydrogen and oxygen.Nucleic acids and proteins contain elements carbon hydrogen, oxygen and nitrogen in common.

Answer: B

99. Which one of the following is incorrect pairing concerning protein structure and its example?
   A. primary structure – insulin
   B. Secondary structure – collage
   C. Tertiary structure – Myoglobin
   D. Quaternary structure – Hemoglobin

Explanation:

Protein structure	Example
Primary structure	Insulin
Secondary structure	Structural protein such as hair, wool nail, beaks, feathers, horn, keratin, etc
Tertiary structure	Globullor proteins (enzymes), collagen
Quaternary structure	Hemoglobin

Answer: B

100. Which one is incorrect concerning proteins?
   A. α- helix proteins are strong and flexible
   B. β - plated sheet protein are more strong and rigid than α- helix.

   C. α-helix forms fibrous protein, while β-plated sheet forms globular protein.
   D. α-helix the degree of stretchedness depends on the amount of disulphide bridge.

Explanation:- α- helix and β-plated forms secondary structure of proteins but not globular proteins.

Answer: C

101. The tertiary structure of protein refers to:
   A. sequence of aminoacids
   B. The presence of α-helix or β - sheets
   C. Unique three diamensional folding of the molecule
   D. Interactions of protein with other sub units of enzymes.

Explanation:- Tertiary structure of proteins:
- ✓ Involves further folding of the secondary structure
- ✓ It has three dimensional structure
- ✓ It has bonds:-
  - Hydrogen bonding
  - Ionic bonding
  - Hydrophobic interactions
  - Disulphide bonds

Answer: C

102. To which of the following groups of proteins do enzymes belong? (UEE, 2001 E.C)
   A. Lipoproteins C. Globular proteins
   B. Fibrous proteins D. Deoxyribose sugars

Explanation:- Enzymes are globular proteins and they have 3-diamension structure which are held together by hydrogen bond, disulphide bond, ionic bond and hydrophobic interactions. Thus, heating breaks these bonds and it loses the conformation on the tertiary structure.

Answer: D

103. Which of the following is a distinguishing feature of aminoacids that is NOT found in carbohydrates? (UEE, 2006 E.C)

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A. Carbon B. Oxygen C. hydrogen D. Nitrogen

Explanation:- See example number 88

Answer: D

104. Which of the following is true about the substance represented by the chemical formula "$C_2H_5NO^2$? (UEE 2006E.C)

A. simple sugar C. fatty acids

B. Inorganic substance D. Aminoacids

Explanation:- The substance indicated is most probably be amino acid. This is because it contains elements carbon, hydrogen, oxygens and nitrogen.

Answer: D

Nuclei acids

Our cells contain two major types of nucleic acids. They are **deoxyribonucleic acids (DNA)** and **ribonucleic acids (RNA)**

Nucleic acids play a huge role in the transmission of hereditary informations and determination of the types of proteins produced by our cells.

Nucleic acids are first isolated in the nuclei of the pus cell and their name stems from the fact that they are acids and were first identified in the nuclei. Nucleic acids are sub polymers of repeating sub units called **nucleotides**. Nucleiotides are biological molecules that are built from three types of molecules. These are:

- Five carbon sugar
- Phosphate group (PO_4^-) Adenine
- Nitrogenous bases
  - Thymine (Uracil in RNA)
  - Guanine
  - Cytosine

If we say this much as an introduction about nucleic acids, lets look at the molecules of DNA and RNA separately.

N.B. We will see more about heredity on Unit 8.

i) Deoxyribo nucleic acid (DNA)

The discovery of the double helix structure of DNA in 1953 is one of the most genuine discoveries in the history of biological science. By the time the team of biologists Maurice Wilkins, Rosiland Franklin, Francis crick and Mark Watson have won the nobel prize which is the top level award on our planet for their discovery.

- The discovery of DNA molecule played a valuable role in different areas. Among these some of them are:
  - Criminal investigation
  - determination of paternity cases on courts
- Above all the discovery of DNA molecule played a key role for the development of the science of heredity. This is because DNA comprises the genes themselves, the hereditary material of the cell and it contains instructions for making all proteins needed by an organism.
- A DNA molecule is found in the chromosomes and genes are short sections of DNA. I hope you remember the meaning of chromosomes and genes from your grade 10 biology lesson.

Chromosomes are thread like structures in the nucleus of the cell and they contain genes.

Genes are units of heredity which are small segments of the chromosome. They are responsible for determining each and every characteristics of an organism.

- A DNA molecule is made up two chains of nucleotides which are joined by their nitrogenous bases. The two strands (chains) of nucleotides wind in to a double helix structure which has a twisted shape.
- Each short section of DNA molecule codes for specific protein and determines a particular feature.
- The nucleotides that build a DNA molecule are composed the following substances:

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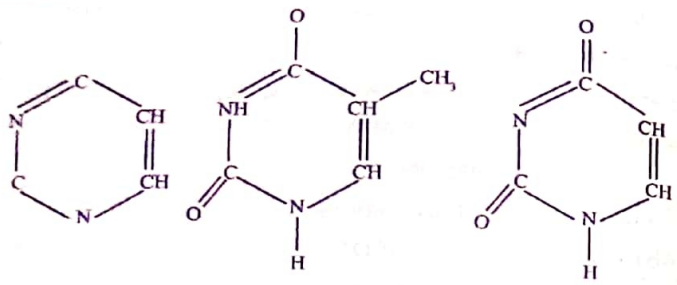
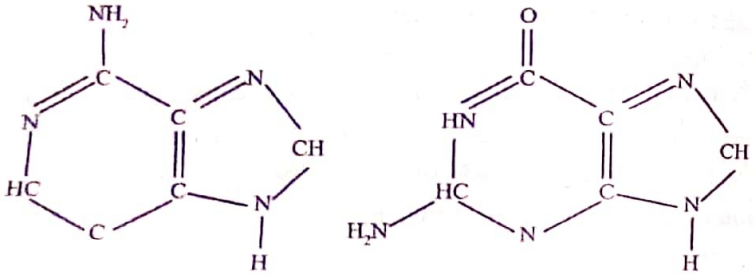
- Deoxyribose sugar (five carbon sugar)
- Phosphate group (PO_4)

➢ Four nitrogenous bases. Namely, adenine, thymine, guanine and cytosine.

a) Pyrimidies



b) Purine s cytosine Thymine(T) Uracil(U)



Adenine(A) Guanine(G)

Note: The nitrogencous base which are double ringed are called **purines**. These are adenine and guanine.

➢ The nitrogeneous bases which are one ringed are called **pyramidines.** These are cytosine, thymine and uracil.

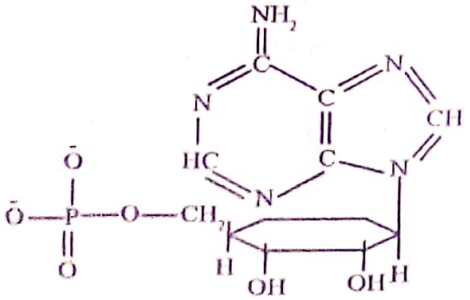
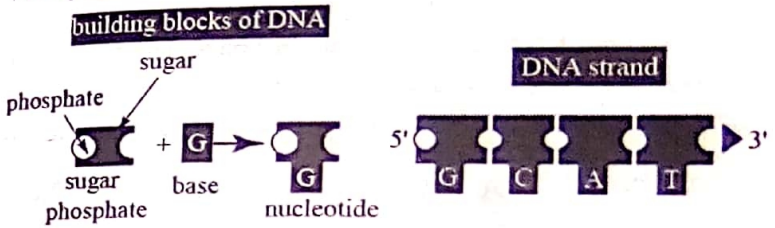
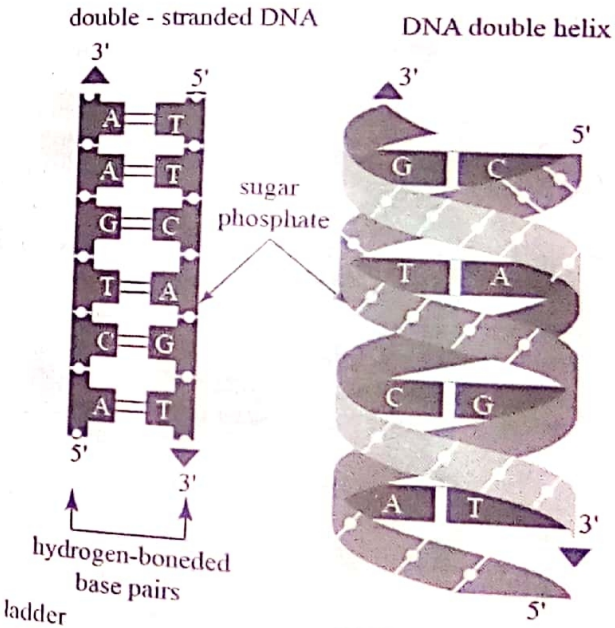


Fig. A nucleotide, adenine Monophosphate (AMP)

➢ In the building of the DNA molecule, if we consider the DNA molecule as a step ladder, the upright parts of the DNA molecule are made up of phosphate and deoxyribose sugar. While the cross bars are composed of the nitrogenous bases. Thus, the two chains of nucleotides in a DNA molecule are joined by their bases. The adenine base always pairs with thymine and guanine base pairs with cytosine base.





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Note: The shape of DNA is the twisted one, while the step ladder structure of DNA is simply for ease of understanding.

ii) Ribonucleic acid (RNA)

RNA is a nucleic acid which is found both in the nucleus and cytoplasm of the cell. RNA play a role in allowing a specific gene (DNA) to produce the protein in codes for and they are also responsible for transmitting the information from the nucleus (DNA) to the ribosome in the cytoplasm where protein synthesis takes place.

There are different types of RNA. These are:

- **Messenger RNA** - carries the information for protein synthesis.
- **Transfer RNA** – collects amino acids in forming protein
- **Ribosomal RNA** – involved in protein synthesis in the ribsomes.

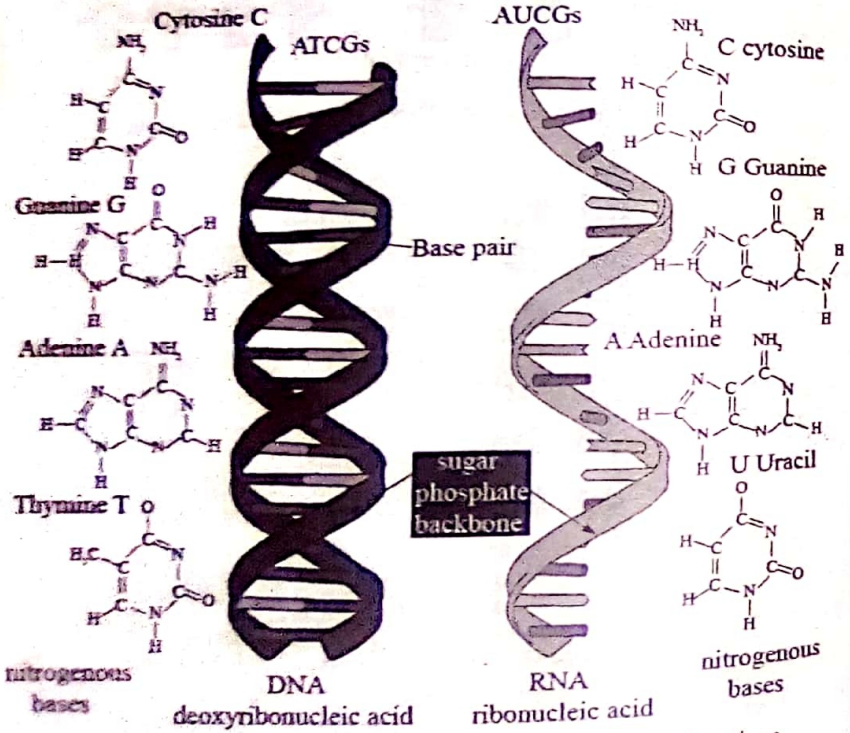

Fig. How the structure of nucleic acids relates the their functions.

Even though, the DNA and RNA have similarities in some aspects, they have various differences based on their size, stability, number of strands, their nitrogenous bases, their ability to duplicate, the type of sugar they contain, etc.

Table: Comparision of DNA and RNA molecules

DNA	RNA
• It is larger in size. This allows it to carry the code for many different proteins in the genes.	• It is relatively smaller in size, thus it codes only for one protein.
• The sugar is deoxyribose.	• The sugar is ribose.
• It is double stranded.	• It is single stranded.
• It contains bases adenine, thymine, guanine and cytosine.	• It contains bases adenine, uracil, guanine and cytosine.
• It is very stable.	• It is less stable.
• It makes its own copy (ruplicated) during the cell division.	• It do not copy itself, while it is formed on a DNA molecule.

Illustrative Examples

104. RNA is composed of:

A. A single polynucleotide chain.
B. Two polypeptide chains.
C. protein.
D. A and C

Explanation:- RNA (Ribonucleic acid) is a nucleic acid that is composed of a single polynucleotide chain. Thus, it is single stranded.

Answer: A

105. Which of the following is true about all nucleic acids?

A. They are polymers of nucleotide.

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B. They are polymers of aminoacid.
C. They are doubled stranded.
D. All are correct except B.

Explanation:- All nucleic acids (RNA and DNA) are the polymers which are composed of chain of monomer units called **nucleotides.**

Answer: A

106. All are the characteristics of DNA, except:
A. They are stable.
B. They carry code for many proteins.
C. They degenerate after the formation of proteins.
D. They are double stranded.

Explanation:- DNA molecules are:
- stable molecules.
- carry codes for many proteins.
- double stranded.
- capable of duplication (copying) themselves.

Answer: C

107. The number of thymine bases in a DNA molecule equals with:
A. Guanine B. Adenine C. Cytosine D. Uracil

Explanation:- In a DNA molecule the number of thymine base is always equal to the number of adenine base and the number of guanine equals with cytosine base.

Answer: B

108. Which one of the following is absent in the nucleoside?
A. phosphate B. sugar C. Nitrogenous base D. None

Explanation:- Nucleosides differ from nucleotides because they do not contain phosphate group.

Answer: A

109. Which of the following is incorrect about purines?
A. They are macromolecules.
B. They are nitrogenous bases.
C. They are double ringed.
D. They always bind with pyramidines.

Explanation:
Purines – are nitrogenous bases which are double ringed.
- Examples are adenine and guanine

Pyrimidines are nitrogenous bases which are single stranded
- Examples are cytosine and thymine.

Answer: A

110. What are the three components of DNA molecule?
A. Glucose, nitrogen and phosphate
B. sugar, phosphate group and nitrogenous base.
C. Protein, carbohydrate, fat
D. Monosaccharide, disaccharide and polysaccharide

Explanation:- The nucleotides which are the building bloks of DNA molecule are composed of:
- Pentose sugar (Deoxyribose)
- Phosphate group
- Nitrogenous bases (Adenine, thymine, guanine, cytosine).

Answer: B

111. The nitrogenous base that is absent in the nucleotides of DNA molecule is:
A. Adenine B. Uracil C. Thymine D. Guanine

Explanation:- DNA molecule consists of nitrogenous bases **adenine, thymine, guanine** and **cytosine** but not **uracil.**

Answer: B

112. What is the bond that exists between nucleotides?
A. phosphodiaster bond C. peptide bond
B. Ester bond D. Glycosidic bond

Explanation:- Phosphodiaster bond is used to join nucleotides together to make a polynucleotide chain.

Answer: A

113. The bond that exists between the complementary nitrogenous bases is:
A. Ionic bond C. coordinate bond
B. peptide bond D. Hydrogen bond

Explanation:- Hydrogen bond is important in linking the nitrogenous bases

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between the complementary nitrogenous bases.

Answer: D

114. Which one of the following statement is true about DNA and RNA?
   A. Both are single stranded
   B. Both have a five carbon sugar
   C. Both are polymers of amino acids
   D. Both contain the same four nitrogenous bases.

Explanation:- Both DNA and RNA contain five carbon sugars which are called deoxyribose sugar and ribose sugar respectively.

Answer: B

115. If is known that the total percentage proportion of purine bases in a DNA molecule is 50% and all the four bases in a DNA molecule have equal proportions, what is the percentage proportion of thymine base?
   A. 25% B. 50% C. 75% D. 100%

Explanation:- Purines (Adenine and guanine) = 50% of the bases
pyrimidines (cytosine and thymine) = 50% of the base.
Therefore since all the four bases have equal proportions as mentioned in the question, the percentage proportion of tymine is 25%. However, the number of all bases is not always expected to be proportional.

Answer: A

116. Which of the following organic substances are correctly matched with their building blocks?
   A. Carbohydrate – glycerol
   B. proteins – fatty acids
   C. Lipids - aminoacids
   D. Nucleicacids – nucleotides

Explanation:

Organic substance	Building block
Carbohydrate	Simple sugars
Proteins	Aminoacids

Lipids	Fatty acid and glycerol
Nucleic acids	Nucleotides

Answer: D

117. In which of their constituents do the nucleotides of DNA and RNA differ?
   A. In their purine base C. In the phosphate group
   B. In the pyrimidine base D. All of the above

Explanation:- The nucleotides of DNA and RNA differ on their pyramidine bases in which DNA has thymine base. While, RNA has uracil base instead.

Answer: B

Conceptual Summary example

1. Compare the physical properties of (mass and charge and locations of electrons, protons and neutrons; distinguish between the atomic number and the mass number of an element.

Explanation:- Each **atom** is composed of a nucleous containing positively charged **protons** and uncharged **neutrons**. Negatively charged electrons encircle the nucleus.

- An atoms is identified by its number of protons (**atomic number**). The atomic mass of an atom is equal to the sum of its protons and neutrons.

2. Describe the chemical properties of carbon that make it central component of organic compounds.

Explanation:

- Each carbon atom forms four covalent bonds with up to four other atoms, these bonds are single, double or triple bonds.
- Carbon atoms form straight or branched chains or join in torings.
- Carbon forms covalent bonds with greater number of different elements than does any other type of atom.

3. Explain the relationship between polymers and macromolecules.

Explanation:

- Long chains of **monomers** (similar organic compounds) linked together through **condensation reaction** are called **polymers**.

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- Large polymers such as polysaccharides, proteins, and DNA are referred to as **macromolecules**. They can be broken down by **hydrolysis reactions**.

4. Distingush among monasaccharides, disaccharides, and polysaccharides; compare storage polysaccharides with structural polysaccharides.

Explanation:

- Carbohydrates contain elements carbon, hydrogen, and oxygen in a ratio approximately one carbon to two hydrogen to one oxygen
- **Monosaccharides** are simple sugars such as glucose, fructose and ribose.
- **Disaccharide** – formed from two monosaccharides joined by glycosidic linkage. They include substances such as sucrose, lactose, and maltose.
- **Polysaccharide** – long chains of repeating units of a simple sugar. They include substances. Such as starch alycogen and cellulose.

5. Distingush among fats, phospholipids, and steroids, and describe the composition, characteristics, and biological functions of each.

Explanation:

- **Lipids** are composed mainly of hydrocarbon-containing regions with few oxygen containing (polar or ionic) functional groups. They have a greasy or oily consistency and relatively insoluble in water.
- **Tria cylglycerol** – the main storage form of fat in organisms, consists of **glycerol** combined with three **fatty acids**.
- **Phospholipids** are structural components of cell membranes. A phospholipid consists of a glycerol molecule attached at one end to two fatty acids are at the other end to a phosphate group linked to an organic compound such as choline.
- **Steroid** molecules contain carbon atoms arranged in four attached rings. Cholesterol, bile salts, and certain hormones are important steroids.

6. Give on overall description of the structure and functions of proteins.

Explanation:

- **Proteins** are large, complex molecules made of simpler sub units, called **aminoacids**, joined by **peptide bonds**.
- Proteins are the most versatile class of biological molecules, serving a variety of functions, such as **enzymes**, structural components, and all regulators.
- Proteins are composed of various linear sequences of 20 different aminoacids. Two amino acids combine to form a **dipeptide**. A longer chain of aminoacid is a **polypeptide**.

7. Describe the features that are shared by all amino acids, and explain how aminoacids are grouped in to classes based on the characteristics of their side chains.

Explanation:

- All aminoacids contain an amino group and a carboxyl group.
- Amino acids vary in their side chains, which dictate their chemical properties – non polar, polar, acidic or basic.
- Aminoacids generally exist as dipolar ions at cell pH and serve as important biological buffers.

8. Distingush among the four levels of organization of protein molecules.

Explanation:

- **Primary structure** is the linear sequence of amino acids in the polypeptide chain.
- **Secondary structure** is a regular conformation, such as a **α - helix** or a **β-pleated sheet**: it is due to hydrogen bonding between elements of the back bones of the aminoacids.
- **Tertiary structure** – the overall shape of the polypeptide chains, as dictated by chemical properties and interactions of the side chains of specific amino acids. Hydrogen bonds, ionic bonds, hydrophobic interactions and disulfide bridges contribute to tertiary structure.
- **Quaternary structure** is determined by the association of two or more polypeptide chains.

9. Describe the components of a nucleotide. Name some nucleic acids and nucleotides, and discuss the importance of these compounds in living organisms.

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Explanation:

- **Nucleotides** are composed of two ring purine or one ring pyrimidine nitrogenous base, a fixe-carbon sugar (ribose or deoxyribose), and one or more phosphate groups.
- The **nucleic acids DNA** and **RNA**, composed of long chains of nucleotide subunits, store and transfer information that governs the sequence of aminoacids in proteins and ultimately the structure and function of the organism.
- **ATP (Adenosine triphosphate)** is a nucleotide of special significance in energy metabolism, NAD^+ is also involved in energy metabolism through its role as an electron (hydrogen) acceptor in biological oxidation and reduction reactions.

Post test

1. Which is a property of water?
   A. The water molecule is non-polar
   B. Solid water is denser than liquid water.
   C. Water molecules repel other water molecules.
   D. Water molecules repel other water molecules.
2. Who first proposed the double – helix structure of the DNA molecule?
   A. Pauling C. Waston and Crick
   B. Mc Clintock D. Harris and Watkins
3. Which is common to both DNA and RNA?
   A. uracil B. thymine C. cytosine D. deoxyribose
4. Which statement best describes the relationship between proteins and polypeptides?
   A. Proteins are composed of polypeptide
   B. Polypeptides are composed of proteins.
   C. Proteins, unlike polypeptides, are composed of amino acids.
   D. Polypeptides, unlike proteins, are composed of amino acids
5. Insulin is a complex protein that is said to have a quaternary structure. This means that insulin
   A. cannot be denatured.
   B. lacks disulfide bridges.
   C. contains all the known amino acids.
   D. has more than one polypeptide chain.

6. What are the subunits of DNA and their function?
   A. nucleotide that store information
   B. monosaccharides that provide quick energy for the cell
   C. lipids that store energy and provide insulation
   D. proteins that provide the building blocks for the structural components of organisms
7. How does DNA code for proteins in a cell?
   A. by creating a new double helix structure
   B. by using its phosphate and sugar molecules
   C. by adding more hydrogen bonds to its structure
   D. by arranging certain nitrogen bases of the cell in a particular order
8. Which statement best describes the relationship between proteins and polypeptides?
   A. Proteins are composed of polypeptides.
   B. Polypeptides are composed of proteins.
   C. Proteins, unlike polypeptides, are composed of amino acids.
   D. Polypeptides, unlike proteins, are composed of amino acids.
9. An example of a catabolic reaction includes the formation of
   A. maltose from glucose.
   B. fatty acids and glycerol from lipids.
   C. ATP and water from ADP and inorganic phosphate.
   D. glucose and oxygen from carbon dioxide and water.
10. Which factor **most** affects the order of amino acids in a protein?
    A. the DNA located in the nucleus of the cell
    B. the cell in which the protein is located
    C. the amount of ATP available for the cell's use
    D. the area in a cell where proteins are produced
11. If you remove all of the functional groups from an organic molecules so that it has only carbon and hydrogen atoms, the molecule becomes a _____ molecule.
    A. Carbohydrate B. Carbonyl C. Carboxyl D. Hydrocarbon
12. All of the following are true of protein denaturation except that it
    A. is a shape change
    B. is always irreversible
    C. may be caused by a pH change
    D. could result from a temperature change

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13. There are several levels of protein structure, the most complex of which is
    A. primary
    B. secondary
    C. tertiary
    D. quaternary
14. the functional group $-NH_2$ is a (n)
    A. carboxyl group
    B. amino group
    C. hydroxyl group
    D. phosphate group
    E. carbonyl group
15. Which of the following is not a lipid?
    A. Chitin
    B. Terpenes
    C. Steroids
    C. Prostaglandins
    D. unsaturated fat
16. Double helix describes the structure of a molecule of
    A. protein B. disaccharide C. starch D. DNA
17. Triacylglycerol contains fatty acids and
    A. glucose B. glycogen C. glycerol D. glycerol
    E. Amino group
18. The globular shape of a protein is called the
    A. Primary structure
    B. Secondary structure
    C. quaternary structure
    D. Tertiary structure
19. The globular shape of a protein is called the
    A. Primary structure
    B. Secondary structure
    C. Quaternary structure
    D. Tertiary structure
20. In a DNA molecule, what holds together nitrogenous bases from the two polymer chains?
    A. Phosphodiester bonds
    B. ionic bonds
    C. peptide bond
    D. Hydrogen bond
21. What type of macromolecule carries out catalysis in biological systems?
    A. proteins called enzymes
    B. carbohydrates called starches
    C. lipids called steroids
    D. nucleic acids called DNA
22. In nucleic acids, the purine nitrogenous bases are
    A. Uracil and thymine
    B. Cytosine and guanine

    C. Thymine and cytosine
    D. Guanine and thymine
23. In proteins, elements of secondary structure combine to form a(n)
    A. Domain B. Motif C. Beta sheet D. Choperone
24. A hydrocarbon is said to be saturated if:
    A. One end of the molecule is hydrophilic while the other end is hydrophobic.
    B. It has one or more double bonds between carbon atoms.
    C. Each internal carbon atom is covalently bonded to two hydrogen atoms.
    D. Its functional groups include at least one aromatic ring.
25. At what level(s) of protein structure would you expect to find disulfide bridges?
    A. Primary B. Secondary C. Tertiary D. Quaternary
26. Polymerization reactions in which pholysaccharides are synthesized from monosaccharides or proteins are synthesized from amino acids:
    A. Result in the formation of water.
    B. Are hydrolysis reactions
    C. Release energy
    D. Result in the formation of covalent bonds between monomers called peptide bonds
    E. All of the above
27. Which of the following is a characteristic of proteins?
    A. Some may enhance the rate of specific chemical reactions.
    B. They may form either long, thin fibrous molecules or compact, rounded globular molecules.
    C. They may form either long, thin fibrous molecules or compact, rounded globular molecules
    D. They store genetic information for cellular metabolism
    E. They form through hydrolysis reactions.
28. The two strands of a DNA double helix are held together by
    A. Ionic bond
    B. Hydrogen bond
    C. Non polar convalent bond
    D. Polar covalent bond
    E. Hydrophobic exclusions

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Critical thinking

1. In what ways are all species a like biochemically? How do species differ from one another biochemically?
2. Like oxygen, sulfur forms two covalent bonds. However, sulfur is far less electronegative. In fact, it is approximately as electronegative as carbon. How would the properties of the various classes of biological molecules be altered if you were to replace all the oxygen atoms with sulfur atoms?

Supplementary Exercise on Unit – 2

Select the most appropriate from the column B for each entry in column A.

Column A	Column B
1. Mono saccharide	a. cellulose
2. Steroid	b. DNA
3. Nucleic acid	c. Glucose
4. Oleic acid	d. Cholesterol
5. Important constituent of cell membrane	e. ATP
	f. Glycerol
6. Sub units of proteins	g. Hydrocarbons
7. Energy currency of cell	h. Amino acids
8. Compounds with the same Molecular formula but different structures	i. Phospholipids
	j. isomers
	k. Amino acids
9. Component of fatty acids	
10. Organic carbons consisting only Of carbon hydrogen, oxygen, nitrogen	

Fill in the blank spaces with appropriate answer.

11. Peptide bonds are found linking _____.
12. The primary structure of a protein refers to _____.
13. When sucrose is hydrolyzed _____ and _____ are formed.
14. _____ is an important component of the cell wall of plant cells.

15. The fatty acid portion of the phospholipd is _____ and will not dissolve in water.
16. The three functional groups which make up the amino acid are the _____, _____ and the R group
17. Animals store glucose in the form of _____.

Multiple Choice

18. Which of the following is inorganic molecule?
    A. Glucose B. Starch C. cellulose D. water
19. An element that is sued for digestion of food is:
    A. Calcium B. Iron C. Chlorine D. Phosphorous
20. The importance of hydrogen bond that exists between hydrogen and oxygen is:
    A. Makes up water molecule strong.
    B. Makes up water molecule long lived
    C. Makes up water molecule weak
    D. Makes up water molecules rigid.
21. The transparency of water molecules is important for:
    A. The process of photosynthesis
    B. Increasing its surface tension.
    C. Making it a universal solvent.
    D. All are possible answers.
22. The amount of heat required to raise the temperature of 1kilogram of water by 1^{9} is:
    A. High heat capacity
    B. High latent heat o9f vaporization.
    C. High surface tension.
    D. Cohesion.
23. The property of water that enables small organisms to walk on the water without breaking its surface is:
    A. Heat capacity
    B. Heat of vaporization
    C. surface tension
    D. None of the above

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24. The measure of how fluid flows easily is:
A. Cohesion B. Adhesion C. viscosity D. capillarity
25. Water is used as a reactant in:
A. Photosynthesis
B. Hydrolysis reaction
C. Respiration
D. A and B are possible answers
26. Organic molecules that contains nitrogen are:
A. Lipids C. Carbohydrates
B. Proteins D. All are possible
27. Which of the following is not a structural component?
A. Cellulose B. Chitin C. starch D. peptide glycan
28. All are the general properties of carbohydrate, except:
A. All are ketoses and aldehydes
B. They all contain several hydroxyl groups.
C. Their chemistry is determined by their hydroxyl group.
D. They are all soluble in water.
29. Which of the following is double sugar?
A. Starch B. Glycogen C. Lactose D. Fructose
30. Which of the following groups of carbohydrate to triose?
A. Glycerol dehyde B. Ribose C. deoxyribose D. Glucose
31. The sugars that make nucleic acids are:
A. Glyceral dehyde and dihyderoxy acetone.
B. Ribose and deoxyribose
C. Glucose and fructose.
D. Starch and glycogen.
32. Glucose differs from galactose in:
A. The position of – OH group on the third carbon.
B. The carboxyl group on the first carbon.
C. The presence of hydroxyl group.
D. All are possible

33. Which of the following differs in its molecular formula?
A. Glucose B. Galactose C. sucrose D. Fructose
34. Which of the following is true about amylsoe?
A. It is branched
B. It is simpler
C. It can easily dissolve in water
D. It is complex.
35. Which of the following is true about cellulose?
A. It is formed by β - 1, 4 – glycosidic linkage.
B. It is soluble in water.
C. It is branched polysaccharide
D. It is composed of chain of fructose
36. The fatty acids and glycerol and joined by:
A. Hydrogen bond C. Glycosidic bond
B. Peptide linkage D. Ester linkage
37. The most commonly known groups of fat are:
A. Mono glycerides C. triglycerides
B. Di-glycerides D. None of the above
38. Which of the following is true about saturated fatty acids?
A. Contain maximum possible number of hydrogen atom.
B. Some of the carbon – carbon bonds are multiples bonds.
C. Tend to be liquids at room temperature.
D. All except 'C'.
39. The bond by which monosaccharide's are joined to make disaccharides is:
A. Ester linkage C. Glycosici linkage
B. Peptide bond D. Hydrogen bond
40. A disaccharide that is formed between two α - glucose molecules is
A. Maltose B. Sucrose C. Lactose D. glucose
41. A reaction in which two molecules are joined to form a large molecule, that produces water as a by product is:

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A. Hydrolysis C. Saturation
B. Condensation D. Saponification

42. _____ is the process of splitting disaccharides into monosaccharides.
A. Hydrolysis C. Saturation
B. Condensation D. Saponification

43. Which of the following is not a polymer?
A. Amino acid B. Protein C. Starch D. Glycogen

44. Which of the following polysaccharide is correctly matched with its function?
A. Cellulose – component of cell membrane
B. Starch – storage form of fat in plants.
C. Glycogen – storage form of carbohydrate in animals.
D. Chitin – structural component of plant cell wall.

45. Which of the following is false about phospholipids?
A. They are amphiphatic lipids
B. They have ghydropobic bond hydrophilic ends.
C. They are important constituent of cell membrane.
D. They are structural components of cell wall

46. The bond between amino acids to make protein are:
A. Hydrogen bond C. Glycosidic bond
B. Peptide bond D. Ester bond

47. A protein structure which is formed when a peptide chain folds or coils in to a helix or α - pleated is:
A. Primary structure C. Tertiary structure
B. Secondary structure D. Quaternary structure

48. Hemoglobin's are examples of:
A. Primary structure C. Tertiary structure
B. Secondary structure D. Quaternary structure

49. Which of the following are globular proteins?
A. Collagen C. Enzymes
B. Keratin D. None

50. Which of the following is NOT a component of DNA?
A. Deoxyribose sugar C. Phosphate group
B. Ribose sugar D. Nitrogenous bases

51. Which of the following bases is purine?

A. Cytosine B. Thymine C. Adenine D. Uracil

52. What makes RNA different from DNA?
A. It is relatively larger.
B. It is double stranded.
C. It contains uracil instead of the mine
D. All are possible

Answer to Supplementary Exercise on Unit – 2

1.C 2.___ 3.B 4. ___ 5.L 6. H 7.E
8.J 9.F 10.K
11. Amino acid 12. Sequence of amino acid
13. Glucose and fructose 14 Cellulose 15. Hydrophibic
16. Amino group and carboxylic group 17. Glycogen

18.D	19.C	20.C	21.A	22.A	23.C	24.C
25.D	26.B	27.C	28.D	29.C	30.A	31.B
32.A	33.C	34.B	35.A	36.D	37.C	38.A
39.C	40.A	41.B	42.A	43.A	44. C	45.D
46.B	47.B	48.D	49.C	50.B	51.C	52. C

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UNIT THREE

Enzaymes

3.1. Nature of Enzymes

Living cells carry out many biochemical reactions. All these reactions take place rapidly due to the presence of enzymes. Enzymes are **globular proteins.** They act as biological catalysts which are used to speed up the rate at which biochemical reactions take place.

There are thousands of chemical reactions in living cells that are catalyzed by enzymes.

What are enzyme molecules like?

Enzymes are composed of globular proteins that have unique tertiary structure which makes them to have a specific shape. The ability of an enzyme to function depends on the specific shape of the protein molecule. The tertiary folding of polypeptides are responsible for the special shape of the active site.

Enzymes have active site on a region where they bind with substrate.

Active site is the part of an enzyme molecule that binds with its substrate. This enables an enzyme to catalyze the chemical reaction.

Substrate is a substance up on which an enzyme acts in a biochemical reaction. An enzyme can only bind with one substrate because of the shape of its active site.

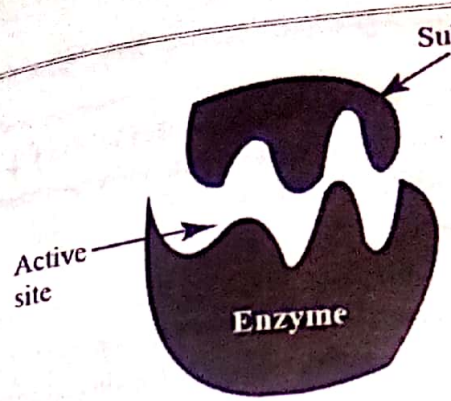
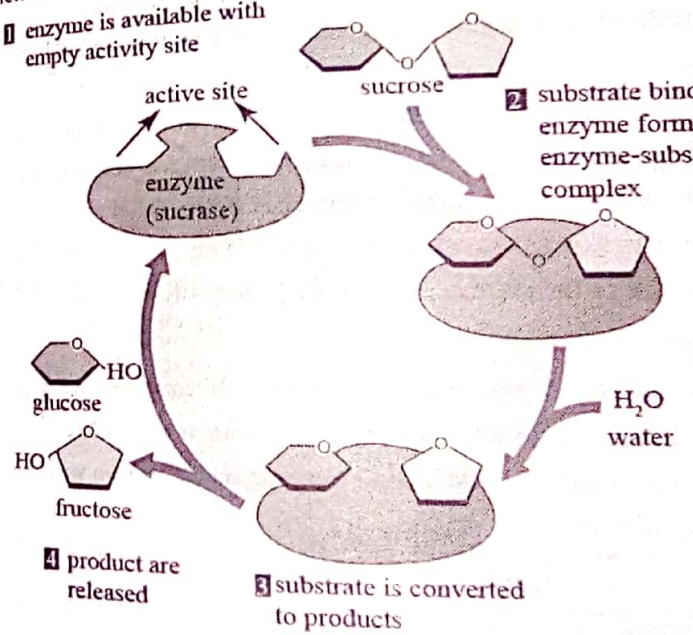


Fig. Enzyme substrate complex

The substrate for enzyme molecule binds with the active site and forms **enzyme-substrate complex.** This can be shown by the following example which show the hydrolysis of sucrose by an enzyme called sucrose.



The active site of an enzyme is used for:

- ✓ Binding a particular substrate only.
- ✓ Minimizing the energy that a reaction requires

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Properties of Enzymes

Even though there are various types of enzymes used to catalyze thousands of chemical reactions in living organisms, all enzymes share certain characteristic in common. The major properties of enzymes are:

- ✓ All enzymes are proteins.
- ✓ Enzymes are biological catalysts. Catalyst is a substance that speeds up the rate of chemical reaction without being used up in the reaction. Because of this reason enzymes can be reused again and again.
- ✓ Enzymes are specific. Every enzyme acts specifically on a particular substrate. The specificity of enzymes is related to the conformation of the active site where only few types of substances binds with.
- ✓ Small amount of enzymes can bring change in large amount of substrate. Therefore, enzymes are effective in small amounts.
- ✓ The efficiency of an enzyme is affected by temperature and PH.

i) PH

The activity of enzymes is markedly changed by any alteration in the acidity or alkalinity of the reaction medium. Full enzyme activity requires specific number of positive and negative charges on the enzyme.

The activity of enzyme requires a narrow range of PH. For example, pepsin is a protease enzyme in the stomach which works actively in acidic media.

ii) Temperature

Enzymes work actively at optimum temperature. If enzymes are exposed to extremely high temperature than their optimum temperature they get **denatured**. At low temperature enzymes become inactive. (we will discuss more about this later on)

Illustrative Examples

1. Which one of the following is true about enzymes?
   A. They are produced in the ribosome.
   B. They can work outside cells.
   C. They get changed at the end of the reaction.
   D. A single enzyme can catalyze various types of reactions.

Explanation: Enzymes
- ✓ Can work out side cells and called extracellular enzymes.
- ✓ Do not get changed at the end of the reaction.
- ✓ A single enzyme catalyzes only one type of reaction due to their specific nature.

Answer: B

2. The region where an enzyme molecule binds with its substrate is:
   A. Allostric site
   B. Active site
   C. peptide bond
   D. Substrate

Explanation: The region of an enzyme which binds to the substrate is called active site.

Answer: B

3. A substrate up on which an enzyme acts in a biochemical reaction is:
   A. Product
   B. Activation energy
   C. Substrate
   D. Enzyme substrate complex

Explanation: Substrate is a substance on which an enzyme acts on.

Answer: C

4. All are the characteristics of biological catalysts as compared to chemistry laboratory catalyst, except.
   A. They have limitless life span
   B. They are made up of organic compound
   C. Their turnover rate is short
   D. They do not exist heat over 40^0c.

Explanation:

Features	Enzyme	Chemistry laboratory catalyst
Life span	Short lived and replacable	Limitless
Chemistry	Organic	Most are inorganic
Resistance to temperature	Mostly denature at temperature above 40^0c	Higher
Turnover rate	Can bind large number of	Less

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	substrate per second	

5. The active site of enzyme
   A. Similar to any other enzyme
   B. Is the part of enzyme where its substrate fit
   C. Has the same shape with the substrate
   D. B and C

Explanation: The active site of an enzyme have the part that fits in to the substrate inorder to catalyze the chemical reaction.

Answer: B

6. The number of substrate molecule affected by enzyme per minute is called:
   A. Active site number
   B. Enzyme activity number
   C. Turn over number
   D. Optimum number

Explanation: Turnover rate is the number of molecules of reactants that form enzyme substrate complex with each molecule of an enzyme per second.

Answer: C

7. Biological enzymes are similar with chemistry laboratory enzymes because:
   A. They are shortlived
   B. They are sensitive to temperature
   C. They remain unchanged after the chemical reaction
   D. None

Explanation: All catalysts do not get altered by the changes in the chemical reaction.

Answer: C

8. Which enzymatic reaction is correct?
   A. $S + E \longrightarrow SE \longrightarrow P + E$
   B. $S + E \leftrightarrows SE \longrightarrow P + E$
   C. $S + E \leftrightarrows SE \longrightarrow P + E$
   D. $S + E \longrightarrow SE \leftrightarrows P + E$

Answer: B

9. The substance that remain unchanged after the chemical reaction is:
   A. Substrate
   B. Enzyme
   C. Product
   D. Substrate and enzyme

Explanation: Enzymes are catalysts. Thus, catalysts are substances that speed up the rate of the chemical reaction and remain unchanged by themselves at the end of the reaction.

Answer: B

10. One of the following molecules is building unit of an enzyme. (UEE, 2005E.C)
    A. Aminoacids
    B. Glucose
    C. Nucleotides
    D. Fatty acids

Explanation: Enzymes are made of glubullar proteins and all proteins are composed of small building blocks called **amino acids.**

Answer:

11. Which of the following pairs of molecules are known to have enzymatic activity: **(UEE, 2005 E.C)**
    A. Lipids and proteins
    B. Carbohydrate and proteins
    C. Protein and RNA
    D. Protein and DNA

Answer: C

12. The enzyme sucrose increases the rate at which sucrose is broken into glucose and fructose. Thus, its works by:
    A. Increasing the amount of free energy of the reaction.
    B. Lowering the activation energy.
    C. Decreasing the equilibrium constant of the reaction.
    D. Applying energy to speed up the reaction

Explanation: Enzymes increase the rate of chemical reaction by lowering the activation energy required for the reaction.

13. Which one is false about enzyme feature?
    A. The conformation of the active site is made up of polysaccharide chain.

Answer: B

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B. Enzyme can bind mineral ions such as zinc and copper.
C. Enzymes bind to their substrate on the active site.
D. All are true.

Explanation: The shape of the active site of proteins is made up of tertiary structure of proteins.

Answer: A

14. Which of the following statement concerning enzymes is incorrect?
A. Each enzyme has an optimal temperature.
B. Each enzyme has an optimal PH.
C. Most enzymes are highly specific.
D. Most enzymes are made up of RNA.

Explanation: Enzymes are biological catalysts which is mainly composed of glubullar proteins. In other ways, enzymes share certain characteristics in common. Among these some of them are:

- ✓ They have optimum temperature and PH.
- ✓ They are specific.
- ✓ They can be reused again and again.

Answer: D

15. Which of the following is the characteristics of enzymes?
A. They are consumed by enzyme mediated reaction
B. They are not altered by enzyme mediated reaction
C. The rise the activation energy
D. They get denatured at optimum temperature

Explanation:

Enzymes

- ✓ Do not get consumed by the reaction they catalyze
- ✓ Lower the activation energy inorder to speed up the rate of chemical reactions
- ✓ Work actively at optimum temperature and PH

Answer: B

Naming and classification of enzymes

Types of enzymes

Enzymes can be basically classified in to two based on their function. They are:

- ✓ **Extracellular enzymes** – are enzymes that function outside the cell.
  - Examples are digestive enzymes in the alimentary canal.
- ✓ **Intercellular enzymes** – are enzymes that function inside the cell.
  - Examples are respiratory enzymes found inside mitochondoria.

Common naming (nomenclature) of enzymes

Enzymes can be named based on various types of criterias. Some these includes:

i) Names given by adding the suffix – ase to the root of the name of the substrate.

Examples: Lipase – lipid hydrolyzing enzyme.

- ✓ **Sucrase** – sucrose hydrolyzing enzymes
- ✓ **Cellulase** – cellulose hydrolyzing enzyme
- ✓ Urease – controllos decomposition of small molecules of urea to carbondioxide and ammonia.

ii) Names given by the type of reaction catalyzed.

Examples: Dehydrogenase – catalyze the removal of hydrogen.

Polymerase – joining similar units together

iii) Names given based on the source from which they were first identified.

Examples: Papayin from papaya

Digestive enzymes like trypsin, pepsin, rennin and ptylin.

iv) Names indicating in at the end shows, they act on protein

Examples: pepsin, trypsin, erepsin, etc.

Enzyme commission a body set up to produce a systematic way of naming enzymes. The naming of enzymes is based on the ways in which enzymes act.

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Enzyme classification an systematic nomenclature of enzymes

There are six groups of enzymes which are classified based on the type of reaction which they catalyze. They are:

i) **Oxidoreductases** – catalyze the transfer of hydrogen and oxygen atoms or electrons from one substrate to another.

Examples: Oxidases – catalyze the transfer of hydrogen molecule to oxygen.

Dehydrogenases - transfer of hydrogen to coenzymes such as NAD^+.

ii) **Transferases** – catalyze the transfer of specific chemical group from one substrate to another.

Examples: Kinases – is type of enzyme that catalyzes the transfer of phosphate groups from high energy phosphate donating molecules to specific substrates.

- ✓ **Transminase** – is an enzyme that catalyzes the reaction between an amino acid and 2 – keto acid.
- ✓ **Phosphor transferase** – catalyze the transfer of phosphate group from one molecule to another.

iii) **Hydrolases** – catalyze hydrolysis reaction which involves the breaking down of large molecules by the addition of water.

Example: Digestive enzymes

- ✓ **Esterases** – splits esters in to an acid and alcohol.

iv) **Isomerase** – catalyze internal structure of substrate molecule.

Example: Phosphogluco – isomerase – rearranges glucose phosphate to fruegtose – 6- phosphate.

v) **Lyases** – Non – hydrolytic removal of a group or addition a group to a substrate.

- They catalyze the addition of chemical groups to a double bond or the removal of pyruvic acid in to ethanol and CO_2.

i) **Ligases (synthetases)** – catalyze joining of two molecules by the formation of new bonds.

Example: DNA ligase is used to repair breaks in DNA molecule.

- Citric acid synthetase

Illustrative Examples

16. Which of the following enzymes are named based on the source from which they were first identified?

A. Sucrase B. Dehydrogenase C. Papayin D. Polymerase

Explanation:

Enzyme	Named based on
Sucrase	adding ase to the root of name of the substrate.
Dehydrogenase	Type of reaction catalyzed
Papayin	Source from which they were first identified
Polymerase	Type of reaction catalyzed

Answer: C

17. Among the following which one is a protease enzyme?

A. Sucrase C. Polymerase

B. Cellulase D. Pepsin

Explanation: Protease enzymes are enzymes that catalyze reactions involving proteins. Some examples of protease enzymes are rennin, pepsin, trypsin, erepsin.

Answer: D

18. Enzymes that catalyze the transfer of specific chemical group from one substance to another are:

A. Oxidoreductases C. Hydrolases

B. Transferases D. Isomerase

Explanation: Transferases like kinase, transminase and phosphor transferase are responsible for transfer of specific chemical group from one substrate to another.

Answer: B

19. There are thousands of enzymes in a cell. This indicates which property of enzymes?

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A. Enzymes are used over and over again.
B. Enzymes are not consumed in chemical reaction.
C. Enzymes are specific in action
D. All

Explanation: Each and every type of reaction has its own specific type of enzyme. Accordingly to catalyze different types of reactions that take place in our body, there are thousands of enzymes.

Answer: C

20. Enzymes speed up a chemical reaction but:
A. They are not specific in action.
B. They do not change at the end of the reaction.
C. They do not change the overall energy of the reaction.
D. B and C

Answer: B

21. Which of the following properties does NOT characterize enzymes?
A. Enzymes are made up of globular and conjugated protein.
B. The presence of enzymes does not affect the nature of the end products of the reaction.
C. Enzymes do not use to start the chemical reaction.
D. Enzymes are effective in low concentration
E. None

Answer: B

22. An enzyme that catalyze the joining of two molecules by the formation of new bonds is:
A. Lyases
B. Ligases
C. Hydrolases
D. Isomerase

Explanation: Ligases (synthetases) are enzymes that catalyze the joining of two molecules.

Answer: B

23. Digestive enzymes are examples of:
A. Isomerase
B. Hydrolases
C. Lyases
D. Syntheses

Explanation: Digestive enzymes which catalyze hydrolysis reactions are

examples of hydrolases.

Answer: B

24. Which of the following is an example of ligases?
A. Citric acid synthetase
B. Phosphoglucose isomerase
C. Esterase
D. Phosphor transferase

Explanation: citric acid synthetase – Ligase
✓ Phosphugluco isomerase – Isomerase
✓ Esterase – Hydrolases
✓ Phosphor transferase – Transferase

Answer: A

25. Which enzymes name is derived from the substrate?
A. Papayin
B. Pepsin
C. Amylase
D. DNA Polymerase

Explanation: Enzymes such as lipase, sucrase, cellulase, urease are enzymes that are derived from the substrate.

Answer: D

26. According to the International Biochemistry Association the name of enzyme based on:
A. The seffix 'ase'.
B. The reaction they catalyze.
C. By the type of product they form.
D. By the substrate they act on.

Explanation: Based on the type of reaction they catalyze enzyme can be classified in to six groups. These are: oxidoreductase, transferases, hydrolases, isomerase, lysases and ligases.

Answer: B

27. Sucrase is an example of:
A. Oxidoreductase
B. Transferate
C. Hydrolase
D. Isomerase

Explanation: Sucrase is a digestive enzyme which is grouped under hydrolases.

Answer: C

28. Which of the following is NOT hydrolase enzyme?

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A. Mutate B. Pepsin C. Trypsin D. Amylase

Explanation: Pepsin, trypsin and amylase are digestive enzymes which are grouped as hydrolase.

Answer: A

29. The enzyme involved this reaction is:

$AH + NAD \longrightarrow A + NADH$

A. Oxidoreductase C. Dehydrogenase

B. Esterase D. Hydrolase

Explanation: Dehydrogenases are involved in catalyzing the transfer of hydrogen to coenzymes such as NAD^+.

Answer: C

30. Distinguish between extracellular enzymes and intracellular enzymes.

Explanation:

Extra cellular enzymes	Intracellular enzymes
✓ Are active outside the cell.	✓ Are active inside the cells
Example: Digestive enzymes	Example: Respiratory enzymes

31. Identify the molecule that is an enzyme?

A. Starch B. Glycogen C. cellulase D. chitin

Explanation: Starch, glycogen, chitin and cellulose are complex sugars. While, cellulase is enzyme.

Answer: C

32. Enzymes in lysosomes are:

A. Etracellular enzymes C. Hydrolase

B. Intracellular enzymes D. B and C

Explanation: Lysosome is an organelle found in the cytoplasm of the cell and it is used to produce enzymes for intracellular digestion.

Answer: B

33. The enzyme aldose is used to split compounds without adding water, so it is grouped in which class of enzyme?

A. Lyase C. Oxido reductase

B. Hydrolase D. Isomerase

Answer: A

34. Which of the following enzyme is named based on the property of its product?

A. Lipase C. Invertase

B. Trypsin D. None

Answer: C

35. Discuss three functions of the following enzymes:

✓ Urease ✓ Polymease

✓ Dehydrogenase ✓ Sucrase

Explanation:

Enzyme	Function
Urease	Controls decomposition of small molecules of urea to carbondioxide and ammonia.
Dehydrogenase	Transfer of hydrogen molecule to coenzymes such as NAD^+
Polymerase	Joining of similar units together
Sucrase	Hydrolysis of sucrose

Importance of Enzymes

Enzymes are very important in determining the survival of human beings on the world. The issue of enzyme utilization seems to be new, however people have been using enzymes for the last thousands of years. One of the typical example are fermentation products such as tej, tella, yoghurt, cheese which need catalytic activity of enzymes.

Now a days enzymes are used in various industrial processes. Some of these include:

✓ **Textile industries** – some of enzymes that are used in textile industries are:

- **Renin**- removes starch used in fabrics as adresive or support during weaving.
- **Protease** – helps to remove hair
- **Lipase** – helps to degrade animal hide

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- **Cellulose** – to bio-polish in cotton fabrics and to biostone to give the stone washed more easily

✓ **Diary industry** – enzymes used in production of diary industries such as cheese and yoghurt are:

- **Renin** – used to coagulate milk protein.
- **Lipase** – helps for ripening of Danish cheese.
- **Lactase** – degrades lactose to glucose and galactose.
- **Biochymyoscysticin** – used in the production of cheese.
- **Lactase** – to produce lactose free milk.

✓ **Medicine**

- **Gluco oxidase** – allows easy diagnosis of diabetes by testing urine.
- **Polmozyme** – used to treat cystic fibrosis.
  - ✓ Cystic fibrosis is a genetic disorder in children which is characterized by abnormal secreations in the body.

✓ **Pharmaceuticals**

- **Streptokinase** are used to dissolve clots of heart attack.
- **Abicavir sulphate** – is an important anti AIDS drug.
- **Pancreatic trypsin** – is used for treatment of inflammation.

✓ **Detergents**

- **Protease and amylases** – are used for washing or removing biological stains at low temperature which is used for saving energy.

✓ **Brewery industry**

- **Amylo glucoxidases** and **proteinases** degrade starch and protein in the barley to sugar and amino acids to feed yeast in alcoholic fermentation.

✓ **Food processing**

- **Opectirase** – used to clarify fruit juice.

✓ **Pulp and paper**

- **Amylase** – reduces the quantity of starch in paper which increases the quality.
- **Xylanase** – used in production of white paper

Illustrative Examples

36. An enzyme that is used in diary industry in coagulation of milk protein is:

A. Lactase B. Lipase C. Renin D. Biochymyosis

Explanation:

Enzyme	Use
Lactase	Degrades lactose to glucose and galactose
Lipase	For ripening of Danish cheese
Renin	To coagulate milk protein
Biochymyosis	For the production of cheese

Answer: C

37. Enzymes that are used to remove biological stains at low temperature are:

A. Protease and amylases
B. Pulmozymes and glucooxidase
C. Lipase and lactase
D. Cellulase and streptokinase

Explanation: In detergent industry enzymes such as protease and amylase are used for washing or removing biological stains at low temperature.

Answer: A

38. An enzyme that is used to treat systic fibrosis is:

A. Glucooxidese C. Cellulase
B. Pulmozyme D. Streptokinase

Explanation:

Enzyme	Use
Glucooxidase	For diagnosis of diabetes
Pulmozyme	To treat cystic fibrosis

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Cellulse	Eo bio-polish in cotton fabrics
Streptokinase	To dissolve clots of heart attack

39. An enzyme that is used in pulp and paper intury is:
A. Pectinase
B. Glucose oxidase
C. Strepto kinase
D. xylanase

Enzyme	Use
Pectinase	To clarify beer
Glucoxidase	Diagnosis of diabetes
Streptokinase	To dissolve clots of heart attack
Xylanase	Production of white paper

Answer: D

40. Abacavir sulphate is:
A. Enzyme used to indicate liver damage
B. Used to clarify fruit juice
C. One of the antiretroviral drugs
D. Used to degrease animal hide

Explanation: Abicavir sulphate is an important anti AIDS drug (antiretroviral drugs)

Answer: C

41. Which enzyme is used in industry to bio-polish in cotton industries?
A. Amylase B. Cellulse C. Lipase D. Protease

Explanation:

Enzyme	Function
Amylase	To reduce quantity of starch in order to increase quality of paper
Cellulase	To biopolish in cotton fabrics To biostone to make the stone washed more easily
Lipase	For ripening of Danish cheese
Protease	Helps to remove hair in textile industries

Answer: B

42. Industrial enzymes are used to prevent global warming because:
A. They hydrolyse the CO_2 produced

B. Reaction carried out at lower temperate
C. No need to add more heat in the system
D. B and C

Explanation: Since enzymes require lowering of activation energy required for the reaction, they work at less temperature (heat). This reduces the amount of CO_2 which contributes for global warming.

Answer: B

43. All are used in diary industry except:
A. Biochymyosin B. Lactase C. Xylanase D. Renin

Explanation: Lactase, renin and biochymyosin are important enzymes that are used in diary industry. However, xylanase is an enzyme which is used in pulp and paper industry for the production of white paper.

Answer: C

44. A pharmaceutical enzyme that is used to dissolve clots of heart attack is:
A. Steptokinase
B. Abacavir sulphate
C. Pancreatic trypsin
D. Pulmuzyme

Explanation: Pharmaceutical industries utilize enzymes in the production of various types of drugs. Among enzymes that are used in this industry streptokinase is an enzyme which is used to dissolve clots of heart attack.

Answer: A

45. Which of the following enzyme is incorrectly matched with its function?
A. Xylanase – production of paper
B. Pectinase – clarifies fruit juice
C. Pulmozyme – for diagnosis of diabetes
D. Abacavir sulphate – used as anti AIDS drug

Explanation:

✓ Pulmozyme is not used for diagnosis of diabetes, rather it is used to treat cystic fibrosis.

Answer: C

46. Lactose free milk is produced by using an enzyme called:
A. Lactase B. Bichymyosis C. Lipase D. Renin

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Explanation:

- An enzyme lactase is used to produce lactose free milk.

Answer: A

47. Which of the following is not important in diary industry?

A. Renin B. Lipase C. Lactase D. Pulmozyme

Explanation: Enzymes such as **renin, lipase, lactase** and **biochymyosis** are used in diary industry while **pulmozyme** is an enzyme which is used in medicine to treat cystic fibrosis.

Answer: D

48. List some enzymes that are important in:

- ✓ Textile industries
- ✓ Medicine
- ✓ Beerwing industry

Explanation:

Industry	Enzymes
Textile	Amylase, protease, lipase, cedllulase
Medicine	Glucoxidase, pulmozyme
Beerwing	Amylo glucoxidases, pectinases

3.2. Function of Enzymes

Enzymes as catalysts

Enzymes are proteins that act as **catalysts.** A catalyst is a substance that increases the rate of chemical reaction but not chemically changed by the reaction. They lower the **activation energy** needed for the reaction to go by bringing substrates together, so they can react with one another by making the substrates more reactive.

The higher the activation energy the slower the reaction.

Because of this, enzymes speed up the rate of the chemical reaction by lowering the activation energy.

Since enzymes act as a catalyst and do not get consumed by the reaction they catalyze, they can be used again and again.

Activation energy – is the energy required to start the chemical reaction.

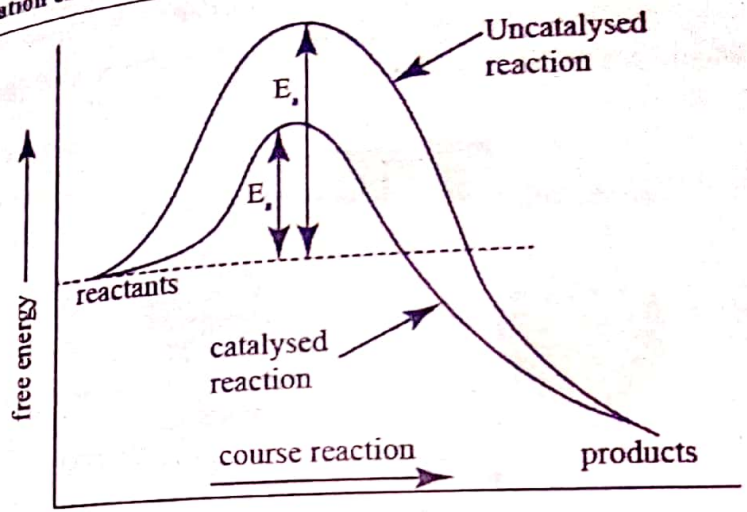

Figure: Activation energy for an uncatalysed reaction and the same reaction with a catalyst.

How do enzymes lower the activation energy?

There are two models of enzyme action which shows how enzymes catalyze chemical reactions by lowering the activation energy. These models are called **the lock and key model** and **induced fit model.**

i) Lock and key model

- An enzyme has a cavity with precise shape called **active site** which a substrate can fit into. In other words, it means that, the shape of the substrate molecule are complementary to that of the active site of an enzyme.
- Enzyme forms temporary chemical compound with their substrates. This forms an **enzyme substrate complex.** As a result the reactants enter the transition state which lowers the activation energy of the reaction.

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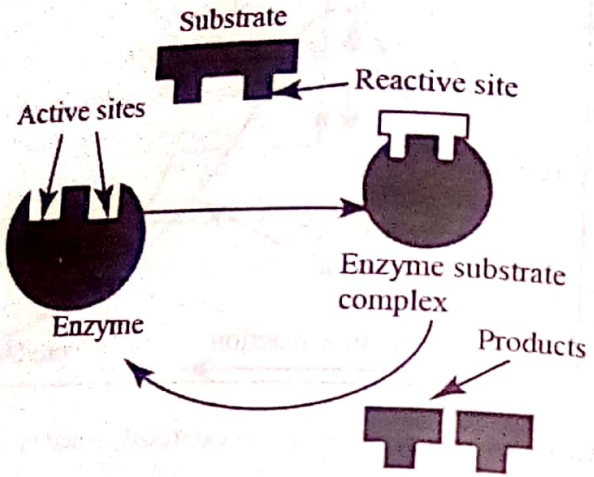
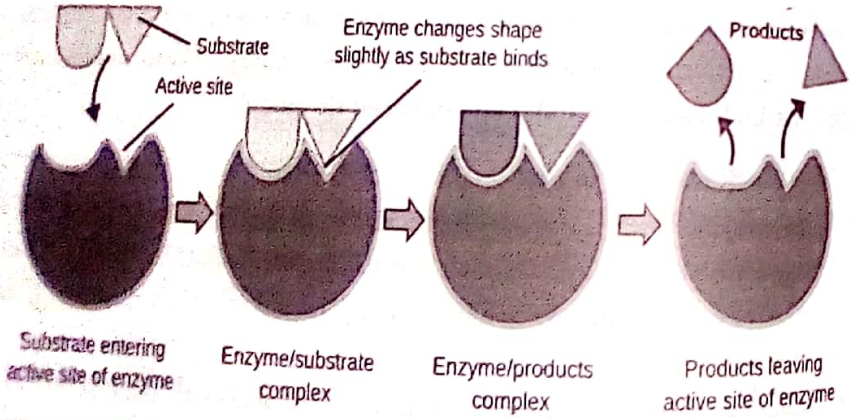
- The lock and key model of an enzyme lowers the activation energy by providing an alternative pathway of the reaction.
- Certain enzymes break a substrate down in to two or more products which is **catabolic reaction.**



- Some enzymes bond two or more substrates together to assemble one product which is **anabolic reaction.**



Note: Enzyme substrate complex is seen as an intermediate where less energy is required than the normal pathway of the reaction

ii) Induced fit model

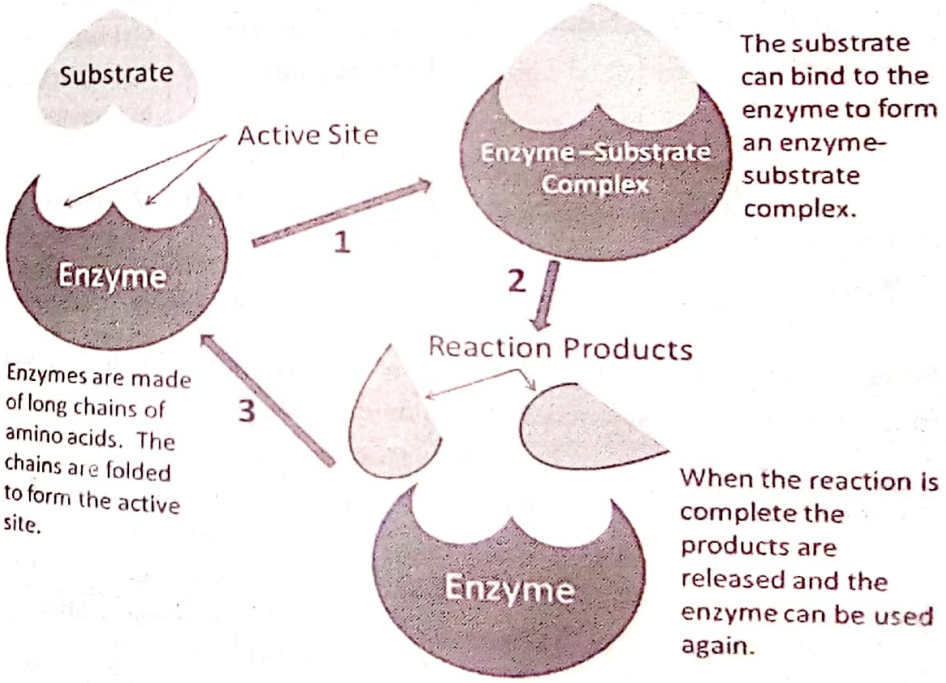
In this model the shape of an enzyme does not seem to be exactly complementary to that of the substrate. Thus, as the substrate approaches the active site, the site produces a conformational change which results in a perfect fit. The change in the shape of the active site of enzymes is possible because the active site of enzymes are not rigid.

As the substrate binds to the active site of an enzyme, they enter in to the transitional state which lowers the activation energy.

The induced fit model fully explains the property of enzymes.

After the reaction has taken place, and the products have gone, the active site of an enzyme returns to its normal shape.

The number of molecules of reactants that form enzyme substrate complexes with each molecule of an enzyme per second is called **the turn over rate.**



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Why do enzymes need cofactors?

- Enzymes are mainly composed of proteins. However, some groups of enzymes contain only proteins while other groups of enzymes contain protein and other part which are non-protein. The protein part of an enzyme is called **apoenzyme** and the non-protein part of an enzyme is called **cofactor.**
- Cofactors may be organic and are known as **coenzymes** or inorganic ions such as Ca^{2+}, Zn^{2+}. Most vitamins are also coenzymes.
- Apoenzymes and cofactors are together known as **haloenzymes.**
- Coenzymes bind with an enzyme and gives it a catalytic activity.

Note: The enzymes work only when the apoenzyme and the cofactor are combined.

- Apoenzyme only have no catalytic activity by itself and a cofactor alone has no catalytic activity by itself.

Table: Common types of coenzymes and their function

Coenzymes	Vitamin	Enzyme	Function
Nicotine amide adenine dinucleotide (NAD)	Niacin	Oxido reducates in respiration	Oxidation of hydrogen transfer in respiration
Flavin adenine dinuleotide (FAD)	Riboflavin	Oxidoreductase in respiration	Oxidation or hydrogen transfer in respiration

Some groups of enzymes that require mineral ions for their proper functioning will be discussed in the following table as:

Enzyme	Mineralion	Function
Carbonican hydrase	Zinc ion (Zn^{++})	Causes CO_2 to react with water to form hydrogen carbonate
Alcohol dehydrogenase	Zinc ion (Zn^{++})	Oxidize alcohol

Cytochrome oxidase	Copper ions (Cu^{++} or Cu^{+})	Transfer electrons to oxygen during respiration

Illustrative Examples

49. Change in shape in induced fit model is stimulated by:
    A. Active site
    B. Allostric
    C. Substrate
    D. Enzyme

Explanation: Substrate induces the active site to change its conformation.

Answer: C

50. In key and lock, the substrate bind on the active site, because they:
    A. Modify their shape according to the active site
    B. Have the same shape with the active site
    C. Have complementary shape with the active site
    D. None

Explanation: key and lock model shows a condition where the active site of an enzyme has a shape complementary to that of the substrate.

Answer: C

51. How do the key and lock model reduce the activation energy?
    A. The method generate heat for collusion of the reactant.
    B. The method align the necessary reactants together.
    C. Reactants round in the cell until they get reacted.
    D. None

Explanation: The lock and key model of an enzyme lowers the activation energy by providing an alternative pathway for the reaction.

52. Enzymes are not involved in:
    A. Proper orientation of the reactant
    B. Increase energy required to start the reaction
    C. Regulate metabolic activities
    D. Catalyzing the chemical reaction

Explanation: Enzymes speed up the rate of chemical reactions by lowering

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the activation energy required to start the chemical reaction.

Answer: B

53. A catalyst speeds up the reaction without affecting the:
   A. The nature of the product
   B. Change in overall energy
   C. Nature of the catalyst
   D. All are possible

Explanation: Catalysts speed up the rate of the chemical reaction without being affected by the changes that occur in the reaction.

Answer: C

54. Enzymes reduce the activation energy by:
   A. Providing starting energy to produce collusion.
   B. Raise the cell energy to product collusion.
   C. They press against the reaction to destabilize the electronic configuration.
   D. Activate the reactants effectively before they start.

Answer: C

55. Which group of organic compounds includes enzymes?
   A. Proteins
   B. Lipids
   C. Starches
   D. Carbohydrates

Explanation: All enzymes are globular proteins. Thus, all enzymes belong to proteins.

Answer: A

56. Which of the following PH values represent the strongest base?
   A. 2 B. 7 C. 13 D. 15

Explanation: The PH limit is between 1 to 14. Thus according to the question above PH value of 13 belongs to strongest bases.

Answer: C

57. Which **region** of human digestive tract contains enzymes that perform well at low PH?

Explanation: Stomach is an organ where digestion of protein starts by the catalytic activity of an enzyme called **pepsin.** Pepsin works actively at acidic PH which is lower.

58. Which of the following is false about the active site of enzymes?
   A. It is the small portion of enzymes
   B. Non-competitive inhibitor changes the shape of the active sites.
   C. Chemical reaction takes place on the active site.
   D. It has catabolic role only.

Explanation: Certain enzymes break down a substrate in to two or more products which is an example of **catabolic reaction.** While some other groups of enzymes bond two or more substrates together in order to form one product which is an example of **anabolic reaction.**

Answer: D

59. In the case of enzymatic chemical reactions, what do you call the substance that is acted up on by an enzyme? (UEE, 2006 E.C)
   A. Polypeptide
   B. Coenzyme
   C. Vitamin
   D. Substrate

Explanation: The substance on which an enzyme acts on is called substrate.

Answer: D

60. Which of the following is the common characteristics of all enzymes? (UEE, 2006 E.C)
   A. They act inside cells only
   B. They are proteins
   C. They require cofactors
   D. They operate at any PH value

Explanation:

- Some enzymes act only inside the cell while others act outside the cell.
- All enzymes are proteins
- Enzymes operate only at optimum PH and temperature

Answer: B

61. In which one of the following points does the induced fit model of enzyme action differ from the lock and key model?
   A. Enzyme lower the energy of activation

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B. Substrate bind at the active site of the enzyme
C. During the reaction, an enzyme substrate complex is formed
D. The shape of the substrate and active site are not complementary

Explanation: In induced fit model the shape of an enzyme is not exactly complementary to that of the substrate while, in key and lock model enzymes active site has a shape which is complementary to that of the substrate.

Answer: D

62. The energy that is required to start the chemical reaction is:
A. Chemical energy
B. Activation energy
C. Mechanical energy
D. kinetic energy

Explanation: Activation energy is the energy which is required to start the chemical reaction.

Answer: B

63. Which of the following is incorrect about lock and key model of enzymes?
A. The substrate fits the active site of enzymes
B. The shape of the substrate molecule is complementary to the active site
C. It lowers the activation energy by providing an alternative path way
D. The binding is of the substrate molecule produces conformational change of the shape of active site.

Explanation:
In key and lock model:

- The shape of the substrate fits into the active site because they are complementary.
- Enzymes lower the activation energy by providing an alternative pathway.

Answer: D

3.3. Factors affecting the function of enzymes

The proper functioning of enzymes can be affected by various factors. Some of those factors include:

✓ Temperature
✓ Substrate concentration
✓ PH
✓ Enzyme concentration
✓ Inhibitors

i) The effect of temperature on the rate of enzyme catalyzed reaction

- Increase in temperature rises the kinetic energy which results in:
  - ✓ Increase in molecular motion which increases the probability of a substrate to collide with an enzyme molecule. The more the collisions, the greater the chance of the substrates to fit in to the active site, up to a specific temperature.
  - ✓ Increase the vibration of molecules which results in breaking of the bonds that keeps the atoms in place. This causes change in the shape of the molecule and active site of an enzyme. It finally results in the lose of tertiary structure of the protein which causes denaturation.
- At lower temperatures than the optimum, the rate of the reaction decrease because of reduced enzyme/substrate collisions.

Therefore, the catalytic activity of enzymes works at normal temperature which is called **optimum temperature.**

Enzymes require different optimum temperatures for different organisms. Most enzymes have an optimum temperature between 30^0c and 40^0c, but there are many exceptions. For instance, enzymes in human beings work actively at the optimum temperature of 37^0c.

ii) Effect of PH on enzyme catalyzed reactions

The PH of the medium can have direct effect on the bonding responsible for the **secondary** and **tertiary structure** of enzymes. If the active site is changed the enzyme action will be affected.

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Each enzyme has an optimum pH. Most enzymes work best at **neutral** or **slightly alkaline** conditions. Some enzymes also work actively at acidic pH.

In appropriate pH affects enzymatic activity by changing the active site where a substrate cannot bind with. As a result, the reaction will not take place. Change in shape is mostly not permanent.

Table: The optimum pH of some enzymes in human body

Enzyme	Optimum pH
Pepsin	2
Salivary amylase	6.7 – 7.7
Catalase	7.6
Pancreatic lipase	7.0
Trypsin	7.5

iii) Effect of substrate concentration enzymes

- When considering the rate of an enzyme catalyzed reaction the proportion of enzyme to substrate molecules should be considered. Accordingly it is better to focus on:
  - ✓ The number of substrate molecules.
  - ✓ The number of available active sites.
- Every substrate molecule fits in the an active site.
- If there are more substrate molecules than enzyme molecules, then the number of active sites available is a limiting factor. In other words, if the substrate concentration is high, the enzyme molecule works at maximum turn over. Thus, the active sites bind with the substrate molecule all the time. However, an increase in substrate concentration beyond the number of active sites have no effect on enzymatic activity, since all the active sites are occupied.
- If the number of substrate molecule is fewer, there are few collisions and few enzyme substrate complexes form.
- The optimum rate of reaction is achieved when all the active sites are in use.

- In general, if more substrate is added, there is no increase in rate of product formation and if there are fewer substrate molecules than enzyme molecules, the reaction will take place very quickly, as long as the conditions are appropriate.

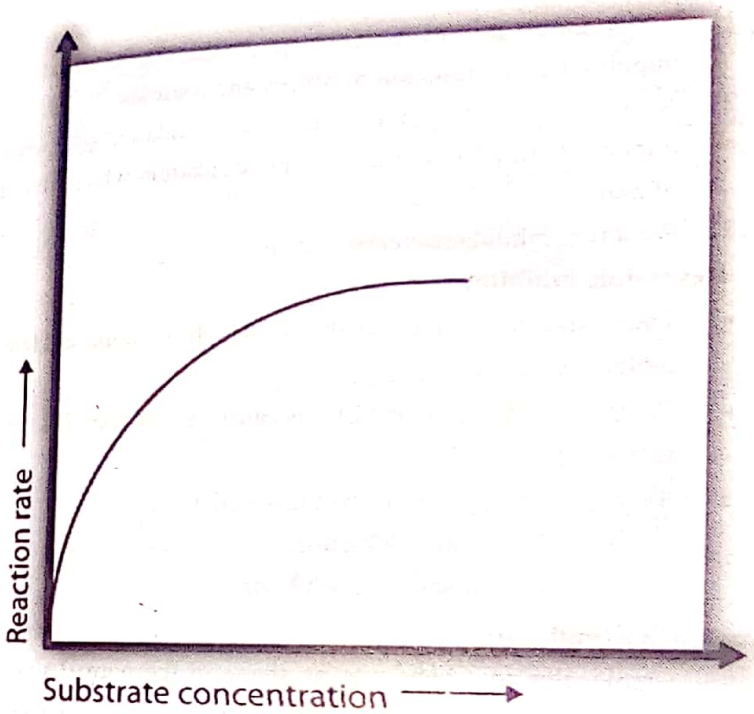

Figure: Relationship between substrate concentration and reaction rate.

iv) Inhibitors

If enzyme reactions inside the cell were to continue without regulation, there would be many problems. Cells posses regulatory chemicals which slow down or stop enzyme catalyzed reactions. This chemicals are called **inhibitors.** Otherwise, inhibitors can be shortly described as, substances that bind to enzymes and prevent them from forming enzyme substrate complexes.

- Enzyme inhibitors may be reversible or irreversible.

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a) Reversible inhibitors

- They combine with the functional group of an enzyme and permanently inactivate or destroy the enzyme.
- Most poisons are irreversible inhibitors.
- Examples:
  1) Nerve gases poison an enzyme acetyl cholinesterase which is important to the function of nerves and muscles.
  2) Asprin binds with an enzyme cyclo-oxidases – 2 which is an important enzyme in producing prostogladnin which give sensation of pain.
  3) Penicillin inhibits bacterial enzymes.

b) Reversible inhibitors

- They bind to enzymes weakly in which the bond holding enzyme molecules breaks easily.
- They are called reversible because allow-enzymes to become active again.
- Reversible inhibitors can be classified as:
  - ✓ Competitive inhibitors
  - ✓ Non-competitive inhibitors

Competitive inhibitors

They are molecules that have similar shape to the normal substrate, thus they have shapes that are complementary to the active site of anzyme.

This inhibitors have a tendency to compete with the normal substrate inorder to bind in to the active site of an enzyme. Therefore, because the inhibitor occupies the active site, enzymatic reaction is reduced.

They occupy the active site of an enzyme temporarily and does not permanently damage the enzymatic reaction.

- Competitive inhibitors can be reversed by an increase in the substrate concentration.
- The higher the proportion of competitive inhibitor, the slower the rate of the reaction.

- Some examples of competitive inhibitors are:
  - ✓ Pain killers act as competitive inhibitors of cyclo-oxidase-2
  - ✓ Cyanide acts as competitive inhibitor of enzyme cytochrome oxidase which is important in the release of energy in respiration.

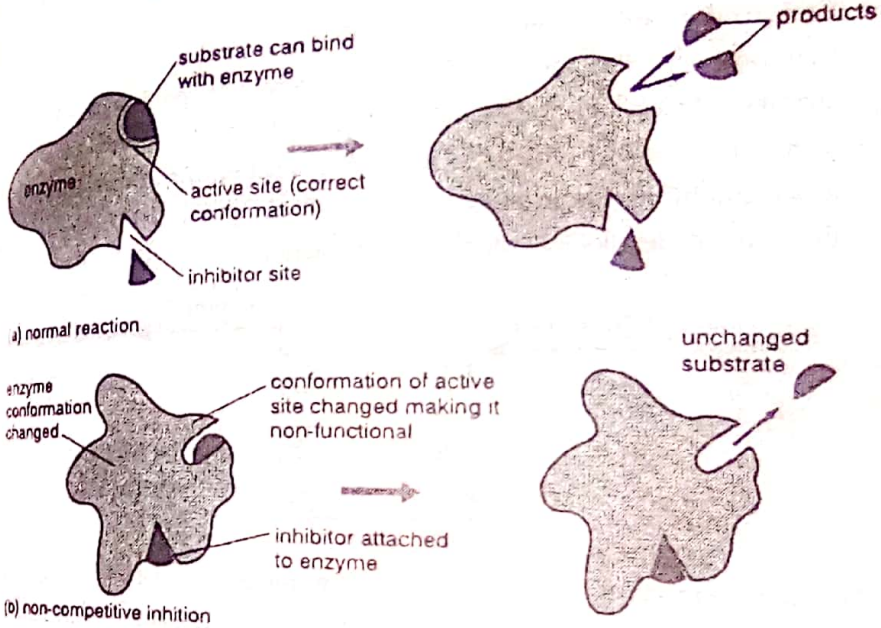

Figure: Competitive inhibition

Non-Competitive inhibitor

- They are molecules that bind to some part of an enzyme other than the active site. They have a different shape to the normal substrate.
- Noncompetitive inhibitor renders the enzyme inactive by altering its shape. Thus, they produce a conformational change in the part of enzyme molecule that includes active site. The change in the shape of the active site no longer allows binding of the substrate.

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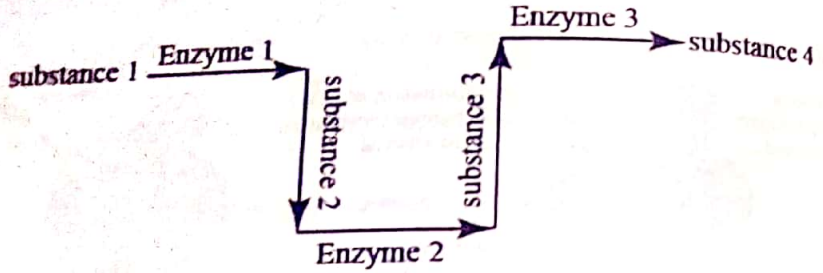
- The effectiveness of non-competitive inhibitors is not affected by the concentration of the substrate. They have greater inhibitory effect because they do not give a chance for a substrate molecule to compete. However, some substrate molecules may reach the active site before the non-competitive inhibitor.

End product inhibition

End product inhibition is a condition when the production of some enzymes that are harmful at high concentration will be controlled. This occurs in substances that are produced in a series of reactions that occur as a result of metabolic pathways.

For example lets look at the following reaction:



The main purpose of the above reaction is to produce a substance 4. Thus, if the need of substance 4 decreases by the cell, concentration of substance 4 in a cell becomes high. Therefore substance 4 which is not required by the cell become harmful and even toxic. In this condition, the path way in which substance 4 is produced must be controlled. This is called end **product inhibition**. This occurs when substance 4 by itself acts as non-competitive inhibitor to stop the conversion of substance1 to substance 2.

iv) Enzyme concentration

As the enzyme concentration increases, the rate of the reaction will increase. Thus, the rate of reaction is directly proportional to the concentration of enzymes. An increase in concentration of enzyme does not increase the activity of enzymes, since each enzyme works at a maximum turn over.

Illustrative Examples

64. Most of the coenzymes are derived from:
A. Vitamins B. Hormones C. Lipids D. Protein

Explanation: Coenzymes are types of cofactors that are the non-protein parts of an enzymes. Most coenzymes are vitamins.

Answer: A

65. The optimum temperature of an enzyme is the temperature at which:
A. there is denaturation.
B. the maximum number of substrate-enzyme-complex is formed.
C. there is maximum number of collusion between enzyme and substrate.
D. The particles have the most kinetic energy.

Explanation: At optimum temperature enzymes work actively and then maximum number of substrate-enzyme complex is formed.

Answer: B

66. An apoenzyme is:
A. A protein with enzymatic activity
B. A non-protein with enzymatic activity
C. A non-protein with no enzymatic activity
D. A protein with no enzymatic activity

Explanation: Apoenzyme is a protein part of an active site of enzyme that combines with a cofactor. Thus, it has enzymatic activity.

Answer: A

67. What do we call the substance up on which an enzyme acts?
A. Product C. Activation energy
B. Substrate D. Enzyme-substrate complex

***Explanation*:** An enzyme acts on a substance which is called substrate.

Answer: B

68. When an enzyme is denatured by heat or extreme pH, which one of the following does it lose?
A. The peptide bond C. Secondary structure

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B. Primary structure D. Tertiary structure

Explanation: Increase in temperature beyond the optimum level causes an enzyme to lose its tertiary structure.

Answer: D

69. If an enzyme has its optimum activity at neutral pH, to which of the following pH values does this correspond? (UEE, 2001)

A. pH 14 B. pH 12 C. pH 7 D. pH 4

Explanation: The neutral pH value is found around pH value of 7.

Answer: C

70. An enzyme known as catalase converts hydrogen peroxide in to water and oxygen. In this reaction, which one is the substrate? (UEE, 2002)

A. Catalse C. Water and oxygen

B. Hydrogen peroxide D. Water only

Explanation: An enzyme catalase acted on a substrate called hydrogen peroxide.

Answer: B

71. All of the following are correct about enzymes except?

A. They are proteins

B. They operate at the same optimum pH

C. They lower the activation energy

D. They require coenzyme to work effectively

Explanation:

Enzymes:

- ✓ are proteins.
- ✓ operate at different pH values.
- ✓ lower the activation energy.
- ✓ require coenzyme to work actively.

Answer: B

72. What is the significance of acidic medium for enzyme pepsin in the stomach?

A. Change in the shape of the enzyme.

B. Maintain ionic and hydrogen bond of the enzyme.

C. Inactivate for a short time.

D. Protect it from disintegration by protolytic enzyme.

Explanation: Change in pH has the following influences:

- ✓ The ionic bond that holds the tertiary structure in place.
- ✓ Affect the charge of aminoacids on the active site which creates a problem on substrates from attaching to the active site.

Answer: B

73. What do we call the situation in which the bonding of the substrate changes the active site of an enzyme to accommodate the substrate?

A. Active site C. Activation energy

B. Allostric inhibition D. Induced fit

Explanation: In induced fit model, since the substrate do not seem complementary to that of the active site, it changes the active site of an enzyme to accommodate the substrate.

Answer: D

74. When a potato cylinder added in solution containing hydrogen peroxides, the reaction will produce water and oxygen. Which one is the substrate?

A. Potato cylinder C. Hydrogen peroxide

B. Water and oxygen D. Water only

Explanation: see example number 70

Answer: C

75. How do you describe the effect of substrate on enzymatic reaction?

A. The rate of chemical reaction increases as we increase the substrate.

B. At higher substrate concentration, the rate of reaction decreases.

C. Substrate concentration does not have effect on the chemical reaction.

D. At higher substrate concentration, the rate of reaction remains constant

Explanation: If the substrate concentration is higher than the enzyme concentration, all the active site of the enzyme saturate with the substrate. So, the rate of reaction becomes constant.

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76. Coenzyme function by:

Answer: B

A. Binding to the enzyme and weakening the bonds the substrate.
B. Binding the substrate and weakening the bonds of the enzyme.
C. Binding two or more substrate.
D. Helping the substrate to find enzyme.

77. If the temperature of enzyme catalyzed reaction is increased from 40°c to 70°c, the rate of the reaction will:

Answer: A

A. Increase indefinitely.
B. Decrease immediately.
C. Decrease to zero because the enzyme denature.
D. Decrease and then increase.

Explanation: At extremely higher temperature than the optimum temperature enzymes get denatured.

78. In negative feedback mechanism, the product attach:

Answer: C

A. On the active site of the first enzyme
B. On the allostric site of the enzyme.
C. On the first substrate molecule.
D. On every part of the first molecule

Explanation: The substrate and end product do not have the same shape. So, they do not compete to hold and the end product to attach on the allostric site.

79. Allostric effector differs from competitive inhibitor in all except:

Answer: B

A. They do not have the same shape with the substrate.
B. They have inhibitory role.
C. The bind on the allostric site.
D. They play regulatory role.

Explanation: Allostric effectors can be inhibitors or activators. They attach on the allostric site and change the shape of the active site. This change can

either activate or inhibit the chemical reaction like competitive inhibitor.

Answer: B

80. Competitive and non-competitive inhibitors are similar in:

A. Both compete to hold the active site
B. Both inhibit enzymatic activity temporary
C. Both inactive enzyme activity
D. Both bring permanent changes

Answer: B

81. Which controlling mechanism is the best to stop chemical reaction until all the accumulated essential biochemical products are used by the cell?

A. End product inhibitor
B. Competitive reversible inhibitor
C. Irreversible inhibitor
D. All

Explanation: End product inhibition or feedback control is the type of enzyme regulation, in which the formation of the product inhibits an earlier reaction in the sequence.

Answer: A

82. Increase in the amount of substrate may solve the problem of:

A. Allostric inhibitor
B. Reversible inhibition
C. Irreversible inhibition
D. End product inhibition

Explanation: As the amount of substrate increases, the capacity of the reversible inhibitors to bind to the active site of on enzyme decreases.

Answer: B

83. Competitive inhibitor inhibit the enzyme by competition with,

A. Enzyme inhibitor
B. Inhibitor for active site
C. Substrate for active site
D. Substrate for allostric sites

Explanation: Competitive inhibitor have similar shape with the substrate. Thus, both compete to hold the active site of succinate dehydrogenase.

Answer: C

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84. Which one of the following is incorrect about allostric regulator?
   A. They have regulatory site rather than the active site
   B. Allostric effector bind with enzyme either to inhibit or promote the chemical reaction.
   C. Allostric activator promote the chemical reaction
   D. The substrate attach on the allostric site

Answer: D

85. In competitive inhibition, which one of the following factor determines the rate of inhibition?
   A. Temperature
   B. Enzyme concentration
   C. Substrate concentration
   D. The rate of inhibitor to enzyme concentration

Explanation: As the substrate concentration decreases, the rate of reaction also increases, because the substrate is more competent to hold the active site.

Answer: C

86. Which of the following mechanism do cells use to regulate enzyme catalyzed reaction in metabolic pathway?
   A. Enzyme denaturation
   B. Irreversible inhibition
   C. End product inhibition
   D. Competitive inhibition

Explanation: When the end product is accumulated, it shuts down the step by binding with the first enzyme.

Answer: C

87. If an enzyme is provided with its normal substrate plus another molecule having the same shape and size as the substrate, which one of the following would happen? (UEE, 2003)
   A. Allostric inhibition
   B. Competitive inhibition
   C. Irreversible inhibition
   D. End product inhibition

Explanation: In competitive reverse inhibitor:

✓ The inhibitor is structurally similar with the substrate.

✓ The inhibitor of the substrate is complementary to the active site of enzyme.

Answer: B

88. In competitive inhibition, which one of the following factors determines the rate of inhibition? (UEE, 2004)
   A. Temperature
   B. Enzyme concentration
   C. Substrate concentration
   D. Ratio of inhibitor to enzyme concentration

Explanation: As the number of inhibitor increases, many of the active sites hold the competitive inhibitor than the normal substrate. So that, the rate of reaction slows down.

Answer: D

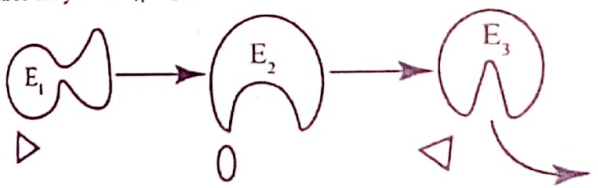
89. Give two factors that are needed to be kept constant when investigating both the inhibited and non-inhibited reactions.

Explanation: Effectiveness of enzymes is highly dependent on:

✓ Temperature

✓ pH

90. The sequence below shows a biochemical pathway which involves three enzymes E_1, E_2 and E_3 to form molecule x.



a) Explain how product x is an allosteric inhibitor of E_1.

Explanation: The molecule fits in to an alternative site on E_1 and the shape of active site is changed, the substrate therefore, cannot bind to the active site.

b) Name the type of inhibition shown by this sequence.

Explanation: End product inhibition

91. Does the substrate compete with a non-competitive inhibitor to bind with the active site of an enzyme?

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Explanation: The substrate do not compete with non-competitive inhibitor to reach on the active site, since the non-competitive inhibitor bind to the part of an enzyme other than the active site.

92. How does a non-competitive inhibitor reduce the rate of an enzyme catalyzed reaction?

Explanation: A non-competitive inhibitor reduce the rate of an enzyme catalyzed reaction by binding to some part of an enzyme other than the active site and change the shape of the active site which prevents an enzyme to bind with the substrate.

93. Which of the following P^H values represents the strongest base? (UEE 2006)

A. 2 B. 7 C. 13 D. 15

Explanation: The strongest bases are those base having the largest p^H value which is in the range of $PH_7 - P_{14}$.

Answer: C

94. Which region of human digestive tract contains enzymes that perform well at low P^H?

A. Mouth B. Stomach C. Small intensive D. Large intensive

Explanation: Most digestive enzymes require neutral and basic P^H. However, enzymes that catalyze digestive process in the stomach require acidic P^H which has lower value.

Answer: B

95. Which of the following is the common characteristics of enzymes? (UEE,)

A. They act inside cells only
B. They are proteins
C. They require cofactors
D. They operate at any P^H.

Explanation: All enzymes have the following characteristics in common. They include:

- ✓ Enzymes act in or outside the cell
- ✓ Enzymes are globular proteins.

- ✓ Enzymes require optimum P^H and temperature.

Answer: B

96. Which group of organic compounds includes all enzymes? (UEE,)

A. Proteins B. Lipids C. Starches D. Carbohydrates

Explanation: All enzymes are composed of proteins.

Answer: A

97. List out some important properties of enzymes.

Explanation: Enzymes have the following properties:

- ✓ They are globular proteins
- ✓ They are biological catalysts
- ✓ They are specific
- ✓ Small amount of enzymes can bring change in large amount of substrate
- ✓ They can be reused several times
- ✓ They are affected by P^H and temperature

98. What are the uses of active site of enzymes?

Explanation: Active site of an enzyme are used to:

- ✓ Bind to a particular substrate
- ✓ Minimize energy required to carry the reaction

99. What is activation energy? What is the relationship of a catalyst to activation energy?

Explanation: Activation energy is the energy required to start up the chemical reaction.

Catalysts are substances that speed up the rate of chemical reaction by reducing free activation energy.

100. Give the function of each of the following:

- ✓ active site of an enzyme
- ✓ coenzyme

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Explanation:

- Active site of an enzyme is used to bind to a particular substrate and lower the activation energy required to start the chemical reaction.
- Coenzymes are organic but non-protein parts of an enzyme. However, coenzyme alone has no catalytic activity.

101. Clearly explain the meaning of enzyme commission

Explanation: Enzyme commission is a team of biologists who decide a systematic way of naming enzymes based on the way enzymes act.

102. Distinguish between the following groups of enzymes.

✓ Hydrolasses and ligases

Explanation:

✓ Hydrolases are groups of enzymes that breakdown large molecules by the addition of water. While, ligases are enzymes that catalyze joining of two molecules by forming new bonds.

103. List out some areas in which enzymes are vital.

Explanation: Enzymes are vital in various industrial areas. Among these some of them are:

- ✓ Textile industries
- ✓ Diary industry
- ✓ Medicine
- ✓ Pharmaceuticals
- ✓ Deteregents
- ✓ Brewery industry
- ✓ Food processing
- ✓ pulp and paper

104. Why do enzyme require less temperature to catalyze chemical reaction?

Explanation: Enzymes mostly require less temperature to catalyze chemical reaction for:

- ✓ Reducing loss of energy
- ✓ Reducing the amount of CO_2 released

105. What makes the active site in key and lock model different from induced fit model?

Explanation: The active site of enzymes in key and lock model has the shape which is complementary to the active site. However, the shape of active site of an enzyme cannot bind to the substrate.

106. Explain the meaning of turn over rate.

Explanation: The over rate refers to the number of molecules of reactants that form enzyme-substrate complex per second.

107. Explain how the substrate concentration affects activity of enzymes.

Explanation: As the substrate concentration increases, enzyme works at maximum turn over and vice versa.

108. What does it mean by enzyme denaturation?

Explanation: **Denaturation** – is alteration of the structural properties of macromolecules such as proteins. This includes enzymes. Enzymes get denatured when they are exposed to extremely high temperature than the optimum one.

109. Distinguish between reversible and irreversible inhibitors.

Explanation:

Reversible inhibitors	Irreversible inhibitors
• Inhibitors that combine on the active site of an enzyme (competitive) or parts other than the active site of enzymes (non-competitive). • Inhibits enzyme activity only temporarily.	Are inhibitors that combine with a functional group of an enzyme and permanently in activate or destroy the enzyme

110. Explain the meaning of end product inhibition?

Explanation: End product inhibition is a condition where the end product inhibits the enzyme controlling the first stage of a reaction sequence. For example:

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Substance 1 —Enzyme 1→ Substance 2 —Enzyme 2→ Substance 3

In above reaction when a substance 3 acts as an end product inhibitor for conversation of substance 1 to substance 2.

111. Clearly distinguish between:

✓ Apoenzyme and cofactor

Explanation:

Apoenzyme	Cofactor
The protein part of an enzyme that requires a cofactor or prosthetic group to be functional.	A non-protein part of an enzyme that is required for proper functioning of enzymes.

112. Give some examples of coenzymes with their functions.

Examples: Some examples of coenzymes and their functions are:

✓ **Nicotine amide adenine dinucleotide (NAD)** – helps for oxidation of hydrogen transfer in respiration.

✓ **Flavin** adenine dinucleatide (FAD) – oxidation of hydrogen transfer in respiration

113. Discuss the meanings of

✓ Holoenzymes

✓ Coenzymes

✓ Activator

Explanation:

Holoenzymes – an enzyme containing apoenzyme and cofactor together.

Coenzymes – non-protein part of an enzyme that is required for enzyme functioning.

Activator – a substance that removes inhibitor.

Conceptual Summary Example

1. What effect does an enzyme have on the required activation energy of a reaction?

Explanation: Enzymes affect the rate of the chemical reaction by lowering the activation energy (Ea) necessary to initiate a chemical reaction. If molecules needless energy to react because the activation barrier is lowered, a larger fraction of the reactant molecules reacts at any one time. As a result, the reaction proceeds more quickly.

2. How does the Function of the active site of an enzyme differ from that of an allosteric site?

Explanation:

- Active site is the region of enzyme which a substrate fits in to.
- Allosteric site is the region the enzyme molecule other than the active site.
- When a substance binds to the enzyme's allosteric site, the conformation of the enzyme's active site changes, thereby modifying the enzymes activity.

3. How are temperature and pH optima of an enzyme related to its structure and function?

Explanation: Most enzymes have an optimum temperature at which they work actively.

- Enzymatic reactions occur slowly or not at all at low temperatures.
- As temperature increases, molecular motion increases, resulting in more molecular collisions. Therefore, the rates of most enzyme controlled reactions increases as temperature increases within limits. High temperature rapidly denature most enzymes when the molecular conformation (3-D shape) of the protein becomes altered as hydrogen bonds responsible for its secondary, tertiary and quaternary structures are broken.

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- The activity of an enzyme may be markedly changed by alteration in pH, which in turn alters electrical charges on the enzyme. Changes in charge affect the ionic charge that contribute to tertiary and quaternary structure.

4. Explain how an enzyme lowers the required energy of activation for a reaction

Explanation: An enzyme is a biological catalyst, which greatly increases the speed of a chemical reaction without being consumed.

- An enzyme works best by lowering the activation energy which is the kinetic energy necessary to get a reaction going.
- The active site of an enzyme is a 3-D region where substrates come in to close contact and react more readily. When a substrate binds to an active site, an **enzyme – substrate complex** forms in which the shape of the enzyme and substrate change slightly. This induced fit facilitates the breaking of bonds and formation of new ones.

5. Describe specific ways enzymes are regulated

Explanation: Enzymes work best at specific temperature and pH conditions. A cell can regulate enzymatic activity by controlling the amount of enzyme produced and by regulating metabolic conditions that influence the shape of the enzyme.

Post test

1. Which describes the relationship in the shape between an enzyme and a substrate?
   A. competitive
   B. regenerative
   C. symmetrical
   D. complementary
2. Why do cells contain many different enzymes?
   A. Enzymes are temperature sensitive.
   B. Enzymes are specific in their action.
   C. Enzymes are sensitive to pH changes.

   D. Enzymes are sensitive to substrate concentration.
3. A reaction takes place because the active site of an enzyme is complementary to:
   A. One type of substrate molecule.
   B. All types of substrate molecule.
   C. One type of product molecule.
   D. All types of product molecules.
4. Which of the following can greatly affect the speed of an enzyme reaction?
   I. pH
   II. Temperature
   III. Presence of inhibitors
   A. I only
   B. II only
   C. III only
   D. I and II only
   E. I, II and III only
5. Which of the following statements best describes an effect of pH on the functioning of salivary amylase?
   A. Salivary amylase functions must effectively at pH of about 4.
   B. Salivary amylase functions most effectively at a pH of about 7.
   C. Salivary amylase cannot break down starch in to maltose at pH values less than 7.
   D. Salivary amylase breaks down protein instead of starch at pH values greater than 9.
6. Why do cells contain many different enzymes?
   A. Enzymes are temperature specific.
   B. Enzymes are specific in their reaction.
   C. Enzymes are sensitive to pH changes.
   D. Enzymes are sensitive to substrate concentration.
7. The kinetic energy required to intiate a reaction is called:
   A. Activation energy
   B. Bond energy
   C. potential energy
   D. free energy

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8. A biological catalyst that affects the rate of chemical reaction without being consumed by the reaction is:
   A. product B. cofactor C. coenzyme D. enzyme
9. The region of an enzyme molecule that combine with the substrate is:
   A. Non – competitive inhibitor
   B. Competitive inhibitor
   C. Irreversible inhibitor
   D. Allosteric regulator
10. In the following reaction series, which enzyme(s) is/are most likely to have an allostric site to which the end product E binds?

    A **enzyme 1** B **enzyme 2** C **Enzyme 3** D **Enzyme 4** E

    A. Enzyme 1 C. Enzyme 3
    B. Enzyme 2 D. Enzyme 4

Critical thinking

1. Given what you have learned in this chapter, explain why an extremely high fever (body temperature above 105^0F or 40%) is often fatal.
2. The optum temperature for enzymatic activity in human body is 35^0c to 40^0c. however, people live at different parts of the world where there is extremely low (below 0^0c) . thus, how can you explain that, this extreme conditions have no effect on enzymatic activity.

Supplementary Exercise on Unit – 3

1. Which of the following is NOT the property of enzymes?
   A. They are globular proteins.
   B. They are biological catalysts.
   C. They have wholstic effect.
   D. They can be reused.
2. If an enzyme gets denatured up on exposure to extreme heat, which property of an enzyme does this explain?
   A. Enzymes are sensitive to P^H.

   B. Enzymes are sensitive to temperature
   C. Enzymes are specific.
   D. Enzymes can be reused again and again.
3. An enzyme that controls decomposition of small molecules of urea to carbondioxide and ammonia is:
   A. Sucrase C. Lipase
   B. Cellulose D. Urease
4. Enzymes that catalyze the joining of similar units together are:
   A. Dehydrogenase C. Lipase
   B. Polymerase D. Pepsin
5. Which of the following enzyme is incorrectly matched with its function in diary industry?
   A. Renin – coagulation of milk protein
   B. Lipase – ripening of Danish bluecheese
   C. Lactase – production of lactose free milk
   D. Biochymyosis – treatment of cystic fibrosis
6. A pharmaceutical enzyme that is used as an important anti AIDS drug is:
   A. Streptokinase C. Pancreatic trypsin
   B. Abacavir sulphate D. Gluco oxidase
7. Which of the following enzyme is NOT useful in pulp and paper industry?
   A. Amylase C. Esterase
   B. Xylenase D. Pectinase
8. Enzymes require less temperature (heat) to catalyze the chemical reaction. This is important for:
   A. Reducing loss of energy.
   B. Reducing the amount of carbondioxide which contributes to the global warming
   C. Keeping an enzyme work at optimum temperature
   D. All are possible answers

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9. Which of the following is incorrect about the key and lock model of enzymes?
   A. The substrate fits the active site of enzyme molecule.
   B. The shape of substrate is complementary to that of the active site.
   C. An enzyme lowers the activation energy by providing the active site.
   D. The substrate molecule produces conformational change of shape in the active site.
10. Alteration of structural properties that causes alteration of the tertiary structure of proteins is:
   A. Inhibition C. Denaturation
   B. Inactivation D. kinesis
11. Organic substances that are derived from vitamins and bind with enzymes to give them catalytic activity are:
   A. Apoenzymes C. Simple enzymes
   B. Coenzymes D. Prostheetic group
12. The mineral ions that are responsible for proper functioning of an enzyme cytochrome oxidase is:
   A. Zinc ions C. niain
   B. Copper ions D. Riboflavin
13. Changes in PH affect an enzyme molecule by:
   A. Breaking ionic bonds that holds tertiary structure
   B. Altering the charge on some amino acids that form the active site
   C. Increasing the activation energy
   D. A and B are possible answers
14. Non-competitive inhibitors:
   A. Competitive with normal substrate
   B. Can be reversed by an increase in the substrate concentration
   C. It renders the enzyme in active by a hering its shape
   D. They have shapes that are complementary to the active site of an enzyme

15. A substance that affects the rate of chemical reaction without being affected by itself is:
   A. Substrate C. Activator
   B. Catalyst D. Inhibitor

Supplementary Exercise on Unit – 3

1. The small portion of an enzyme molecule that combines with the substrate is _____.
2. _____ are biological catalysts produced by cells.
3. _____ inhibitor alters the shape of the enzyme, rendering it inactive.
4. A substance that affects the rate of a chemical reaction without being consumed by the reaction is a _____.
5. The energy that is required to initiate a chemical reaction is called _____.
6. Enzyme and their substrate combine temporarly to form _____.
7. _____ is the non-protein part of haloenyme.
8. _____ have similar shape with the substrate and they complete to hold the active site to inhibi5 enzymatic activity temporarly.
9. Enzymes that work inside the cells are called _____
10. Exess heat affects the _____ structure that held the active site inplace.
11. _____ is co-enzyme make up of riboflavin.
12. _____ is used to bleech pulp to white paper in pulp and paper industry.
13. The number of substrate that attach or the active site of an enzyme-per second is called _____.

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II. Write 'TRUE' if the statement is correct and "FALSE" if the statement is incorrect.

14. Hydrolase enzymes are sued for condensation reaction.
15. Enzymes are made up of globular proteins.
16. Apoenzyme alone has catalytic role.
17. Allostric site is the part of enzyme other than the active site.
18. Non-competitive reversible inhibitor bind on the allostric site of enzyme.
19. Inhibitor can also activate a chemical reaction.
20. Reversible inhibitor bind on enzyme permanently

III. Multiple choice

21. Which of the following NOT the property of enzymes?
    A. They are globular proteins
    B. They are biological catalysts
    C. They have wholstic effect.
    D. They can be reused.
22. If an enzyme gets denatured up on exposure to extreme heat.
    A. Enzymes are sensitive to pH
    B. Enzymes are sensitive to temperature
    C. Enzymes are specific
    D. Enzymes can be reused again and again.
23. An enzyme that controls decomposition on small molecules of urea to carbon dioxide and ammonia is:
    A. Sucrose C. Lipase
    B. Cellulase D. Urease
24. Enzymes that catalyze joining of similar units together are:
    A. Dehydrogenase C. Papayin
    B. Polymerase D. pepsin
25. Which of the following enzyme is incorrectly matched with its function in diary industry?
    A. Rennin-cougulation of milk protein
    B. Lipase – ripening of Danish blue cheese.

C. Lactase-production of lactose free milk.
D. Biochymyosis-treatment of cystic fibrosis

26. A pharmaceutical enzyme that is used as an important anti-AIDS drug is:
    A. Streptokinase C. Esterase
    B. Abacavir sulphate D. Pectinase
27. Which of the following enzyme is NOT use full in pulp and paper industry?
    A. Amylase C. Esterase
    B. Xylenase D. Pectinase
28. Enzyme require less temperature (heat) to catalyze the chemical reaction. This is important for:
    A. Reduce loss of energy
    B. Reduce the amount of carbondioxide which contributes to the global warning.
    C. Keeping an Enzyme work at optimum temperature.
    D. All are possible answers.
29. Which of the following is incorrect about the lock-and-key model of enzymes?
    A. The substrate fits the active site of the enzyme molecule
    B. The shape of the substrate is complementary to that of the active site.
    C. An enzyme lowers the activation energy by providing an a native pathway of the reaction.
    D. The substrate molecule produces conformational change of shape in the active site.
30. Alteration of structural properties that causes alteration of the tertiary structure of protein is:
    A. Inhibition C. Denaturation
    B. Inactivation D. Kinesis
31. Organic substances that are derived from vitamins and bind with enzymes to give them catalytic activity.

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A. Apoenzymes
B. Coenzymes
C. Simple enzyme
D. Prosthetic group

32. The mineral ions that are responsible for proper functioning of an enzyme cytochrome oxidase is:
A. Zinc ions
B. Copperions
C. Niacin
D. riboflavin

33. Changes in pH affect an enzyme molecule by:
A. Breaking ionic bonds that holds tertiary structure.
B. Altering the charge on some of the amino acids that form the active site.
C. Increasing the activation energy
D. A and B are possible answers.

34. Non-competitive inhibitors:
A. Compete with the normal stbstrate
B. It can be reversed by an increase in the substrate concentration.
C. It renders the enzyme in active by altering its shape.
D. They have shapes that are complementary the active site of an enzyme.

35. A substance that affects the rate of chemical reaction without being affected by itself is:
A. Substrate
B. Catalyst
C. Activator
D. Inhibitor

Matching

Column A	Column B
36. Transferases	A. Catalyze internal structure of substrate molecule
37. Oxidoreductases	B. Involves the breaking down of large molecules by the addition of water
38. Hydrolases	C. Catalyze the transfer of hydrogen and oxygen atoms from one substrate to another
39. Isomerase	D. Catalyze the transfer of specific chemical group from one substrate to another
40. Lyases	E. Catalyze the joining of two molecules by the formation of new bonds.
41. Lipases	F. Non-hydrolytic removal of a group or addition of a group to a substrate.

Supplementary Exercise on Unit – 3

1. Active site
2. Enzymes
3. Non-competitive
4. Catalyst
5. Activation energy
6. enzyme – in substrate – complex
7. Co-factor
8. competitive inhibitor
9. Intracellular enzymes
10. Tertiary
11. Flain adenine di nucleotide
12. Zylanase
13. Turnover rate
14. False
15. True
16. False
17. True
18. True
19. False
20. False

21. C	22.B	23.D	24.B	25.D	26.B	27.D
28. D	29.D	30.C	31.B	32.B	33.D	34.C
35. B	36.D	37.C	38.B	39.A	40.F	41.E

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UNIT FOUR

Cell Biology

The word "cell" derived from a Latin word called '*cellula*' which means '*little rooms*' and this name is first given by an English biologist Called Robert Hooke in 1665. Then after people have been using this name and believed to use it for the future.

Cells are called the *building units* of all organisms. This is true, because all living things are made up of cells. Some groups of organisms are made up of only a single cell and they are called *unicellular organisms*, while other groups of organisms are made up of many cells and they are called *Multicellular organisms*. In multicellular organisms a single cell called zygote divides to form new cells again and again. This results in forming complex tissues, organs and systems of an organism

The cell is a basic functioning unit of an organism which is capable of carrying on all the activities necessary for life. This is because the chemical reactions that take place in cells involve energy release needed to support life and build structures.

4.1. Cell theory

Cell theory is an idea which states that cells are the fundamental units of life. This theory involves the collection of statements developed through the contribution of many biologists. Accordingly, the cell theory has the following key points:

- ✓ All living things are composed of cells and cell products.
- ✓ New cells are formed only by the division of pre-existing cells.
- ✓ The cell contains inherited information (genes) that are used as instructions for growth, functioning and development.

- ✓ The cell is structural and functional unit of life. Structural because the whole structure of an organisms body is built from cells and functional because all the chemical reactions of life take place within cells.
- ✓ As mentioned above cell theory is the collection of ideas of different biologists who contributed a lot for our knowledge about cell biology. According to some writers say chose biologists are called 'Milestones in cell biology' and let us use this word to express those biologists.

Milestones in Cell Biology

Robert Hooke – (1665) – introduced the term cell in describing the microscopic structure of cork. He believed that the cell walls are important parts of the cell and other parts are empty.

This was published in his book entitled micrographia.

Antonivan Leeuwen Hoek (1674) – the is the first person to see living cells when he observed a moving cell in a drop of water.

- He produced over 500 single lens microscope sand responsible for invention of a microscope that can magnify up to 300× for the first time.
- He is also responsible for the discovery of human red blood cells, spermatozoa, bacteria and protozoa. He was also a Friend of Robert Hooke.

Rene Dutrochet (1824) – He is the first person who stated that all organisms are composed of cells. He is also responsible for the discovery of various aspects in biology. Among these, some of them are:

- ✓ Stomata of the leaf.
- ✓ The process of Osmosis.
- ✓ Chlorophuyll and its importance for photosynthesis
- ✓ Occurrence of respiration in plants and animals.
- ✓ He also stated that growth is the result of increase in volume of cells or addition of more little cells.

Mathias Schielden and Theodor Schwann (1839)

- The two German scientists were working under separate areas of specification. Accordingly, M.J. Schielden was a botanist and

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Theodor Schwann was a zoologist. They separately proposed a cell theory for plants and animals. Finally, they came together and stated that:

- Plants and animals are composed of groups of cells and that the cell is the basic unit of living organisms:

- In addition, Theodor Schwann is the first person to introduce Schwann cells wrap themselves around the axon of nerve cells (neurons).

Rudolf Virchow (1858) - He is a German doctor who extended the cell theory by stating that:

New cells are formed by the division of previously existing cells.

He is also responsible for stating that

- ✓ All organisms are made up one or more cells.
- ✓ Cell is structural, physiological and organizational unit of living things.

Oscar Hertwig (1892) – He proposed a protoplasmic theory which states that all living things are formed from protoplasm.

Note: Protoplasm is the living portion of cell as distinguished from the non – living cell wall.

August Weismann (1880) – He added to Virchows' idea by stating a '**germplasm theory**' which states that the germ cell (reproductive cell) is a stable hereditary material which is transmitted unchanged from generation to generation in the reproductive process.

The major events in the study of cell biology will be discussed in the table below.

Year	Event
1505	Convex lenses with a magnification greater than x5 became available
1595	Zekarias Jansen built the first compound Microscope. This microscope uses 2 convex lenses to make objects look larger. It suffered badly from color distortion, an effect called 'spherical aberration.'
1626	Fransisco Redi Postulated that, living things do not

	arise from spontaneous generation.
1661	Marcello Malpighi used lenses to study insects.
1665	Robert Hooke described cells in cork.
1672	Nehemiah Grew wrote the first two well illustrated books on the microscopic anatomy of plants.
1674	A.V. Leeuwenhoek discovered protozoa.
1833	Brown described the cell nucleus in cells of an orchid.
1838 – 1839	Schielden and Schwann proposed the cell theory.
1857	Kolliker described mitochondoria
1858	Rudolf Virchow stated 'Omnis cellula cellula which means, all cells arise from pre-existing cells.
1869	Niescher isolated DNA.
1879	Fleming described chromosome behavior during mitosis.
1880	August Weirmann added to virchows idea by pointing out that, new cells are formed only from their ancestory.
1898	Golgi described the golgi apparatus in cells.
1939	The first transmission electron microscope
1953	Watson and Crick proposed the double helix structure of DNA.
1965	The first scanning electron microscope
2000	Human genome DNA sequence draft.

Illustrative Examples

1. Cell theory was first stated by:
   A. Zekarias Jansen
   B. Rene Dutrochet
   C. Schielden and schwann
   D. Robert Hook

   Explanation: The cell theory was first stated by the two German scientists, Mathias Schelieden and Theodre Schwann. They stated that cell is the unit of structure, physiology and organization of living things. **Answer: C**

2. The name cell is first coined by:

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A. Robert Hooke
B. Anton Van Leuwenhook
C. Rene Dutrochet
D. Schwann and Schielden

Explanation: The name cell is first coined by an English biologist called **Robert Hooke**, after he examined the cork, cells and called them **cellolar** which means little rooms.

Answer: A

3. Robert Hooke used:
   A. Electron microscope to see cork cells.
   B. Simple microscope to see cork cells.
   C. Compound microscope to see cork cells.
   D. None

Explanation: Robert Hooke used a compound microscope, even if the lens were not good. Thus, they show blurred and do not show much detail.

4. What is the total magnification power of a microscope with 10x ocular lens and 25x objective lens? (UEE, 2003)
   A. 2.5x
   B. 35x
   C. 250x
   D. 2,500x

Explanation:

Total magnification power = magnification of eye piece × magnification of objective lense

$= 10 \times 25$

$= 250x$

Answer: C

5. Which one of the following objective and ocular combinations of the light microscope respectively gives the largest field of vision?
   A. 10x and 10x
   B. 4x and 10x
   C. 40x and 12x
   D. 100x and 10x

Explanation: The objective lens power is inversely proportional to the field of vision. Thus, the lower the objective lens power, the higher the field of vision.

Answer:

6. Which term is used to express the ability of microscope to show the detail clearly? (UEE, 2003)
   A. Enlargement
   B. Reduction
   C. Magnification
   D. Resolution

Explanation: Microscope has two major abilities. They are:
- ✓ **Resolution** ability to show fine scattered detail of an object.
- ✓ **Magnification** ability show in increased size of an object.

Answer: D

7. Which of the following statements is in agreement with the modern cell theory? (UEE, 2004)
   A. Cells came from nothing
   B. Cells came from existing cells
   C. Cells came from non-living material
   D. Cells arise by means of spontaneous generation

Explanation: The major points included in a cell theory are:
- ✓ All cells came from pre-existing cell.
- ✓ Cells are the basic structural and functional units of life.

Answer: B

8. The study of cells started after the discovery and improvement of microscope. Explain why?

Explanation: The study of cells started after the discovery and improvement of microscope, because most cells are smaller and cannot be seen by our naked eyes without aid of the microscope.

9. What would be the approximate image size of a cell with the size of 0.03. microns viewed under a compound microscope with magnification powers marked as 10x on eye piece and 100x on the objective lens?
   A. 10 microns
   B. 20 microns
   C. 30 microns
   D. 40 microns

Explanation:

Total magnification power $= 10 \times 100$

$= 1000x$

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Thus, since an object have original size of 0.03 microns, the magnified size of an object by 1000 times will be 30 microns.

Answer: C

10. Which one of the following types of microscope is the best to show the details of the surface of an object? (UEE, 2004)
    A. Optical microscope
    B. Compound microscope
    C. Scanning electron microscope
    D. Transmission electron microscope

***Explanation*: There** are two major types of electron microscope. They are:

- ✓ **Transmission electron microscope** – show the fine detail of the cell structure
- ✓ **Scanning electron microscope** – views only the structure of an object.

Answer: D

11. The first person who observed a living cell is:
    A. Robert Hooke
    B. Anton Van Leeuwenhook
    C. Theeordre Schwann
    D. Rudolf Virchow

Explanation

Name of biologist	Contribution
Robert Hooke	Discovered cells and introduced the name cell
Anton Van Leeuwenhook	First person to see a living cells like spermatozoa, RBCS
Theodre Schwann	Formulated the cell theory
Rudulf Virchow	Stated that one cell comes from another cell

12. Which of the following point is not included in a cell theory?
    A. All organisms are made up single cell.
    B. Cells are structural and functional units of life.
    C. Cells came from pre-existing cells.
    D. None of the above.

***Explanation*:**

See example number 7

Answer: A

13. How are cells similar to each other?
    A. They can reproduce
    B. They can exchange material with their environment
    C. They have the same size and shape
    D. All except 'C'

***Explanation*:**

All cells are similar in that:

- ✓ They can reproduce.
- ✓ They can exchange materials with their environment.

Answer: D

14. The surface area to volume ratio of a cell is important because it determines:
    A. How efficient the cell releases energy in respiration
    B. How efficient the cell in conserving energy
    C. How large the cell is
    D. How efficient the cell is obtaining oxygen it needs for respiration

***Explanation*:** The amount of oxygen delivered to the cells is decided largely by how much surface is there. Thus, large surface area to volume ratio means that it is likely that the surface will be able to supply the oxygen demands of the cell for cellular respiration.

Answer: D

15. As a cell becomes larger:
    A. The surface area to its volume ratio increases.
    B. Material distribution becomes more efficient.
    C. Cells use energy less efficiently.
    D. None

***Explanation*:** As the cell becomes larger, its surface area to volume ratio decreases. This reduces the rate of exchange of materials between the cell and its environment. It also affects the uptake of oxygen by the cell which also affect the efficiency of respiratory process.

Answer: C

16. Smaller cells are metabolically active because:

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A. They have large surface area which enables fast material distribution
B. Cellular products reach to each organelle first
C. Waste material is removed fast and the cell keeps its internal conditions normal for enzymatic activity.
D. All

Explanation: Smaller cells have relatively large surface area to volume ratio, this enables them to have efficient and faster distribution of materials. This includes:

- ✓ Distribution of useful substances faster.
- ✓ Cellular products reach organelles faster
- ✓ Waste material can be removed fast to enable internal condition of the cell normal for enzymatic activity

∴ Thus, smaller cells are metabolically active than larger cells.

Answer: D

17. Cells increase their surface area by:
    A. Growth (elongation)
    B. Cell division
    C. Cell fractionation
    D. All

Explanation: Cell elongation or growth is the stage in the life cycle of cells where they increase in size. Thus, as the size of cell increases, the surface area for exchange of materials also increases.

Answer: A

18. Cells are limited in size by the:
    A. The rate at which substance needed by the cell can enter through its surface.
    B. Rate at which the cell can manufacture genetic information.
    C. Amount of materials that the cell can collect to fill itself.
    D. Amount of materials the cell can produce

Answer: A

19. If a cell is measured 10μm, what will be its size in mm?

Explanation:

1milimeter = 1000micrometer (μm)
?(x) = 10 micrometer (μm)

$$x = \frac{1mm \times 10\mu m}{1000\mu m} = \frac{1}{100}mm = 0.01mm$$

20. In which of the following cells the rate of exchange of material is very fast? In a cell with diameter of _____mm
    A. 0.4
    B. 0.04
    C. 4
    D. None

Explanation: An organism with large surface area to volume ratio has fast rate of exchange of materials. Thus, smaller organisms (objects) have large surface area to volume ration.

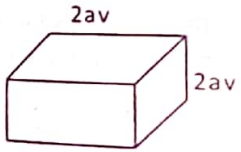
Answer: B

21. Suppose we consider four hypothetical cells (designated A, B, C and D) having cubic shape with their sides measuring 2,4,6 and 8 arbitrary units respectively, which of these cells has the largest surface are to volume ratio? (UEE, 2004).
    A. Cell A
    B. Cell B
    C. Cell C
    D. Cell D

Explanation:

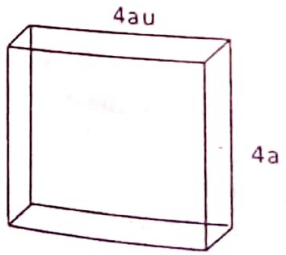
Cube	Arbitrary unit (av)
A	2
B	4
C	6
D	8

i) **Cube A**



Area = 2au × 2au = $4au^2$
Volume = 2au×2au×2au = $8au^3$
SA=6×$4av^2$ = $24au^2$
$\frac{SA}{V} = \frac{24}{8} = \frac{3}{1}$

ii) **Cube B**



Area (A) = 4au×4au
= $16av^2$
Volume (v) = 4au×4au×4au
= $64au^3$
SA=6×$16au^2$

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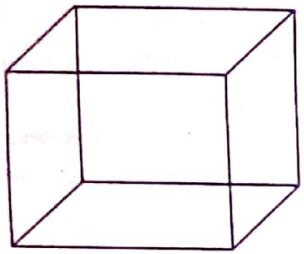
$= 96au^2$

$\frac{SA}{v} = \frac{96}{64} = \frac{3}{2} = 1.5$

iii) Cube C



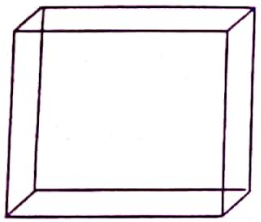
Area (A) = 6au×6au
$= 36au^2$

Volume (v) = 6au×6au×6au
$= 216au^3$

SA = 6×36
= 216

$\frac{SA}{V} = \frac{216}{216} = 1$

iv) **Cube D**



Area (A) = 8au × 8au
$= 64au^2$

Volume(v) = 8au×8au×8au
$= 512au^3$

SA = 6×64au²
$= 384au^2$

$\frac{SA}{V} = \frac{384}{512} = \frac{3}{4} = \underline{0.75}$

∴ Therefore, the smaller cube has large surface area to volume ratio.

Answer: A

22. A cell was examined under a compound light microscope with an eye piece lens marked 13x and objective lens marked 40x. How many times larger would the cell appear to the viewer?

A. 40times C. 400times
B. 53times D. 520times

Explanation: The magnification power = 13 × 40
= 520 times

Answer D

23. Which unit is the best tool for measuring the smallest cells and organelles (UEE, 2005)

A. Micrometer C. Meter
B. Millimeter D. Nanometer

Explanation:

- 1 meter = 1000 milimeter
- 1 milimeter = 1000 micrometer
- 1 micrometer = 1000 nanometer

Answer: D

24. If the size of a cell increases, which one of the following gets smaller? (UEE, 2005)

A. The volume of cell
B. The surface area of the cell.
C. The surface area to volume ratio of the cell.
D. Volume to surface area ratio of the cell.

Explanation: Increase in a size of the cell causes:

- ✓ Increase in the volume of the cell
- ✓ Increase in the surface area of the cell.
- ✓ Decrease in surface area to volume ratio of the cell.
- ✓ Increase in volume to surface a ratio of the cell.

Answer: C

25. If you observe a piece of paper on which several small squares are drawn, under which of the following magnifications of the microscope do you count more number of squares? (UEE, 2002)

A. 40x B. 100x C. 400x D. 1000x

Explanation: Magnification power of a microscope is inversely proportional to the field of vision. The lower the magnification power, the wider the field of vision and large number of squares are under focus.

Answer A

26. Choose the name of the researcher/scientist who introduced the term cell for the first time? (UEE, 2003)

A. Aristotle
B. Anton Van Leeuwenhook
C. Robert Hooke
D. Rudolf Virchow

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Explanation:

- **Robert Hooke** is the first person who discovered cells and coined the name cell.
- **Antonie Van Leeuwenhoek** the first person who saw a living cell.

Answer: C

27. Describe the discoveries of Rene Dotrochet.

Explanation:

- Rene Dutrochet has the following contributions and discoveries in the field of biology.
  - ✓ He is the first to conclude that all organisms are composed of cells.
  - ✓ He discovered:
    - Stomata of the leaf
    - The process of osmosis
    - Chlorophyll and its importance for photosynthesis
    - Occurrence of respiration in plants and animals

28. What is germ plasm theory?

Explanation: A germ plasm theory is stated by **August Wiseman** and it states that reproductive cell is a stable hereditary material transmitted unchanged from generation to generation in the reproductive process.

29. Which one of the following cellular forms did Robert Hooke observe under his crude microscope?

A. Bacteria
B. Protozoa
C. Yeast
D. Empty cellwall

Explanation: Robert Hooke first observed dead corck cells having an empty cell wall. However, cells like bacteria and protozoa were first discovered by Anton Van Leeuwenhook.

Answer: D

30. Suppose a hypothetical cube shaped cell has sides of 10 micrometers, what is the surface area to volume ratio of this cell?

A. 6:10
B. 10:10
C. 3:6
D. 4:8

Explanation:

$$\text{Area (A)} = 10\mu m \times 10\mu m = 100\mu m^2$$

$$\text{Surface area (SA)} = 6 \times 100\mu m^2 = 600\mu m^2$$

$$\text{Volume (v)} = 10\mu m \times 10\mu m \times 10\mu m = 1000\mu m^3$$

$$\frac{S}{V} = \frac{600}{1000} = \frac{6}{10}$$

∴ Thus, surface area to volume ratio is 6:10

Answer: A

4.2. Types of Cells

Prokaryotic and eukaryotic cells

Cells can be grouped in to two types based on their structure and complexity. Accordingly, they are divided as **prokaryotic** and **eukaryotic cells.**

i) Prokaryotic cells

Prokaryotic cells are cells that were evolved before eukaryotic ones. However, when they are compared with eukaryotic cells they lack various structures that are present in eukaryotic organisms. Prokaryote means **'before the nucleus'.** Prokaryotic cells have simpler structure as compared to eukaryotic cells. They lack nuclear membrane and double membrane bounded structure. However, all cells have chromatin (genetic) materials including prokaryotic cells.

Groups of organisms that posses prokaryotic cells are those which are grouped in kingdom monera (bacterias and blue green algaes).

ii) Eukaryotic cells

Eukaryotic cells are cells that have complex structures as compared to prokaryotic cells. The meaning of eukaryote is **true nucleus:**

Eukaryotic cells have features which makes them different from prokaryotic cells. They are:

- ✓ Eukaryotes have nuclear materials which are enclosed by nuclear membrane.

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- ✓ Eukaryotes have distinct internal membrane system which surrounds cell organelles. Because of these the organelles in eukaryotic cells are said to have double membrane bounded. The membrane bounded structures in eukaryotic cells are **nucleus, mitochondria, lysosomes, golgi apparatus, chloroplast** (if present).
- ✓ Eukaryotic cells have complex membrane system that are found inside cells and called **endoplasmic reticulum.**

- Having membrane bounded structures increase the efficiency of cells because it reduces the interruption of the activities of different organelles of the cell by other cellular activities.
- Eukaryotic cells have three major parts. They are:
  - ✓ Cell membrane / plasma membrane
  - ✓ Nucleus
  - ✓ Cytoplasm

i) Cell membrane/plasma membrane – is the outer semi-permeable layer of the cell. It is also called a **lipo-protein layer** because it is composed of protein sandwiched in between lipid bilayers. Cell membrane helps for:

- ✓ Regulation of the material movement in and out of the cell.
- ✓ Maintaining the cell shape.
- ✓ Communication with other cells.

ii) Nucleus is large structure which is surrounded by double membrane and it contains **nucleolus** and **chromosomes.**

Nucleus is important because:

- ✓ It contains hereditary informations of an organism in the chromosomes.
- ✓ It controls and directs the cellular activities.

iii) Cytoplasm is the part of protoplasm found outside the nucleur membrane. It is filled with jelly fluid and it encompases the largest proportion of the cell.

Cytoplasm is also known as the center for the chemical reactions of the cell. It consists of thousands of little structures called **organelles.** Each

organelle in the cytoplasm has specific activity to perform. Among various organelles found in the cytoplasm, some of the major ones are:

- ✓ Mitochondria
- ✓ Golgiapparatus
- ✓ Endo plasmic reticulum
- ✓ Ribosomes
- ✓ Chloroplasts
- ✓ Lysosomes
- ✓ Centriole
- ✓ Vacoule

Note: Protoplasm – is the living portion of the cell as distinguished from the non-living cell wall. Thus, it consists of cell membrane, cytoplasm and nucleus.

Origination of eukaryotic cell

The first to originate on the earth were the primitive prokaryotes. Thus, it was the prokaryotic cells which evolved in to eukaryotic cells through the following major stapes according to **endo symbiotic theory** which was first proposed by **Lynn Margulis.**

1) The prokaryotic cells became memebraneous over millions of years.
2) The plasma membrane around the cell become more and more in folded which finally evolved in to endoplasmic reticulum of eukaryotic cells.
3) The evolved memebraneous cells engulfed other smaller cells. This enabled the cell to be more efficient in respiration.
4) The engulfed prokaryotic cells evolved in to mitochondria eukaryotic cells. These cells were evolved in to heterotrophs like animals, fungi and protists.
5) Some of the cells also engulfed smaller prokaryotic cells that could photosynthesize. Those engulfed photosynthetic prokaryotes were evolved to chloroplasts. These cells were developed in to autotrophic cells which further evolved into plant cells.

Do you remember?

- **Heterotrophs** – are organisms that cannot synthesize their own source of energy.
- **Autotrophs** – are organisms that are capable of synthesizing their own source of energy.

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As you have learned at grade 9 biology, organisms living on the earth are classified in to five kingdoms according to the modern classification system which is based on the evolutionary relationship between organisms. Therefore, let us try to see each kingdom and the type of cells they contain in table below.

Kingdom	Cell Feature
Moneera	Prokaryote
Protista	Eukaryote
Fungi	Eukaryote
Plantae	Eukaryote
Animalia	Eukaryote

Differences between eukaryotic and prokaryotic cells

Prokaryotic cells	Eukaryotic cells
Have no true distinct nucleus which is bounded by nuclear membrane	Have true distinct nucleus which is bounded by nuclear membrane.
Their size is smaller which ranges between 1-10 μm	Relatively they are larger in size which ranges between 10-100μm
They have hereditary materials DNA, which is not in association with protein to form chromosomes.	Have DNA which is their hereditary material which is associated with histone proteins in chromosomes.
Some organelles such as mitochondoria and chloroplast are absent. However, some have a kind of chlorophyll and can photosynthesize.	They have most of the organelle including mitochondoria and chloroplast which is absent in animal cells.
Have ribosomes which are smaller.	Have ribosomes which are relatively larger the in prokaryotes.
Have cell wall which is made up of other polysaccharides.	Have cell wall which is usually present in plants, algaes and fungal cells.
Examples are bacterias (eubacterias) and blue green algaes (cyanobacterias)	Examples are all organisms except bacteria and blue green algaes.

Illustrative Examples

31. Which group of organisms has a system of protein synthesis in which transcription and translocation take place at separate times? (UEE, 2004)
A. In all eukaryotic organisms
B. In multicellular organisms only
C. Only in prokaryotic organisms
D. In both prokaryotic and eukaryotic organisms.

Explanation:

- **Transcription** is the process of copying the DNA code to form a strand of RNA which can be transfer RNA, messenger RNA or ribosomal RNA. It takes place in the nucleus.
- **Translation** is the process of assembling aminoacids into proteins at the ribosome according to the instructions carried by messenger RNA. It occurs in cytoplasm.

Answer: A

32. Which of the following ideas in the cell theory was contributed by Rudolf Virchow? (UEE, 2004)
A. All plants are made up of cells.
B. All animals are made up of cells.
C. Cells are the structural units of life.
D. Cells came from pre-existing cells.

Explanation:

- **Rudolf Virchow** is a German doctor who developed many surgical techniques. He also declared that, "***Omnis cellulla cellula***", which means that, a cell can only a rise from another cell.

Answer: D

33. Select the kingdom of life in which the cellular organelles are without membranes around them (UEE)
A. Plantae
B. Protista
C. Monera
D. Fungi

Explanation: Cells in which cellular organelles are without membranes are **prokaryotic cells.** Such kinds of cells are only found in

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all organisms under kindom monera. They are bacteras and blue-green algaes.

34. In which kingdom of life are the unicellular eukaryotes grouped? (UEE, 2004) **Answer: C**

A. Monera
B. Protista
C. Plantae
D. Animalia

Explanation: To further understand the detail let us look at the table below.

Kingdom	Feature
Monera	Unicellular, prokaryotic
Protista	Unicellular, eukaryotic
Fungi	Unicellular or multicellular, eukaryotic
Plantae	Multicellular, eukaryotic
Animalia	Multicellular, eukaryotic

Answer: B

35. Compare and contrast eukaryotic and prokaryotic cells

Explanation:

Eukaryotic cells	Prokaryotic cells
Evolved from prokaryotes	Evolved 3.5 billion years ago
Cell division is by meiosis mitisiis or both	Cell division is by mitosis
Lenear DNA contained in the nucleus and associated with proteins and RNA to form chromosomes.	Circular DNA free in the cytoplasm
Many organelles with membrane bounded nucleus	Few organelles are present
Cell walls are present in plant cells, algal cells and fungal cells	Cell wall is always present

36. In which of the following features are eukaryotic cells distinguished from prokaryotic cells? (UEE, 2005)

A. They have mitochondria
B. Their nuclei lack membranes

C. They have no DNA
D. They have smaller ribosmes.

Explanation:
Eukaryotic cells differ from prokaryotic cells, because they:

- ✓ Have memebrane bounded nucleus and cellular organelles.
- ✓ Have some organelles like mitochondria that are absent in prokaryotic cells.
- ✓ They have DNA attached to the chromosomes.
- ✓ They have large ribosomes.

Answer: A

37. Which of the following structures is not present in animal cells?

A. Cell wall
B. Nucleus
C. Protoplasm
D. Nucleic acid

Explanation: Structures that are present in plant cells but not in animal cells are:

- ✓ Cellwall
- ✓ Chloroplast
- ✓ Large central vacuole

Answer: A

38. What is the general term for the part of the protoplasm that lies outside the nucleus?

A. Cytosol
B. Cytoplasm
C. Central vacuole
D. plasma membrane

Explanation: Protoplam is the living portion of the cell that mainly consist of cytoplasm and nucleus.

Answer: B

39. Which of the following cellular structures is possessed by all cells?

A. Cell membrane
B. Nucleus
C. cell wall
D. Golgi apparatus

Explanation: Structures that are possessed by all cells including eukaryotes and prokaryotes is the cell membrane.

Answer: A

40. Which of the following carbohydrates has structural function?

A. Cellulose
C. Strach

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B. Glycogen D. Sucrose

Explanation: Cellulose is a complex sugar that serves as a structural component of plant cell wall

Answer: A

41. Which one of the following eukaryotic cell organelles was a free living cell before eukaryotic cells evolved?
    A. Nucleus C. Chloroplast
    B. Ribosome D. Nucleous

Explanation: Ribosomes are the only structures that are possesed by both prokaryotes and eukaryotes.

Answer: B

42. One of the following organelles of the eukaryotic cells does protein synthesis takes place?
    A. The nucleus C. The chloroplast
    B. The ribosome D. The mitochondoria

Explanation:

Structure	Function
Nucleus	Controls and directs cellular activities. Contains genetic materials.
Ribosomes	Involved in protein synthesis
Chloroplast	Site of photosynthesis
Mitochondria	Site of cellular respiration

Answer: B

4.3. Parts of the cell and their function

Cell membrane

All cells have a plasma (cell) membrane that forms the outer limit of the cell. In addition, membranes are found in eukaryotic cells as part of membranous organelles. Some of those membranous structures which are found inside eukaryotic cells include:

- ✓ **Nuclear membrane** surrounds the nucleus and controls the passage of genetic information to the cytoplasm. It also serves to protect the DNA.
- ✓ **Mitochondria** have an outer membrane that controls the entry and exit of materials involved in aerobic respiration. Inner membranes provide attachment site for enzymatic activity.

- ✓ **Golgi apparatus** comprises stack of membrane bound sacs. It is involved in packaging materials for transport or export for the cell as secretory vesicles.
- ✓ **Plasma membrane** surrounds the cell and controls the movement of most substances in to and out of the cell.

Importance of cell membrane

Cell membrane which is referred as a 'living layer' of the cell is important mainly due to the nature of its selective permeability. Accordingly, cell membrane is important in various ways. These are:

- ➢ It controls the movement of substances in and out of a cell. This is because of its **selectively permeable membrane**. This occurs in the following major processes.
  - Simple diffusion
  - Facilitated diffusion
  - Osmosis
  - Active transport
  - Endocytosis
  - Exocytosis
- ➢ It allows the cell to recognize and identify various types of substance like hormones, immune system, growth hormones such as auxin in plants. This is called **cell signaling.**

Structurally plasma membrane of a cell is composed of two layers of phospholipids

Each phospholipid is composed of a glycerol molecule attached to two chains of fatt acids. The head portion of the phospholipid is **hydrophilic** (water loving) and points out ward while the tail portion is **hydrophobic** (water hating) and points in ward.

The structure of plasma membrane is explained in different ways according to different models. Thus, let us look at these models as follows.

Events in developing the current model of plasma membrane

The current model of membrane is not something developed over night. It is developed through a series of works done by different biologists over the last three to four centuries. Thus, the development of the current membrane model is as follows:

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Year	Event
1665	Robert Hooke discovered cells, but only saw dead cells and has no idea of a cell membrane.
1895	Charles Overton showed that the cell membrane is composed of some kind of lipids.
1905	Langmuir proposed a lipid monolayer as the basic membrane structure.
1910-1920	Evidence accumulates to show that the lipid in the membrane must be a phosopholipid.
1925	eGorter and G Grendel suggested that the plasma membrane is a phospholipid bilayer.
1935	Davison and Danielli know that proteins are also found in plasma membranes and suggest a sandwich model.
1959	Based on electron microscope evidence that appears to support the Davison – Danielli model, J.D Robertson proposes the membrane model-he suggested that all membranes are essentially the same.
1972	S.J. Singer and G.L.Nicholson propose the fluid mosaic model of membrane structure. As more and supporting evidence accumulates, this is essentially the model which is acceptable by the time.

Two models of membrane structure

1) The Davson-Danielli 'Sandwich" model
2) Fluid mosaic model

1) Davison – Danielli Model (1935)

Davison-Danielli model explains that phospholipids are sandwich between two protein layers. By the time two substances in a membrane. Accordingly, proteins are like the bread of 'sandwich' and the phospholipids form the filling in the sandwich.

Again in 1954, this model is revised and added the protein lined pores on it.

Davison – Danielli model however does not properly explain how molecules move across the membranes.

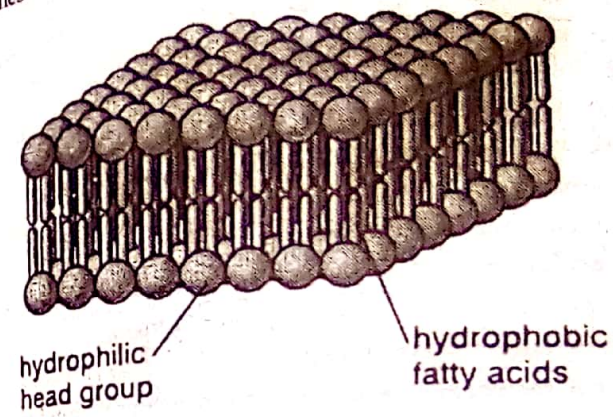

Fig. Davison-Danielli mosaic model

2) Fluid Mosaic Model (1972)

Fluid mosaic model is introduced by **S. Singer** and **G. Nicolson**. This model states that the membrane is a mosaic of protein dispersed in a fluid bilayer of phosphlipid.

- According to the fluid mosaic model a cell membrane is a fluid lipid bilayer with constantly changing "mosaic pattern" of associated proteins.

Fluid mosaic model rejects the sandwich arrangement and it suggests that protein were stud in to the bilayer at different points. In addition, this model explains that the arrangement is fluid and constantly changing. The fluid mosaic model is the one which is currently in use. According to this model, the phosphlipid bilayer which is the basis for a membrane consists of the following.

1) Membrane proteins

There are two major types of membrane proteins; **integral membrane protein** and **peripheral membrane proteins.**

Integral membrane proteins

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i) **Integral proteins** are proteins that have regions which are inserted into the hydrophobic regions of the lipid bilayer. Some of these proteins pass all the way through the membrane, so that large parts of them are on either side of the membrane; these particular integral proteins are called **transmembrane proteins.**

Integral proteins play role in moving substances across the membrane. There are two main types of integral proteins:

A) **Channel proteins** – have a channel through them in which specific ions can pass.

B) **Carrier proteins** – are proteins which are used to move larger molecules through the membrane by facilitated diffusion or active transport.

ii) **Peripheral membrane proteins** – are membrane proteins that can be easily removed from the membrane without disrupting the structure of the bi-layer they usually bind to exposed regions of integral proteins. They act as enzymes and anchor integral proteins to the cytoskeleton.

- Peripheral proteins are also known as **extrinsic proteins.**

2. **Glycoproteins and glycolipids**

They act as receptor sites for hormones and drugs. They also allow cell to cell recogonition. Glycoproteins also work in sticking the correct cells together in tissues.

3. ***Cholesterol*** – reduces the fluidity of the membrane. They also work in reducing further escape or entry of polar molecules through the membrane.

Movement of substances across the plasma membrane

Plasma membrane has a nature of selective permeability because it largely contains lipids. Accordingly, the passage of molecules of any given substance is determined by:

- ✓ Structure of the membrane
- ✓ Size and charge of molecules.

Based on the nature of the membranes to allow the passage of substance through them, they can be called:

i) **Permeable** – if it permits all the substances to pass through.

ii) **Impermeable** – if it will not permit the substance to pass through

iii) **Semi-permeable** – if it allows some but not other substances to pass through it. Such a membrane is found in cells.

Because of the selective/semi-permeability of the membrane particles that pass through the plasma membrane need to be:

- ✓ Smaller in size
- ✓ Lipid insoluble
- ✓ Non-charged, however some ions, sugars and aminoacids which are lipid insoluble and not smaller in size can be considered as exceptions.

The passage or movement of substances across plasma membrane occurs through the following major processes:

i) **Passive transport** – involves the movement of substance along concentration gradient.

- It involves the movement of substances from the region of higher concentration to the region of lower concentration.
- Passive transport do not utilize or expend energy from the metabolism of cell.

ii) **Active transport** – involves the movement of substances against concentration gradient.

- It involves the movement of substances from the region of lower concentration to the region of higher concentration.
- It expends or utilizes extra energy (ATP) from the metabolism of cells to transport materials.

Saying, this much about the two of transport systems across the plasma membrane as an introduction, let us look at the two systems of transportation in detail.

1. **Passive transport**

The passive transport system mainly involves two major processes. They are:

i) Diffusion **ii) Osomosis**

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i) Diffusion

Diffusion can be:

a) Simple diffusion

- Simple diffusion is the movement particles from an area of higher concentration to an area of lower concentration until the molecules tend to distribute uniformly.
- It does not necessarily require membrane to move particles.
- It enables substances to spread out and occupy all the spaces.
- For diffusion to take place, there must be a concentration difference between two regions. Thus, molecules move equally in both direction in order to create an equilibrium.
- The rate of diffusion can be affected by various factors. Among these some of the major factors that affect the rate of diffusion are:
    - **Concentration gradient** – is the gradual change in the number of molecules per unit of volume over distance or it is the difference in concentration between the two areas. Thus, as the concentration difference increases, the rate of diffusion also increases.
    - **Thickness of the membrane** – the thinner the membrane (barrier), the faster the rate rate of diffusion which involves the passage of particles through it.
    - **Surface area of the membrane** – the larger the surface area of the membrane, faster the rate of diffusion.

      Temperature – as the temperature rises the rate of diffusion also increases.

$$\text{Rate of diffusion} = \frac{\text{surfaace area of the membrane} \times \text{concentration differencec}}{\text{Diffusion distance}}$$

b) Facilitated diffusion – is transport of substances across the membrane of a cell in which proteins speed up the movement. The substances can only flow down their gradient. The proteins that speed the movement of substances across the membrane are known as **carrier proteins.**

- Facilitated diffusion is required because some substances such as glucose would not beable to diffuse quickly across the cell membrane by their own. Thus, they combine with globular proteins, in the cell membrane to carry them across. Thus, it is sometimes called a special form of diffusion.
- Each carrier protein will only bind with specific molecule and binding changes the shape of the carrier which deposits the molecule into the cytoplasm.
- Facilitated diffusion is much faster than the regular diffusion because of the carrier molecules and no energy is used in the process.

Facilitated diffusions is powered by concentration gradient.

Facilitated diffusion has the following characteristics:

- ✓ It is specific. This means, only certain molecules can pass through the channel.
- ✓ It is passive. This means, the direction of net movement being determined by the relative concentration the transported molecule inside and outside the membrane.

The rate of facilitated diffusion is also affected by the following major factors:

- ✓ the number of carrier proteins present.
- ✓ Concentration gradient.
- ✓ Thickness of the membrane.

ii) Osmosis

Osmosis can be defined in a broader sense in the following major ways:

- It is a special case of diffusion which involves the movement of water or solvent from less concentrated or weaker solution to more concentrated or stronger solution through a semi-permeable membrane.
- It is the movement of water molecules across selectively permeable membrane:
    - ✓ From a lower concentrated solution to a higher concentrated solution.

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- ✓ From where water molecules are at a higher concentration to where they are at a lower concentration.
- ✓ From a hypotonic solution to a hypertonic solution.
- ✓ From hyper osmotic solution to a hypo-osomotic solution.
- ✓ From an area of higher water potential to lower water potential.

- During osmosis the net movement of water molecules continue until equilibrium of the water molecule is reached.

Water potential – is the sum of two opposing forces (osmosis and turgor pressure) that can cause the movement of water into and out of a cell.

- Water potential is used as a measure of water movement from one place to another in a plant.
- It is measured in terms of pressure and the units are either kpa **(killo paskal)** or **MPa (mega paskals).**
- It is indicated by the symbol ψ(pronounced psi).
- The following equation allows us to work out the water status of a plant cell.

φ (cell)		φ_s		φ_p
Water potential (of a cell)	=	solute potential (of ions inside cell)	+	pressure potential (of cell wall)

Osmotic pressure

Measure of the pressure driving the transfer of solvent molecules across a membrane from a solution of low soluted concentration to a solution of high solute concentration. Thus, osmotic potential of a solution is a measure of the tendency of water molecules to diffuse out of it. A very concentrated solution, having relatively few water molecules, has low osmotic potential, while dilute solution with larger proportion of water molecules has high osmotic potential.

Note: Pure water has the highest possible osmotic potential.

Weak (dilute) solution – is a solution that consists of low concentration of solute molecules. Such a solution has:

- ✓ High concentration of water molecules.
- ✓ Low osmotic pressure.
- ✓ High osmotic potential.

Strong (concentrated) solution – is a solution that consists of high concetnraiton of solute molecules. It has:

- ✓ Low concentration of water molecule.
- ✓ High osmotic pressure.
- ✓ Low osmotic potential.

If a cell is placed in solutions of different concentrations, the following conditions will occur:

- **Hypotonic** – if the surrounding fluid has a concentration of dissolved substances less than the concentration with in the cell. This occurs when the osmotic pressure of the external solution is lower than that of the cell.
  - If the cell is placed in hypotonic solution which has less concentration of solute than the cell, water moves in to the cell as the cell **swells up.**
- **Hypertonic** – if the surrounding fluid has a concentration of dissolved substances (solute) greater than the concentration within the cell. It occurs when the osmotic pressure of external solution is greater than that of the cell.
  - If the cell is placed in hypertonic solution which has greater concentration of solute cell, water moves out of the cell and it becomes **shrink** or **flaccid.**
- **Isotonic** – is a condition where the cell has the same water potential as its environment. Thus, there is no net movement of water molecules in to and out of the cell.
  - The blood plasma and other fluids in our body are isotonic to our cells.

Note: If animal cells are placed in hypertonic solution, they lose water by osmosis and they shrink. This condition is called **crenation.**

- If animal cells are placed in hypotonic solution, they gain water by osmosis and they swell up and finally get brust. This condition is known as **haemolysis.**

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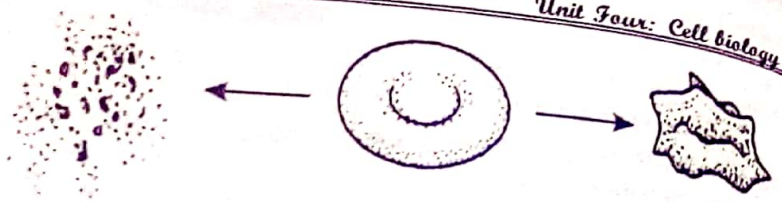

Cell in more dilute solutions gains water and bursts; it becomes **haemolysed**

Cell in same concentration of solution maintains its shape

Cell in more concentrated solution loses water and shrinks; it becomes **crenated**

Osmosis in plan cells

i) When plant cells are placed in hypotonic solution.

- Plant cells have cell wall which is freely permeable to water and other dissolved substances. As water enters the cell vacuole by osmosis, the vacuole swells and starts to push the cytoplasm against the cell wall. This creates the pressure known as **turgor pressure** against the cell wall. In such a case the cell is known as **turgid**.
- In such a case, the cell would not get brust because of the wall **pressure** which is exerted by the cell wall to resist the turgor pressure.
- This is the optimum condition for plant cells.

ii) **When plant cells are placed in hypertonic solution**

- ✓ Plant cells loose water by osmosis when placed in to a hypertonic solution, it looses water and shrinks. Thus, there is less pressure against the cell wall and the cell is said to be **flaccid**.
- ✓ Too much shrinking of the cytoplasm, makes the cell to loose contact with the cell wall and the cell becomes **plasmolysed**.
- ✓ **Plasmolysis** – is the loss of turgor pressure in plant cells.

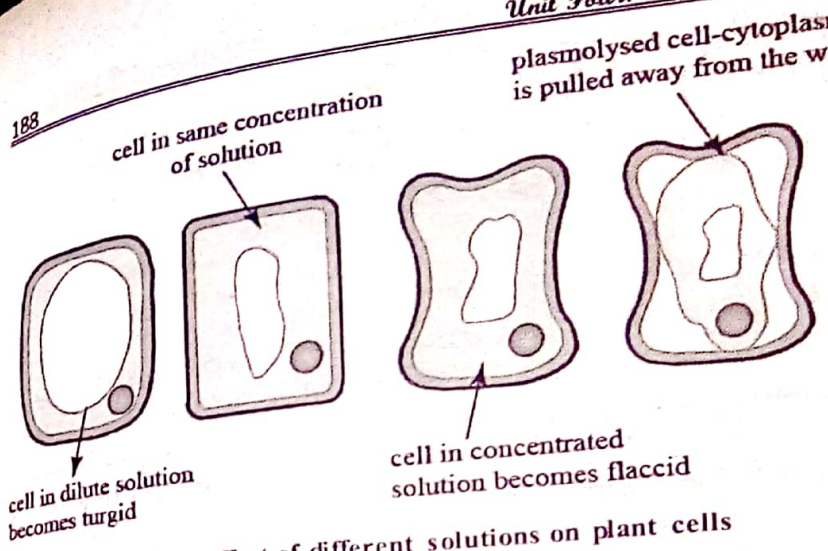

Fig: The effect of different solutions on plant cells

Note: **Turgidity** is important because it provides support for non woody plants and avoids wilting of a plant.

Solute concentration in solution A	Solute concentration in solution B	Tonicity	Direction of net movement of water
Greater	Less	• A hypertonic to B • B hypotonic to A	B to A
Less	Greater	• B hypertonic to A • A hypotonic to B	A to B
Equal	Equal	Isotonic	No net movement

Active Transport

- Active transport is the movement of substances from the region of lower concentration to higher concentration against concentration gradient.
- It requires energy from the metabolic process to transport substances. Thus, a cell expends ATP. Proteins are used to carry substances across the cell membrane and are called **transport** or **carrier proteins.**

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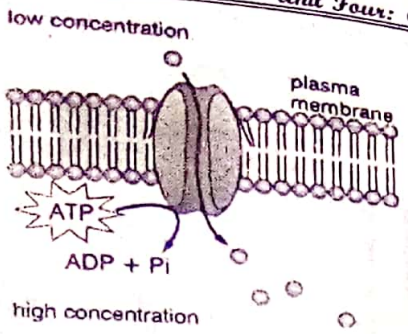

Endocytosis and Exocytosis

Endocytosis – is the process in which large particles are engulfed by a cell. Thus, it is responsible for movement of materials into the cell. The process endocytosis allows entrance of large particles into the cell as follows:

- ✓ The substance contacts with the cell surface membrane which indents.
- ✓ The substance is surrounded by the membrane, forming vacuole or vesicle
- ✓ Each vacuole contains the substance and an outer membrane which has detached from the cell surface membrane

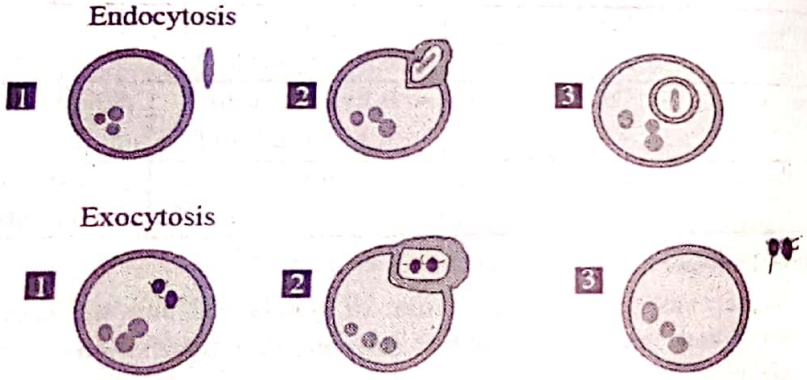

Fig. Endocytosis

- There are several types of endocytotic mechanism that operate in biological system. These include, phagocytosis, pinocytosis and receptor mediated endocytosis.

i) Phagocytosis (cell eating)

The cell ingests large solid particles such as bacteria or food. This process is used by organisms such as protozoa and several classes of white blood cells to ingest solid particles such as bacterium. For example, ameabas use an extension from their plasma membrane called **pseudopodia** which is used for enclosure of large particles outside the cell.

ii) Pinocytosis (cell drinking)

Pincocytosis is the process by which the cell takes in dissolved materials or liquid. Droplets of fluid are trapped by folds of the plasma membrane, which pinch of fluid into cytoplasm as tiny vesicles.

iii) Receptor mediated endocytosis

This is the process by which specific proteins or particles combined with receptor proteins embedded in the plasma membrane of a cell. The embedding stimulates in folding. It is an important process by which cholesterol in the blood stream is taken up by animal cells.

Exocytosis

This is the process by which a cell ejects waste products or specific secretion products such as hormones by the fussion a vesicle on the plasma membrane of the cell.

Table: Summary of the major transport process of the cell with its environment

Process	Influence of concentration	ATP required	Type of substance moved	Transport proteins needed
Simple diffusion	Occurs from high to low	No	Lipid soluble, small, non-polar	No
Facilitated diffusion	Occurs from high to low	No	Ions and medium sized particles	No
Osmosis	Occurs from	No	Water	No

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	high to low (water potential)		molecule	
Active transport	Occurs from low to high	Yes	Ions and medium sized non-lipid soluble particles	Yes
Endocytosis	Can occur either way	Yes	Very large particles	Yes
Exocytosis	Can occur either way	Yes	Very large particles	Yes

Cellular structures and their function

- **Nucleus** is the part of the cell which occupies around 1/10 of the cells composition. It is surrounded by neuclear membrane and its inner fluid filled portion is called **necleoplasm.**
  - ✓ The inner portion of the nucleus consist of chromatins of DNA molecules with protein called **histone**.
  - ✓ **Nucleous** is little organelle that is found in the nucleus and synthesizes components of ribosmes into the cytoplasm.
  - ✓ Nucleus is mainly responsible for controlling and directing cellular activities.

Mitochondria

- They are organelles which are the site of most of the chemical reactions that convert the chemical energy present in certain foods to another form of chemical energy, ATP. The conversion of food energy to ATP is called **cellular respiration**. Cells that are metabolically active contain large number of mitochondoria. There are cells such as liver cell, muscle cells, heart cells, etc.
- Mitochondorias are surrounded by two set of membranes. The inner folded membrane is called **cristae.** It helps to increase the surface area for enzymatic actions and other proteins involved in transforming energy in food molecules into different forms of chemical energy stored in ATP.

- The outer membrane of mitochondorian is smooth and it allows many small molecules to pass through it.

Ribosomes

Ribosomes are found both free in the cytoplasm and bound to the endoplasmic reticulum membrane. They are mainly involved in protein synthesis. Ribosomes are comprised of sub units of RNA and protein which are produced in the nucleus.

Endoplasmic Reticulum (ER)

They are network of internal membranes extending through the cytoplasm. There are two forms of endoplasmic reticulum.

i) **Rough endoplasmic reticulum** – have ribosome on its membrane surface. It is involved in protein transport which is synthesized by ribosmes.

ii) **Smooth endoplasmic reticulum** – has no ribosome on its surface and it is concerned with lipid synthesis, carbohydrate metabolism and detoxification.

Golgi apparatus (Golgi body)

- Golgi complex was first described by an Italian microscopist called Camilo Golgi.
- It consists of membrane bounded sacks where proteins are modified. It's function is principally protein processing, sorting and modifying apparatus.

Lysosomes

- They are small sacs of digestive enzymes that are dispersed in the cytoplasm of animal cells. They are surrounded by single membrane and are formed in the golgi apparatus. The enzymes break down complex molecules including fats, proteins, carbohydrates and nucleic acids.
- Lysosomes are also used to degrade foreign molecules that have been ingested by cells.
- They are involved in disintegration of damaged and worn out cells. Accordingly, they are also called **self destructing organelles** or **succide** bags of a cell.

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Some organelles that are present in plant cells but not in animal cells

Cell wall – is the most outer layer of plant cells which is involved in providing support, shape and protection for the cell.

- ✓ It is composed of a complex polysaccharide substance called **cellulose.**

Vocoule – is a fluid filled structure and central componenet in plant cells. The vocoules are single, membrane – bound structures that store substances for the cell. Fresh water protista like paramecium and amoeba have **contractile vacuoles** that pump excess water out of the cell. Plant cells have large central vacuole for storage.

The set of fluid that fills the vacuole is called **cell sap.**

Plastids

Plastids have double membrane and are found only in plants and algae. There are three types:

i) **Chloroplasts** are green because they contain chlorophyll. They are the sites of photosynthesis. In addition to a double outer membrane, they have an inner one that forms a series of structures called **grana.** The grana lie in the **stroma.** Chloroplasts, like mitochondoria, contain their own nuclear material and can self-replicate.

ii) **Leucoplasts** are colorless and store starch.

iii) **Chromoplasts** – store carotenoid pigments and are responsible for the red orange – yellow colouring of carrots, tomatoes and many other plants. The pigments in petals attracts pollinating insects to flowers.

Some of the major cellular organelles and their functions

Organelle	Description	Function
Endoplasmic reticulum	Network of internal membranes extending through cytoplasm.	Synthetic site of membrane lipids and many membrane proteins; origin of intracellular transport vesicles carrying proteins to be secreted.
Ribosomes	Grgnules composed of RNA and proteins; some	Modifies proteins, packages secreted proteins sorts other

	attached to endoplasmic reticulum some free in cytoplasm.	proteins to vacuoles and other organelles.
Golgi complex	Stacks of flattened membrane sacs.	Modifies proteins, packages secreted proteins sorts other proteins to vacuoles and other organelles
Lysosomes	Membranous sacs (in animals).	Contain enzymes to break down ingested materials, secretions, wastes.
Vacuoles	Membranous sacs (mostly in plants, fungi, algae).	Transport and store ingested materials, wastes, water
Mitochondoria	Sacs consisting of two membranes, inner membrane is folded to form cristae.	Site of most reactions of cellular respiration; transformation of energy originating to form glucose of lipids in to ATP.
Centrioles	Pair of hallow cylinders located near the center of cell; each centriole consists of nine microtubule triple, etc	Mitotic spindle forms between centrioles during animal cell division; may anchor and organize microtubule formation in animal cells; absent in higher plants.
Chloroplast	Contain chlorophyll in internal thylakoid membrane	Chlorophyll captures light energy; ATP and other energy rich compounds are formed and then used to convert CO_2 to glucose.

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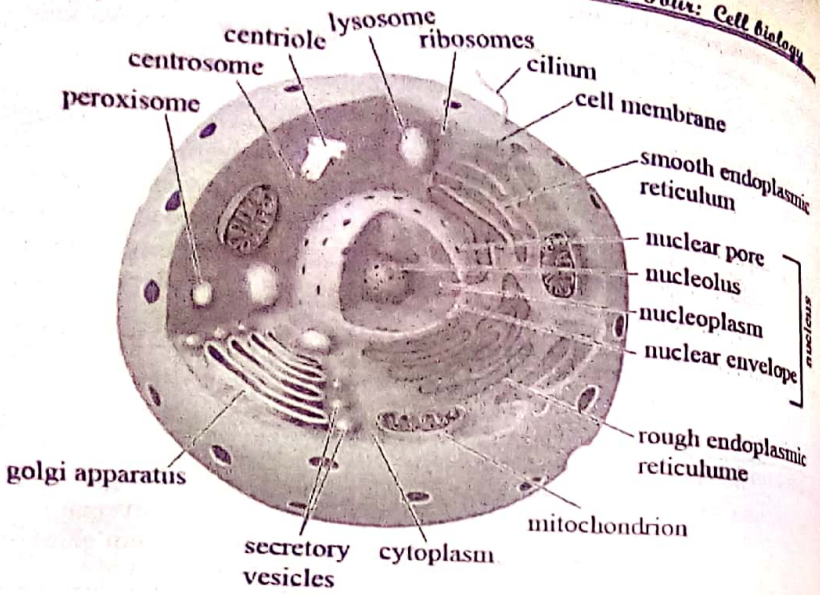

Fig. Generalized animal cell

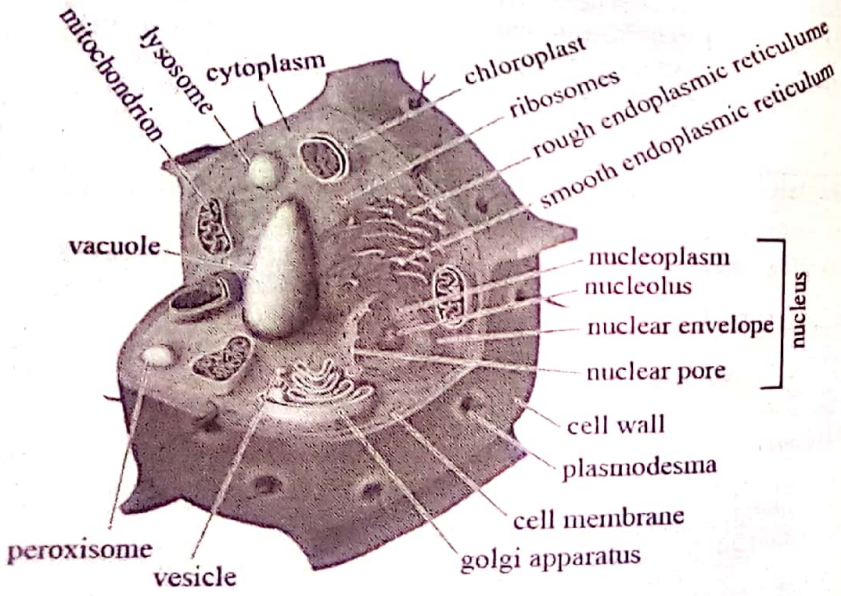

Fig. Generalized plant cell

Illustrative Examples

43. Among the following which one can be an example of prokaryotic organisms? (UEE)
A. Paramecium
B. Amoeba
C. Liverwort
D. Bacteria

Explanation: Bacterias are prokaryotic organisms, while paramecium, liverwort and amoeba are eukaryotic organisms.

Answer: D

44. Which one of the following is largely made up of phospholipids? (UEE)
A. Cell wall
B. Cell membrane
C. Nucleus
D. Chromosomes

Explanation: Cell membrane is composed of phospholipids bilayer and protein. Phospholipids are composed of lipids and phosphate group.

Answer: B

45. Sodium ions pass from a region where they are at lower concentrations to a region where they are found at higher concentration in human cells. This an example of which of the following process?
A. Osmosis
B. Simple diffusion
C. Passive transport
D. Active transport

Explanation: Active transport is the movement of solutes across a membrane from low to high concentration; active transport requires energy from cellular metabolism.

Answer: D

46. The main component of plant cell wall is (UEE)
A. Starch
B. Cellulose
C. protein
D. chitin

Explanation: In plant cells, the most outer layer which is used to provide shape, support and protection for the cell is cell wall. This structure in plant cells is mainly composed of a complex sugar called **cellulose.**

Answer: B

47. Which of the following is the main constituent of biological membranes?

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A. Phospholipids
B. Glycoproteins
C. Glycolipids
D. Cholestrols

Explanation: See example number 40

Answer: A

48. Most cell membranes are primarily composed of which compounds?
A. Protein and lipids
B. DNA and ATP
C. Chitin and starch
D. Nucleotides and aminoacids

Explanation: Cell membrane is sometimes called **lip-protein** layer because it is mainly composed of lipid bilayer and protein (glycolipids, glycoproteins and phospholipids).

Answer: A

49. During chemiosmosis, what substance diffuses from one side to the other side of the membrane?
A. Water molecules
B. Protons
C. Electrons
D. ATP molecule

Answer: B

50. Which of the paired organelles are membrane bounded?
A. Ribosomes and peroxisomes
B. Chloroplasts and ribosomes
C. Mitochondoria and ribosomes
D. Chloroplasts and mitochondoria

Explanation:

- The main feature of eukaryotic cells is that they have membrane bounded organelles. Thus, the major organelles of a cell that have double membrane bounded structure are nucleus, mitochondoria, Golgi apparatus, endoplasmic reticulum, chloroplasts and vacuole.
- Cellular organelles such as ribosome and peroxisomes lack membrane bounded organelles.

Answer: D

51. What would occur if a plant cell were placed into a strong hypertonic solution?
A. Nothing would happen because the cell would prevent any movement of water or salt.
B. It would shrink because water would flow toward the more concentrated solution.

C. It would swell because salt would flow in to the cell.
D. If would swell because water and salt would flow down the gradient and into the cell.

Explanation: Water flows outwards at high concentration of salt, that is, toward a hypertonic solution. The cell would shrink and the process called **plasmolysis**. Hypertonic means contain a greater concentration of salt.

Answer: B

52. Which process requires energy?
A. Passive diffusion
B. Facilitated diffusion of calcium through a channel.
C. Water flowing into a paramecium in a lake.
D. A contractile vacuole removing water from an amoeba.

Explanation: Contractile vacuole pumps water against a gradient. Active transport requires energy and involves pumping of substances against a gradient. Passive transport requires no energy. Passive transport includes simple diffusion, facilitated diffusion and osmosis.

53. Cellular structure that is found in large quantities in white blood cells and other phagocyting cells is:
A. Nucleoulus
B. Ribosomes
C. Lysosomes
D. Golgibody

Explanation: Lysosomes contain hydrolytic enzymes that digest cell parts or entire microbes.

Answer: C

54. All of the following are true about prokaryotes except:
A. They have one chromosome
B. They contain ribosomes
C. They are smaller than eukaryotic cells
D. They can be either anaerobic or aerobic
E. They contain small mitochondoria

Explanation: Prokaryotes contain no mitochondoria. They have no internal organelles made of membranes, such as mitochondoria, nuclei or vacuoles. They do, however, contain ribosomes, which are smaller than ribosomes in eukaryotic cells.

55. Which of the following cell types can be rich in lysosomes? (UEE)

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A. Red blood cells
B. Nerve cells
C. Phagocytic cells
D. Muscle cells

Explanation: Lysosomes are organelles that contain mixture of digestive enzymes. Thus, they are responsible to digest material taken in to the cell by the process of phagocytosis. Examples of phagocytic cells are which blood cells.

Answer: C

56. If the red blood cells shrink when placed in a certain solution, what is the strength of the solution relative to the strength of the protoplasm of the cells?
A. Hypotonic
B. Isotonic
C. Hypertonic
D. Isoosmosis

Explanation: If RBC's are placed in hypertonic solution, the loose their water content by the process of osmosis. Thus, the RBCs get shrink when placed in to hypertonic solution.

Answer: C

57. Pieces of fresh potato are kept for some times in sugar solutions of 20%, 10%, 5% and distilled water, which piece will gain the highest percentage of weight? (UEE, 2002)
A. The one in 20% solution
B. The one in 10% solution
C. The one in 5% solution
D. The one in distilled water

Explanation: The solution with highest percentage of sugar (solute) contains less amount of water. While, distilled water is pure water which has no sugar. Therefore, because of osmosis more water molecule moves into the solution with 20% sucros solution.

Answer: A

58. Singer and Nicholsons model is:
A. Fluid mosaic model
B. Monoploid model
C. Sandwich model
D. Key and lock model

Explanation: Singer and Nicolson's model which is stated in 1972 is known as **fluid Mosaic model.** As the name indicates:

- ***Fluid*** – is the phosphlipid bilayer has fluid property because the fatty acid is made up of unsaturated hydrocarbon.
- ***Mosaic*** – refers to the position of protein which are inserted, and dispersed in the fluid (phospholipid bilayer).

Answer: A

59. What happens to a human red blood cell when it is placed in a hypertonic solution? (UEE,)
A. It becomes turgid
B. It might swell and brust
C. It might loose water and shrink
D. It will remain unchanged

Explanation: Look at example number 56.

Answer: C

60. What is the process by which cells like amoeba and white blood cells engulf and internalize particles such as bacteria? (UEE, 2002)
A. Osmosis
B. Pinocytosis
C. Active transport
D. Phagocytosis

Explanation:

- **Phagocytosis** is the process through which food particles are surrounded by cell membrane and incorporated in to the cell cytoplasm, forming a food vacuole. Such, process is used by unicellular organisms like amoeba and white blood cells in multicellular organisms.

Answer: D

61. When does a hypotonic condition exist in the environment of the cell?
A. When there is equal concentration of solutes outside and inside the cell.
B. When the solute concentration outside the cell is higher than inside the cell.
C. When the solute concentration inside the cell is greater than outside the cell.
D. When the net movement of water is from the cell to outside environment.

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Explanation: Hypotonic condition exist in the environment of the cell when the concentration inside the cell is less dilute (more solute) than its environment. Thus, water moves into a cell from its environment. This makes a cell to swell up.

Answer: C

62. Which of the following modes of materials transport across the cell membrane is NOT governed by the concentration gradient of the transport material? (UEE, 2004)
    A. Simple diffusion
    B. Facilitated diffusion
    C. osmosis
    D. Active transport

Explanation:

- Active transport is the movement of molecules from the region of lower concentration to higher concentration against concentration gradient. However, passive transport including simple diffusion, facilitated diffusion and osmosis occurs along the concentration gradient.

Answer: D

63. Which organelle of the cell has a function of modifying protein for secreation?
    A. Golgi body
    B. Ribosome
    C. Food vacuole
    D. Lysosome

Explanation:

Cell structure	Function
Glogi body	Modifies proteins, packages secreated proteins, sorts other proteins to vacuoles and other organelles.
Ribosome	Involved in protein synthesis
Food vacuole	Stores ingested materials
Lysosomes	Contains enzymes to break down ingested materials, secreations and wastes.

Answer: A

64. Which one of the following is true about Daniel and Davison membrane model?
    A. The phospholipid bilayer is in between the two protein is longer.

B. The globular proteins are dispersed in the lipid bilayer
C. The fatty acid tail is directed in the lipid bilayer.
D. The water living part of the membrane sandwiched by fatty acid tail

Explanation:

- Daniel and Davison membrane model is called **sandwiched model** because phospholipid bilayer is sandwiched by globular proteins.

Answer: A

65. Which of the following organelles are likely to be more aboundant in active cells such as muscle cells of human heart? (UEE)
    A. Lysosome
    B. Chromosome
    C. Mitochondoria
    D. Golgi bodies

Explanation: Mitochondoria is an organelle of a cell where the process of respiration takes place

- ✓ It is largely found in cells like muscle cell, liver cell, heart cells and others which are metabolically active.
- ✓ It consists of its own DNA and RNA.
- ✓ It multiples without the aid of the nuclies

Answer: C

66. Which of the following pairs are both organelles concerned with energy transformation? (UEE)
    A. Nucleus and nucleolus
    B. Mitochondoria and nucleus
    C. Chloroplasts and vacuole
    D. Chloroplast and mitochondoria

Explanation:

Energy transducting organelles ofle a cell are:

- ✓ **Mitochondoria** – site of most reactions of cellular respiration, transformation of energy originating from glucose or lipids in to ATP.
- ✓ **Chloroplast** consists of chlorophyll to capture light energy. ATP and other energy rich compounds are formed and then used to convert CO_2 to glucose.

Answer: D

67. Choose the structure that is usually present only in the cells of animals.

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A. Vacoule
B. Cell wall
C. Nucleus
D. Centriole

Explanation:

Structures that are only present in animal cells	Structures that are only present in plant cell
✓ Centriole	✓ Chloroplast
✓ Lysosome	✓ Cellulose cell wall
✓ Central nucleus	✓ Large central vacuole

Answer: D

68. All of the living components of the cell are collectively known as (UEE)
A. Cytoplasm
B. Nucleus
C. cell membrane
D. protoplasm

Explanation:

- **Protoplasm** is the living portion of a cell including the cell membrane, cytoplasm and nucleus with exception of cell wall.

Answer: D

69. Which of the following is found in both plant and animal cells?
A. Cell wall
B. Chromoplasts
C. chromosomes
D. leucoplasts

Explanation: Structures that are commonly found in both plant and animal cells are:

✓ Mitochondoria
✓ Nucleus
✓ Endoplasmic reticulum
✓ Golgi body
✓ Ribosome
✓ cell membrane, etc

Answer: D

70. One of the following would be harder to see under the ordinary light microscope that is more likely to be available in the school laboratories:
A. Nucleus
B. Bacterium
C. Mitochondoria
D. Ribosome

Explanation: Ribosomes are the smallest of all the structures mentioned above. Thus, it is difficult to see them by using an ordinary light microscope

Answer: D

71. The most modern cell membrane modal was suggested by
A. Needham and Koch
B. Leewenhook and Danel
C. Singer and Nicholson
D. Daniel and Davison

Explanation: The fluid mosaic model of the membrane which is introduced by Singer and Nicolson is the one which is acceptable and in use by the current time.

Answer: C

72. In which part of the cell do Grham-positive and Grham-negative bacteria differ regarding their staining property with Grhams stain?
A. Cytoplasm
B. Cell membrane
C. Cell wall
D. Nucleus

Answer: C

73. Identify the correct statement about the feature of cell membrane:
A. Fluidity of the cell membrane would increase the tail of the fatty acid.
B. Chlestrol in the cell membrane reduces the fluidity of the cell membrane
C. Two sides of the membrane differ in composition and function
D. All are possible answers

Answer: D

74. What is the process by which water passes across the cell membrane? (UEE, 2006)
A. Active transport
B. Osmosis
C. Facilitated diffusion
D. Pinocytosis

Explanation: Osmosis is the movement of water molecules from the region of higher concentration to lower concentration through a semi-permeable membrane.

Answer: B

75. Which of the following is NOT the characteristics of the fluid mosaic model of cell membrane?
A. The phospholipid bilayer is flexible.
B. Amphiphatic protein is dispersed in the bilayer

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C. Non-polar part of the amphiphatic protein are in contact with the non-polar part.
D. None

Explanation: In fluid mosaic model:

- Phospholipid is made up of unsaturated fat.
- The protein is **amphipathic**. Tthus, they have hydrophilic part in contact with phosphate and hydrophobic interior of the fatty acid.

Answer: B

76. Which means of particle transport requires input of energy by the cell? (UEE)
A. Simple diffusion
B. Facilitated diffusion
C. Osmosis
D. Active transport

Explanation: The movement of substances in to the cell takes place through the following major processes:

- ✓ **Passive transport** – is the movement of molecules along concentration gradient.
- ✓ **Active transport** – is the movement of molecules against concentration gradient.

Answer: D

77. Which of the following requires expenditure of ATP?
A. Osmosis
B. Facilitated diffusion
C. Simple diffusion
D. Endocytosis

Explanation: The movement of substances in and out of the cell occurs either by passive transport or active transport:

i) Passive transport involves:
- ✓ Diffusion
- ✓ Osmosis

ii) Active transport-requires expenditure of energy
- Endocytosis
- Exocytosis

Answer: D

78. What type of molecules cannot pass across the cell membrane by simple diffusion?
A. Charged molecules
B. Non-polar molecules
C. Liquid soluble molecules
D. Molecules of very small size

Explanation:
- Molecules that can easily pass through the cell membrane by diffusion are those molecules which are:
  - ✓ Non-polar
  - ✓ Non-charged

Answer: A

79. What will happen if human red blood cells are kept in a hypotonic solution?
A. Lose water by osmosis and burst
B. Lose water by osmosis and shrink
C. Taking in water by osmosis, swell and brust
D. Take in water by osmosis, swell and remain turgid

Explanation: If red blood cell (animal cell) is kept in hypotonic solution, it gains water and may get brust. This is a turgor pressure. However, since red blood cells (animal cells) have no cell wall that produces wall pressure to resist turgor pressure, they can easily brust.

Answer: C

80. Which types of solution is the water potential more negative than the cells?
A. Hypotonic
B. Hypertonic
C. Isotonic
D. Equal solute and solvent concentration

Explanation: The solution with highest water potential has less amount of solute and more amount of solvent. Thus, hypertonic solution which has relatively more solute has a water potential which is more negative than the cells.

Answer: B

81. If a suspension of mixture of cellular organelle is spun in a centrifuge which organelle settles to the bottom first? (UEE, 2005)
A. Mitochondoria
B. Nuclei
C. Chloroplast
D. Ribosomes

Explanation: Cell fraction is a technique that is used to separate different organelles depending on their size spinning in a centrifuge. The organelles separate out in a particular order, according to

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their density and shape. Thus, because the nuclei has higher density than the rest, it will settle down at the bottom first.

Answer: B

82. Which of the following is an important function of Golgi apparatus?
A. Protein synthesis
B. Packaging of protein for export out of the cell
C. Removing of debris from the cell
D. Storage of waste materials not needed by the cell

Explanation: Golgi apparatus is a cellular organelle which is used in packaging, collecting and concentrating substances ready for transport.

Answer: B

83. Compare the process of passive transport and active transport.

Explanation:

Passive transport	Active transport
✓ Involves the movement of molecules from the region of higher concentration to lower concentration	✓ Involves the movement of molecules from lower concentration to higher concentration.
✓ It occurs along the concentration gradient	✓ It occurs against concentration gradient.
✓ It does not expend energy	✓ It requires expenditure of energy from the cell
✓ Examples:	✓ Examples:
• Simple diffusion	• Endocytosis
• Osmosis	• Exocytosis
• Facilitated diffusion	

84. List the types of endocytosis with their description.

Explanation: The major types of endocytosis are:

✓ **Phagocytosis** – is the process by which a cell ingests larger or solid particles.

✓ **Pinocytosis** – is the process in which the cell takes in dissolved or liquid materials.

✓ **Receptor mediated endocytosis** – the process in which specific proteins embed on the plasma membrane to stimulate infolding.

85. Diffusion differs from active transport in all except:
A. Movement of material is from high concentration to low concentration.
B. Used to distribute materials evenly in the medium
C. Used to concentrate particles in one place
D. None

Explanation: Diffusion differs from active transport because:
- It is a passive transport system
- It is used to distribute materials equally in the medium.
- It does not require expenditure of energy.

Answer: C

86. All increase the rate of diffusion, except:
A. Thickness of the membrane
B. Small surface area
C. Increase in temperature
D. None

Explanation: Increase in temperature and thickness of the membrane decreases the rate of diffusion. However, having small surface area decreases the rate of diffusion.

Answer: B

87. Which component of the plasma membrane is mismatched with its role?
A. Antigen – determine the characteristic feature of the cell.
B. Glycolipids – cell to cell recognition.
C. Cholesterol – carrier or channel molecule on the cell membrane.
D. None

Explanation:

Membrane component	Description
Antigen	Identity marker
Glycolipids	Pass signal for cell to cell recognition
Cholesterol	Do not use as carrier or channel protein

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88. Cell membrane:
   A. Control the passage of material in and out of cells
   B. Compartmentalize cell organelles.
   C. Are place for certain chemical reaction.
   D. Are selectively permeable.
   E. All are possible.

Answer: (

Explanation:

Cell membrane:

- ✓ Controls the movement of materials in and out of the cell
- ✓ Cover or encloses organelles and chemical reaction that take place inside the cell.
- ✓ Is selectively permeable membrane.

Answer: (

89. The role of cholesterol is to:
   A. Increase the semi-liquid nature of the cell membrane
   B. Increase the flexibility of the membrane.
   C. Increase the solid nature of the membrane.
   D. Give colour to the membrane.

Explanation: Cholesterol which is inserted in between the phospholipid used to prevent the escape of organelles out of the cell and increases the solid nature of the membrane.

Answer: (

90. Which of the following is not the nature of phospholipid membrane?
   A. They are associated together to form a bilayer.
   B. They are permeable to lipid soluble substances.
   C. They have hydrophilic head.
   D. They shift from one side of the membrane to other side.

Explanation: The phospholipids are fluid and diffuse rapidly on their own layer but not from one side to another.

Answer: I

91. The rate of movement of particles through cannel protein and carrier protein is unaffected by:
   A. Increase in temperature
   B. Increase in concentration gradient.
   C. Increase in the surface area of the membrane.

D. Number of channel and carrier protein.

Explanation: It is not increase or decrease in surface area that affects the rate of movement on the channel and carrier proteins, rather it is the number of channel or carrier protein on the surface of the membrane that affects the rate at which particles move.

Answer: C

92. Compare and contrast simple diffusion and facilitated diffusion.

Explanation:

Feature	Simple diffusion	Facilitated diffusion
Direction of molecular movement	From higher concentration to lower concentration	From higher concentration to lower concentration
Diffusion is assisted by	Small size of particles	Carrier and channel protein
Factors that affect the rate	• Concentration gradient • Surface area of the membrane • Thickness of the membrane	Concentration gradient Temperature

93. Osmosis differ from diffusion in:
   A. It is passive transport system
   B. It takes place through a semi-permeable membrane
   C. In involves the movement of water molecule only
   D. B and C

Explanation:

Osmosis	Diffusion
• Occurs through semi-permeable membrane. • Involves only the movement of water molecules	• Does not necessarily require semi permeable membrane. • It is the movement of any molecules.

Answer: D

94. Which of the following is hypertonic to the rest?
   A. 1% salt
   B. 0.1% salt
   C. 0.01% salt
   D. 0.5% salt

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Explanation: Hypertonic solution is a solution which consists of more solute (salt) than the rest.

Answer: A

95. In which of the following solution the RBC will get hemolyzed?
   A. 1% salt
   B. 0.1% salt
   C. 0.01% salt
   D. 0.5% salt

Explanation: When an animal cell (RBC) is placed in a hypertonic solution it gains water and finally it will brust. This condition is called **hemolysis.** Therefore, 0.01% salt solution is hypotonic to all the rest.

96. RBCS gets crenated, if they are kept in:
   A. Hypotonic solution
   B. Isotonic solution
   C. Hypertonic solution
   D. None

Explanation: If RBCS are kept in hypertonic solution, they loose their water content and shrink. This condition is called **crenation.**

Answer: C

97. A potato cylinder is kept in a solution and does not show any change in weight. So, the solution is _____ to the potato cylinder.
   A. Hypertonic
   B. Hypotonic
   C. Isotonic solution
   D. A and B

Explanation: Isotonic solution has the same concentration with the cell. Thus, there is no movement of molecule in and out of the cell.

Answer: C

98. Which of the following is true based on the above diagram?
   A. The level of 'B' increases.
   B. The level of 'A' increases.
   C. 'B' is hypotonic to 'A'.
   D. Both increase in level.

Explanation: 'B' is hypertonic to A. Thus, 'A' looses its water content and decrease in level, while 'B' gains more water and increases in its level.

Answer: A

99. Water potential of a solution is more negative than the cell means:
   A. The solution is hypertonic
   B. The solution is hypotonic
   C. The cell is hypertonic
   D. A and C

Answer: A

100. Which of the following organelle contains its own DNA?
   A. Ribosome
   B. Lysosome
   C. Mitochondoria
   D. Golgi apparatus

Explanation: Chloroplast and mitochondoria are cellular organelles that have their own DNA, ribosomes and enzymes.

Answer: C

101. Choose the structure found only in animal cell.
   A. Phagocytic vesicles
   B. Centrioles
   C. Microtubules
   D. All except 'C'

Explanation: Some animals cells have pseudopodia which is used to produce phagocytic cells.

Example: White blood cells.

- Centrioles are used to form spindle fiber in animal cells during cell division.
- Microtubules are present in both plant and animal cells.

Answer: D

102. What is the difference between rough ER and smooth ER?

Explanation:

Rough ER	Smooth ER
• Have ribosome • Involved in the transportation of protein	• Have no ribosomes • Concerned with lipid synthesis, carbohydrate metabolism and detoxification.

103. Which of the following mechanisms moves digested amino acids and glucose across the plasma membrane of the cells lining the wall of the small intestine?
   A. Osmosis
   B. Simple diffusion
   C. facilitated diffusion
   D. active transport

Explanation: Digested aminoacids and glucose can move across the plasma membrane of the cells lining the wall of the small intestine by facilitated diffusion. Because the plasma membrane has the carrier proteins or channel proteins with ion pores.

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104. Which one of the following characteristics is unnecessary for object to be considered as living things? (UEE, 2006)
A. Ability to respond to stimuli
B. Ability to reproduce
C. Ability to grow
D. Ability to move

Answer: C

Explanation: None of the options can be taken as an answer.

105. If a cell fails to clear its cellular debris, which one of its organelles is most likely not functioning?
A. Nucleus
B. Mitochondoria
C. Endoplasmic reticulum
D. Lysosomes

Explanation: Lysosomes are cellular structures that are used to distentigrate damaged and wornout cells and their products. Because of this reason they are called '**sucide bags**' of the cell.

Answer: D

106. Among the following identify the organelle in which nucleic acid is not found?
A. Mitochondoria
B. Chloroplast
C. Ribosome
D. Golgi apparatus

Explanation: Nucleus, mitochondoria and chloroplast have their own DNA (Nucleic acid). Ribosomes contain RNA (nucleic acid). However, golgi complex has no nucleic acid contained in it.

Answer: D

107. In which of the following solution does an animal cell undergo haemolysis?
A. In hypotonic solution
B. In hypertonic solution
C. In isotonic solution
D. In both hypotonic and hypertonic solution

Explanation: When animal cells are putted in to a hypotonic solution they gain water by osmosis and the may get brust. This condition is called **haemolysis.**

Answer: A

108. Which of the following will happen if a plant cell is kept in a solution is stronger than its protoplasm?
A. The cell wall become liquid.
B. The central vacuole ill expand.
C. The protoplasm will get plasmolyzed.
D. The cell will swell and brust.

Explanation: If a plant cell is placed in a strong (concentrated) solution, it looses its water content by osmosis and get plasmolyzed.

Answer: C

Cell size, shape and function

Cells vary from one another in various ways. Some cells have exceptional features that are found in all other cells. However, the variation between different types of cells can be considered mainly with respect to their size, shape and function.

Cell shape

The shape of the cells is highly related to their functions. Some cell have oval shape, circular shape, elongated shapes, irregular shapes or etc.

Cell	Shape
Neuron	Thin and elongated
Sperm cell	Thin and long tail with oval shaped head
Ovum	Oval
Red blood cell	Disc shaped
White bbodcell	Irregular

Cell function

As you have learned on grade nine biology, cells get specialized in order to perform different functions in our body. Accordingly, our cells are specialized to perform all the life activities in an organism. Unicellular organisms such as bacteria and protozoa are single celled by those cells are self sufficient to perform all the life activities that are performed by this organisms.

In multicellular organisms there are variety of cells that are used to perform specific tasks for which they are specialized. Accordingly, let us see some of the cells with their functions.

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Cell	Function
Neuron	To transmit nervous impulse within the body.
Red blood cell	To transport various kinds of substances with in the body
Ovum sperm cell	For reproductive purpose
Muscle cell	The enable body movement, working of different organs such as hear
White blood cell	To defend the body against disease

Cell size

Even though, there are some groups of cells that are visible and can be easily seen by our nacked eye, most cells are microscopic. Microscopic cells are those cell which cannot be easily seen lay our nacked eye and can only be seen by the aid of microscope.

The size of cells varies from 1μ to 17500μ (175mm). the largest cell ever is an ostrich egg cell. In order to compare the size of some cells, chicken egg cell is as big as 5cm is length, while bacterias are only about 100nm in length.

The size of cells can be measured mainly by using the three smaller units that are commonly used. They are:

- ✓ $Millimeters\,(mm) - \frac{1}{1000}\,of\ meter$
- ✓ $Micrometers\,(\mu m) - \frac{1}{1000}\,of\ a\ millimeter\ and\ \frac{1}{1000000}\,of\ a\ meter$
- ✓ $Nanometer\,\square(nm) - \frac{1}{1000}\,of\ a\ micrometer, - \frac{1}{1000000}\,millimeter - \frac{1}{1000000000}\,of\ a\ meter$

Illustrative Examples

Convert a cell having a size of 0.5mm to μm

Explanation:

1mm=1000μm

$$0.5mm = x$$

$$x = \frac{0.5mm \times 1000\mu m}{1mm}$$

$$= \underline{500\mu m}$$

Convert a cell having a size of 6250nm in to μm

Explanation:

1μm = 1000nm

x = 62500nm

$$x = \frac{1\mu m \times 62500nm}{1000nm}$$

$$= \underline{62.5\mu m}$$

Surface area to volume ratio of a cell and its consequences

Eventhough cells vary in size, shape and organization, all cells exchange materials with the environment. Accordingly, they get raw materials from their environment and remove waste material out to their environment.

Every cell reproduce and respond to external environment. Thus, nutrients, oxygen and other materials enter the cell through its surface. Therefore, surface area to volume ratio of cells is highly determinant for the exchange of material between the cell and its environment.

The larger the surface area to volume ratio, the higher their ability to pass substances in and out. Therefore, cells do have different types of modification in increase their surface area which is vital for efficient exchange of material. Some of the modifications are:

The smaller the cell, the larger its surface area to volume ratio. Thus,

- ✓ Their ability to pass nutrient into and outside the cell is faster.
- ✓ The cell is metabolically active

The bigger the cell, smaller their surface area to volume ratio. Thus,

- It will not allow sufficient nutrients to pass into the cell.
- The rate of metabolic activity of the cell is relatively slow.
- Cellular products distribute slowly.

Arbitrary units (au) - are units that we use when we do not know the actual diamension, but we know the mathematical relationship between different conditions.

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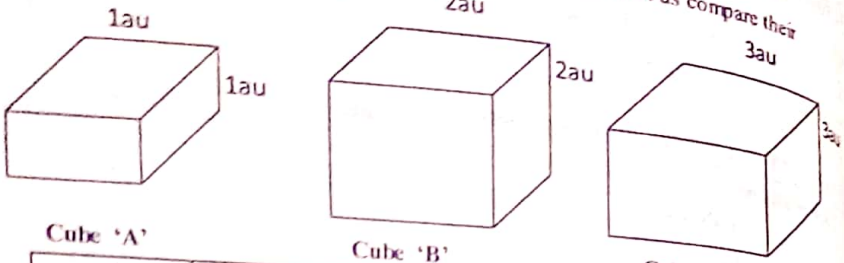
Figure below shows three cubes of different size. Thus, let us compare their surface area to volume ratio.



	Cube 'A'	Cube 'B'	Cube 'C'
Area (A)	=1au×1au =1au^2	=2au×2au =4au^2	=3au×3au =9au^2
Surface are (SA)	=6×1au^2 = 6 au^2	=6×4au^2 = 24 au^2	=6×9au^2 = 54 au^2
Volume (V)	=1au×1au×1au = 1au^3	=2au×2au×2au = 8au^3	=3au×3au×3au = 27au^3
Surface area to volume ratio (SA/V)	$=\frac{6au^2}{1au^3}$ = 6	$=\frac{24au^2}{8au^3}$ = 3	$=\frac{54au^2}{27au^3}$ = 2

∴ From this table we can easily conclude that, smaller objects have large surface area to volume ratio and vice versa.

Let us discuss the effect of cell size in relation to various features by using the following table.

Feature	Large sized cell	Small sized cell
Surface area	Small as compared to its volume	Large as compared to its volume
Volume	Large	Small
Distribution of material in the cell	Slow	Fast
Rate of exchange of material	Slow	Fast
Metabolism	Less active	Less active

Measuring the size of cells by using microscope

Stage microscope – a microscope slide with very precise scale and helps to measure small distances or thickness.

Eye piece gratitude – is a piece of plastic with less accurate scale then the gratitude that fits inside the eye piece of the microscope.

Conceptual Summary Example

1. Evaluate the importance of membranes to the homeostasis of the cell, emphasizing their various functions.

Explanation:

- Biological membranes:
  - ✓ Physically separate the interior of the cell from the extracellular environment.
  - ✓ Form compartments within eukaryotic cells, allowing a variety of separate functions.
  - ✓ Regulates the passage of materials in to and out of the cell.

2. Describe the fluid mosaic model of cell membrane structure.

Explanation: According to the fluid mosaic model, membranes consist of a fluid phospholipid bilayer in which a variety of proteins are embedded.

3. Explain how the properties of the lipid bilayer govern many of the properties of a cell membrane and of the cell.

Explanation:

4. Describe how membrane protein's associate with the lipid bilayer, and discuss the functions of membrane proteins.

Explanation:

- **Integral membrane proteins** – are embedded in the bilayer with their hydrophilic surface exposed to the aqueous environment and their hydrophobic surfaces in contact with the hydrophobic interior of the bilayer.
- **Transmembrane proteins** – are integral proteins that extend completely through the membrane.
- **Peripheral membrane proteins** – associated with the surface of the bilayer, usually bound to exposed regions of integral proteins,

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and are easily removed without disrupting the structure of the membrane.

- Generally membrane proteins have the following functions:
  - ✓ Transport of materials.
  - ✓ Acting as enzymes or receptors.
  - ✓ Cell recognition
  - ✓ Structurally linking cells together.

5. Summarize the relationship between cell organization and homeostasis.

Explanation: plasma membrane helps to maintain homeostasis by allowing the cell to exchange materials with its external environment and to maintain internal conditions that may be different from those of the outer environment.

- Cell have organelles, internal structures that carryout specific functions that help maintain homeostasis.

6. Describe methods that biologists use to study cells, including microscopy and cell fraction.

Explanation:

- Biologists learn about cell structure by studying cells with light and electron microscope and by using a variety of chemical methods.
- The electron microscope has superior resolving power, enabling investigators to see details of cell structures not observable with conventional microscopes.
- Cell biologists use cell fraction methods for purifying organelles, to gain information about the function of cell structures.

7. Compare and contrast the general characteristics of prokaryotic and eukaryotic cells.

Explanation:

Prokaryotic cells	Eukaryotic cells
• Bounded by plasma membrane but have little or no internal membrane organization.	• Have membrane – enclosed nucleus and cytoplasm, which contains variety of organelles.
• They have nuclear materials that freely suspend in the cytoplasm rather than a membrane enclosed nucleus. • Typically have a cell wall, and ribosomes and may have propeller like flagella.	

8. Describe the structure and functions of the nucleus.

Explanation: The nucleus, the control center of the cell, contains genetic information coded in DNA.

- It is bounded by nuclear membrane consisting of a double membrane perforated with nuclear pores that communicate with the cytoplasm.
- DNA in the nucleus associates with protein to form chromatia which is organized in to chromosomes. During cell division the chromosomes condense and become visible tread like structures.
- The nucleus is the region in the nucleus that is the site of ribosomal RNA synthesis and ribosome assembly.

9. Compare functions of mitochondria and chloroplasts.

Explanation:

Mitochondria	Chloroplasts
✓ Sites of aerobic respiration ✓ enclosed by double membrane ✓ the inner membrane is folded forming cristae that increase its surface area. ✓ Contains DNA that codes for some proteins. ✓ Play an important role in a apoptosis (programmed cell death). ✓ The inner membrane matrix	✓ are plastids that carry photosynthesis. ✓ the inner membrane of chloroplasts fluid filled space called **stroma**. ✓ **Grana** stacks of disk like membranous sacs called **thylakoids**. ✓ contain a green pigmented substance that absorbs light during photosynthesis

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contains enzymes for the reactions of aerobic respiration.	

10. Contrast the physical processes of simple diffusion and osmosis with carrier – mediated physiological processes by which materials are transported across the cell membrane.

Explanation:

- **Diffusion** – is the net movement of a substance down its concentration gradient.
- **Osmosis** – is a kind of diffusion in which the molecules of water pass through a selectively permeable membrane from a region where water has higher effective concentration to a region where its effective concentration is lower.
- Diffusion and osmosis are physical processes that do not require the cell to directly expend metabolic energy.
- Membrane transport proteins facilitate the passage of certain ions and molecules through biological membranes.
  - ✓ **Channel proteins** – are transport proteins that form passage ways through which water and certain ions travel through the membrane.
  - ✓ **Carrier proteins** – are transport proteins that undergo a series of conformational changes as they bind and transport specific solute.

11. Differentiate between the process of facilitated diffusion and active transport. Identify the energy sources for each processes.

Cell (Post test)

1. Which serves as the outer boundary of a cell?
   A. Golgi apparatus C. nuclear membrane
   B. Plasma membrane D. endoplasmic reticulum
2. Which cellular structure produces proteins using coded instructions from DNA?
   A. ribosome C. Golgi apparatus
   B. Lysosome D. endoplasmic reticulum

3. Which is present **only** in plant cells?
   A. nucleus C. mitochondrion
   B. large vacuole D. cell membrane
4. What is the function of mitochondria?
   A. to make new DNA through DNA replication
   B. to make protein through the process of translation
   C. to convert sunlight into glucose through photosynthesis
   D. to convert pyruvate into ATP through cellular respiration
5. The Golgi apparatus is responsible for the
   A. Manufacture of lipids.
   B. Production of energy for the cell.
   C. Destruction of unwanted cellular molecules.
   D. Modification and packaging of protein molecules
6. If energy is needed to move materials into or out of a cell, what is ***most likely*** occurring?
   A. active transport C. osmosis
   B. passive transport D. diffusion
7. Which would be the ***best*** evidence that a cell is using active transport to move a substance across its cell membrane?
   A. Substances are moving rapidly across the cell membrane.
   B. ATP is being rapidly consumed near the cellular membrane.
   C. Substances are moving from high to low concentrations.
   D. Substances are moving through channels in the cell membrane.
8. Intravenous saline injections are often given as treatment for severe dehydration. The concentration of saline (0.9% NaCl) in these injections is that same as that present in human cells. What would happen if pure water was introduced into the body instead of saline?
   A. The cells would gain water and swell.
   B. The cell would lose water and shrivel.
   C. The cell would become impermeable to sodium (Na^+) ions.
   D. The cells would become impermeable to chlorine (Cl-) ions
9. Which organelle produces proteins?
   A. nucleus C. ribosome
   B. lysosome D. Golgi body

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10. What is the primary purpose of the receptor proteins located within a cell membrane?
    A. diffusion
    B. communication
    C. active transport
    D. energy production
11. What will ***most likely*** be the result if all of the mitochondria are removed from a plant cell?
    A. It will be unable to carry out respiration.
    B. It will lose water through osmosis.
    C. It will break down the ribosomes in the cell.
    D. It will be unable to photosynthesize.
12. How are prokaryotic and eukaryotic cells similar?
    A. Both contain a nucleus.
    B. Both contain ribosomes.
    C. Both contain membrane-bound organelles.
    D. Both contain cell walls.
13. Which would be the ***best*** evidence that a cell is using active transport to move a substance across its cell membrane?
    A. Substances are moving rapidly across the cell membrane.
    B. ATP is being rapidly consumed near the cellular membrane.
    C. Substances are moving from high to low concentrations.
    D. Substances are moving through channels in the cell membrane.

Critical thinking

1. Most adjacent plant cells are connected by plasmodesmata, whereas only certain adjacent animal cells are associated through gap junctions. Why?
2. Evaluate the importance of membranes to the cell, discussing their various functions.
3. Why can't larger polar molecules and ions diffuse through the plasma membrane? Would it be advantageous to the cell it they could? Explain
4. Why does eukaryotic cells need both membraneous organelles and fibrous cytoskeletal components?

Supplementary exercise on Unit 4

I. Match the descriptions under 'column A' with the name of scientists under 'column B'

A	B
1. The first compound microscope	a. Robert Hook
2. Coined the name cell first	b. Rudolf Virchow
3. Discovered protozoa	c. Anton Van Leeuwenhook
4. Stated all cells came from pre existing cells	d. Watson and Crick
5. Formulated the cell theory	e. Brown
6. Described the nucleus	f. Jensen
7. Isolated	g. Schielden and Schwann
8. Described mitochondoria	h. Meisher
9. Proposed the double helix structure of DNA	i. Kollier

II. Multiple Choice

10. The first living cell is identified by:
    A. Louis Pasteur
    B. A. Van. Leeuwenhook
    C. Robert Koch
    D. Rachel Carson
11. Which of the following is NOT included in the cell theory?
    A. All living things are made up of cells.
    B. Cells arise spontaneously from other objects.
    C. All cells have basically the same chemical composition.
    D. Cells are structural and functional units of life.
12. The protoplasmic theory is stated by:
    A. August Wicmann
    B. Oscar Herwig
    C. Rudolf Virchow
    D. Mathias Schielden
13. If an object having a size of 0.003mm is magnified by an eye piece of 10x and objective lens of 10x. What is the magnified size of an object in mm?
    A. 3mm
    B. 30mm
    C. 0.3mm
    D. 0.03mm
14. If a cell have a size of 0.04mm. What is its size in μm?
    A. 40μm
    B. 0.004μm
    C. 0.0004μm
    D. 0.4μm

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15. Which of the following is an example of prokaryotic cells?
    A. Algal cell
    B. Fungeal cell
    C. Bacterial cell
    D. Plant cell
16. Which of the following is incorrect about eukaryotic cells?
    A. They have DNA in the form of chromosomes.
    B. They have distinctive nucleus.
    C. Organelles are double membrane bounded
    D. They are first to evolve on the earth.
17. Cellular structure that consists of organelles is:
    A. Cell wall
    B. Cell membrane
    C. Cytoplasm
    D. Nucleus
18. Hereditary informations are contained in the:
    A. Nucleus
    B. Ribosome
    C. Cytoplasm
    D. cell membrane
19. Which of the following process requires energy?
    A. Simple diffusion
    B. Endocytosis
    C. facilitated diffusion
    D. Osmosis
20. Which of the following is incorrect about fluid-mossaic model?
    A. It is recently working membrane model.
    B. It rejects the sandwich arrangement of phospholipid bilayer.
    C. Its arrangement is fluid and constantly changing.
    D. It do not properly explain how molecules move across membranes.
21. The fluidity of the membrane is reduced by:
    A. Integral proteins
    B. Glycolipids
    C. Glycoproteins
    D. Osmosis
22. The process in which particles move from low concentration to higher concentration using carrier proteins is:
    A. Surface area of the membrane.
    B. Concentration difference.
    C. Thickness of the membrane.
    D. None of the above.
23. Which of the following transportation process occurs along concentration gradient?
    A. Facilitated diffusion
    C. Pinocytosis

    B. Endocytosis
    D. Exocytosis
24. Which of the following happens if plant cell is placed in hypotonic condition?
    A. Brusting
    B. Plasmolysis
    C. Turgidity
    D. Shrinking
25. Which of the following transportation process requires carrier proteins?
    A. Simple diffusion
    B. Exocytosis
    C. Osmosis
    D. Facilitated diffusion
26. Cellular organelle that is responsible for protein synthesis is:
    A. Mitochondoria
    B. Ribosome
    C. Golgi body
    D. Vacoule
27. Transportation of proteins in the cell is carried by:
    A. Rough ER
    B. Smooth ER
    C. Golgi body
    D. Ribosomes
28. Cellular activities are controlled and regulated by:
    A. Cell membrane
    B. Cell wall
    C. Nucleus
    D. Vacoule
29. An organelle that maintains the turgidity in plant cells is:
    A. Chloroplast
    B. Lysosome
    C. vacuole
    D. protoplasm
30. One difference between a plant cell and animal cell is:
    A. Only plant cells have chloroplasts, while only animal cells have mitochondoria.
    B. Only plant cells have large vacuoles.
    C. Only animal cells have nucleoli.
    D. Only animal cells have plasma membranes.
31. All of the following are correct about prokaryotes except:
    A. They have one chromosome.
    B. The contain ribosomes.
    C. They are smaller than eukaryotic cells.
    D. They can be either aerobic or anaerobic.
    E. They contain small mitochondoria.
32. What would occur if a plant cell were placed in to a strong hypertonic solution?

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A. Nothing would happen because the cell wall would prevent any movement of water or salt.
B. It would skink because the cell wall would prevent any movement of water or salt.
C. It would swell because salt would flow into the cell.
D. It would shrink because water would flow into the cell.

33. Which process requires energy?
A. Passive diffusion
B. Facilitated diffusion of calcium through a channel
C. Water flowing into a paramecium in a lake
D. A contractile vacuole removing water from an amoeba.
E. Osmosis of water in to a cell.

34. All of the following are true about smooth endoplasmic reticulum except
A. It connects the rough endoplasmic reticulum to golgi apparatus.
B. It detoxifies the cell
C. It synthesizes steroids
D. It manufactures proteins
E. It synthesizes lipids

35. According to the theory of endosymbiosis,
A. Authotropic cell were first to evolve
B. Heterotrophic cells were first to evolve
C. Chloroplasts evolved from mitochondoria
D. Mitochondoria evolved from chloroplasts
E. Chloroplasts and mitochondoria evolved when free-living prokaryotes permanently took up residence inside larger prokaryotic cells

36. All of the following are correct about mitochondoria EXCEPT:
A. They are site of cellular respiration.
B. They contain their own DNA.
C. They are the site of photosynthesis.
D. They can self-replacate
E. They contain an outer double membrane.

III. Blank space items

37. The fluid in the vacuole is called _____.
38. The folded inner membrane of mitochondoria is _____.
39. Organelles in a cell for photosynthesis are _____.
40. _____ and _____ are the main components of ribosome.
41. The model of plasma membrane structure that produced by singer and Nicholson is 1972 is called _____.
42. _____ is a solution that have lower water potential than the cell.
43. The solution having the same water potential as the cell is called _____.
44. _____ are primarily responsible for the physical properties of biological membrane.

Answer to supplementary exercise on Unit 4

1. F	11. B	21. B	31. E
2. A	12. C	22. D	32. B
3. C	13. C	23. A	33. D
4. B	14. C	24. C	34. D
5. G	15. C	25. D	35. E
6. E	16. D	26. B	36. C
7. H	17. C	27. A	37. Cell sap
8. J	18. A	28. C	38. Cristae
9. D	19. B	29. C	39. Chloroplast
10. B	20. D	30. B	40. Protein and RNA
			41. Fluid mosaic
			42. Hyportonic
			43. Isotonic
			44. Phospholipids

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UNIT FIVE

Energy Transformation

Introduction

In biological processes one of the most important concept is energy. **Energy** is shortly defined as the capacity to do work. In biology energy is expressed in a unit called **kilojoules** (kJ) or **kilocalories** (kCal). Remember, one kcal is equal to 4. 184kJ.

Most actions of an organism involve a complex series of energy transformations that occur as kinetic energy is converted to potential energy or as potential energy is converted to kinetic energy.

Chemical energy, potential energy stored in chemical bonds, is of particular importance to organisms. The chemical energy of food molecules is converted to kinetic energy is the muscle cells. The contraction muscle cells to perform the life activities of an organism is an example of mechanical energy.

The study of transformation of one form of energy in to another is called **thermodynamics**. There are two basic laws in thermodynamics. They are:

i) **The first law of thermodynamics** – states that energy cannot be created nor destroyed, but it can be transferred or converted from one form to another.

According to this law,s a system may absorb energy from its surroundings or it may give up some energy to its surroundings, but the total energy context of that system plus its surroundings is always the same.

As this law states, organisms cannot create the energy they refuire inorder to live. Instead, they must capture energy from the environment and transform it to a form that can be used for biological work.

ii) **The second law of thermodynamics** – states that when energy is converted from one form to another some usable energy available to do work is converted in to heat energy that disperses in to its surroundings.

As stated in this law no process requiring an energy conversion is ever 100% efficient, because much of the energy is dispersed as heat.

Accordingly, during energy transfer from sun to plants and to animals, there is loss of energy at each step of transfer.

Energy and Metabolism

Metabolism is the sum of all the chemical reactions that take place in living organism. It involves various types of chemical reactions that are vital to perform all the life activities which occur in living organisms.

Metabolism involves two main types of reactions. They are:

i) **Anabolism** – involves the process by which simpler molecules synthesize complex molecules.

Some examples of anabolic reactions are:

- ✓ Joining of amino acids to form protein.
- ✓ Joining of glucose units to form complex sugars such as starch and glycogen.
- ✓ The process of photosynthesis in which carbondioxide, water and light energy are used to synthesize organic molecule glucose.

ii) **Catabolism** – involves the process by which larger molecules are broken down or disintegrate in to smaller substances.

Some examples of catabolic reactions are:

- ✓ Degradation of starch in to monosacharides like glucose.

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- ✓ Oxidation of glucose in to carbondioxide, water and energy during respiration.

The transfer of energy in a community of organisims can be shown clearly in the following diagrammatic representation.

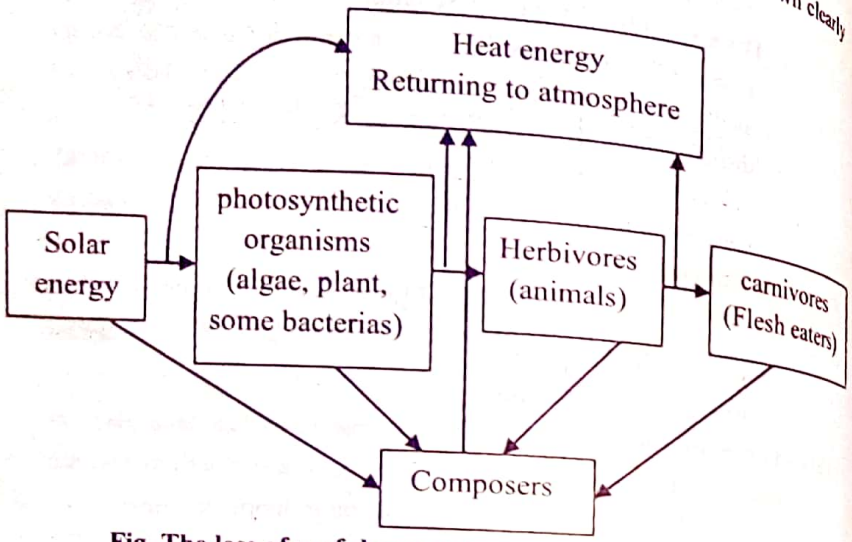

Fig. The loss of useful energy in a community of organisms

5.1. Respiration

Respiration is the process by which the chemical energy in organic molecules is released by oxidation. This energy is then made available to living cells in the form of ATP and the biochemical process which occurs in the cells is called **cellular respiration**.

The structure of ATP Molecule

The full name for ATP is **Adenosine Tri – phosphate**. ATP molecule is composed of **adenine, ribose sugar** and **three phosphates**. Thus, ATP is a nucleotide.

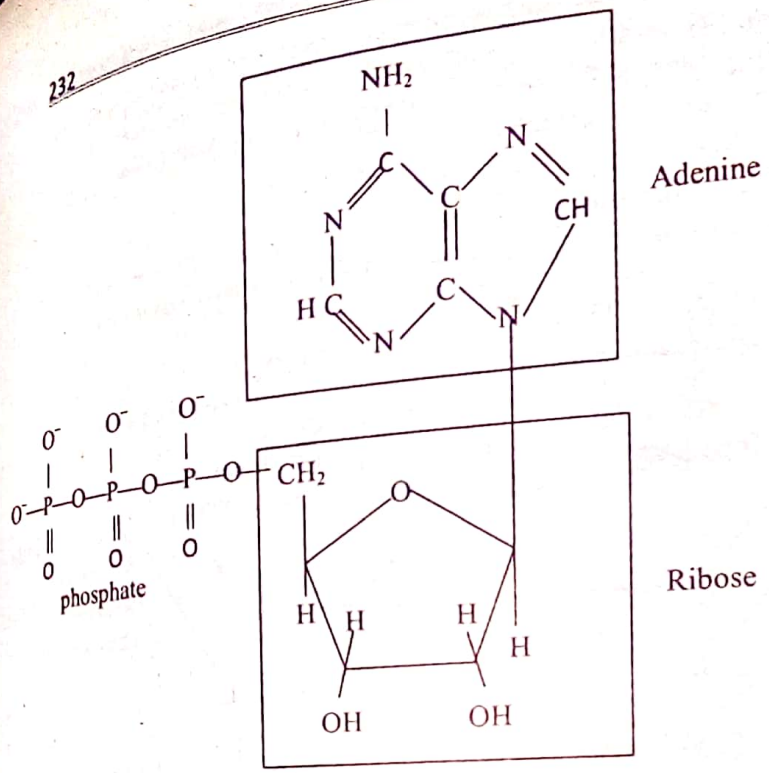

Fig. 5.2.The structure of ATP

Note: ATP is a nucleotide that has energy – rich bonds joining the two terminal phosphate groups to the nucleotide.

- Therefore, ATP molecule is a nucleotide that consists of
  - A nitrogenous base (Adenine)
  - A pentose sugar (deoxyribose)
  - A phosphate group
- Since ATP molecule has phosphate groups added on it, it is also called a **phosphorylated nucleotide.**
- ATP is adenine nucleotide with two extra phosphate groups added on it to make it three phosphates. Thus, addition of the extra phosphates requires energy, specially when the third phosphate is added.

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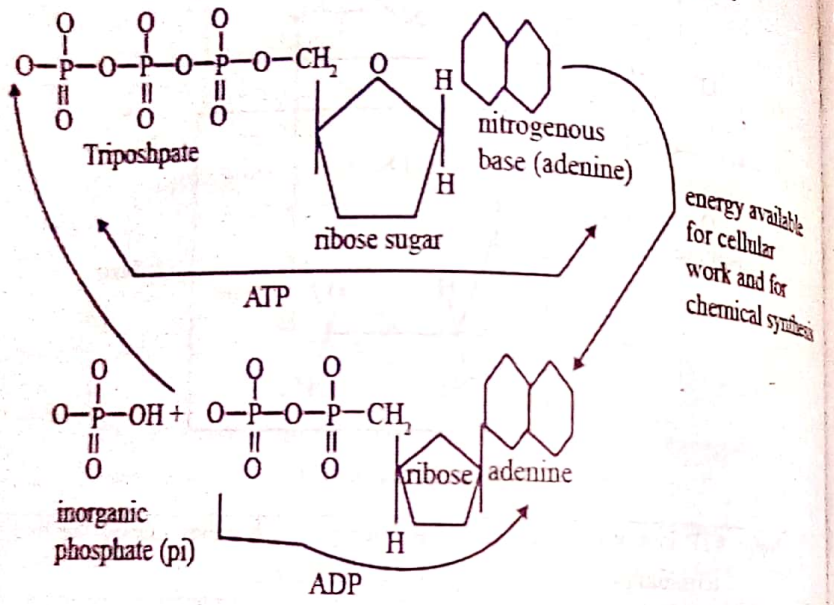
Accordingly, the bond between the second and the third phosphate contains energy which is released again when this bond breaks down. This splits **ATP** (adenosine triphosphate) in to **ADP** (adenosine di-phosphate) and **pi** (phosphate).

ATP ⟶ ADP + pi + energy

The energy which is stored in ATP comes primarly from sunlight which forms ATP during **photosynthesis** and **cellular respiration.**



ATP as energy transfer molecule in cells

ATP serves as energy transfer molecule for cellular processes, since organisms cannot directly utilize the energy from sunlight (For photosynthetic organisms) and the energy stored in glucose molecule ATP is used as energy transfer molecule turning the process of cellular respiration. Accordingly, ATP is called the **direct/immediate source of energy** for the cellular activities.

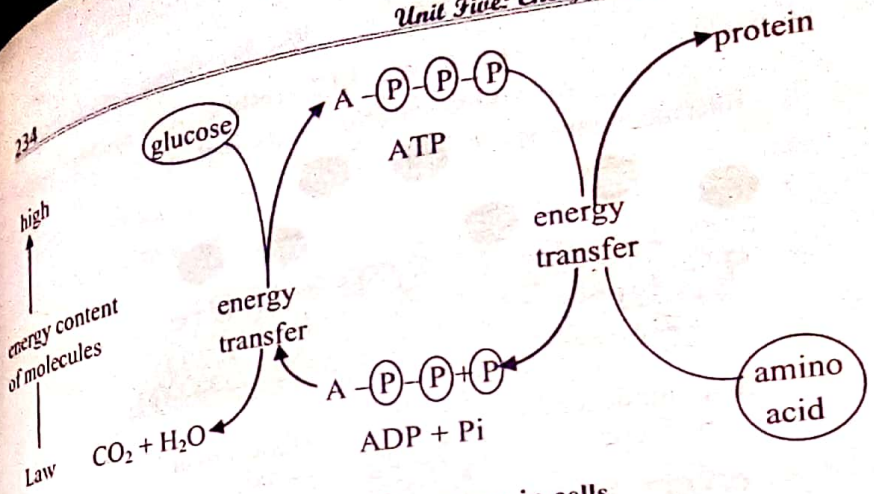


Fig. 5.4. Coupled reactions transfer energy in cells

ATP is adapted to its role because:

- It releases energy in relatively small amount that are closely matched to the amounts of energy required in many biological processes Occurring inside cell.
- It releases energy in a single – step hydrolysis reaction, so that energy can be released quickly.
- It is able to move around the cell easily, but cannot escape from the cell.

ATP Molecule is required in various processes. Among these, some of them are:

- Synthesis of macromolecules like proteins.
- Active transport across the plasma membrane.
- Muscle contraction.
- Conduction of nerve impulse.
- The initial reactions of respiration

Production of ATP in a cell

As it is described earlier the formation of ATP involves joining of ADP and pi which also requires an input of energy. This process by which ATP is

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formed involves an enzyme called **ATP synthese**. ATP synthase is within the membranes of **mitochondrion** or **chloroplast**

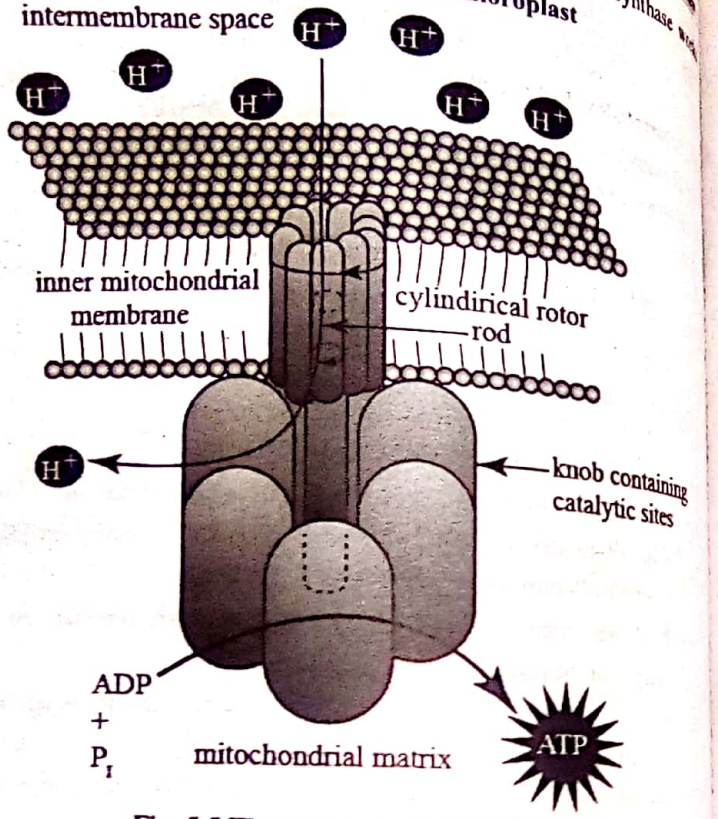


Fig. 5.5 The structure of ATP Synthase

Production of ATP in respiration

There are two main pathways by which ATP molecule can be produced in the process of respiration. This are:

1) **Aerobic pathway** (aerobic respiration) – requires the presence of oxygen.
2) **Anaerobic pathway** (Anaerobic respiration) – takes place in the absence of oxygen.

ATP Production in aerobic respiration

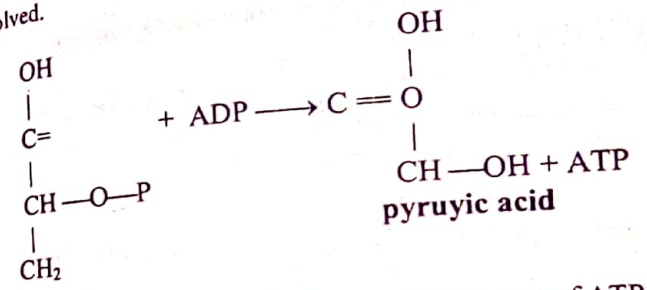
There are two ways of ATP production in aerobic respiration.

These are:

a. Substrate level phosporylation
b. Oxidat level phosphorylation

a. **Substrate level phosphorylation**

This level involves the transfer of phosphate group directly to **ADP** to produce **ATP**. In this process an enzyme **ATP synthase** is not involved.



During substrate level phosphorylation small amount of ATP which is only about 10% is produced. It takes place during **glycolysis** and **krebs cycle.**

b. **Oxidative phosphorylation**

- It is oxygen dependent production of ATP with the help of an enzyme **ATP synthase**. ATP synthase acts as a turbine that converts one form of energy in to another. This occurs when the hydrogen ions (H^+) pump through the catalytic knob of carrier protein rotates and the active site of ATP synthase exposed to bind ADP and pi to produce ATP.
- About 90% of the ATP is produced by oxidative phosphorylation.
- Organic substances that can be respired are called **respiratory substrates**. The **respiratory** substrates work by:
  - Decreasing the oxidation state of a substance **(reduction)**.
  - Increasing the oxidation state of a substance **(oxidation)**
- The most commonly respired substrate is **glucose molecule.**

The transfer of hydrogen ions from glucose to ATP synthase

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The two molecules that are involved in the transfer of hydrogen from glucose to ATP synthase are:

1. Nictinamide Adenine Dinucleotide (NAD)
2. Flavin Adenine Dinucleotide (FAD)

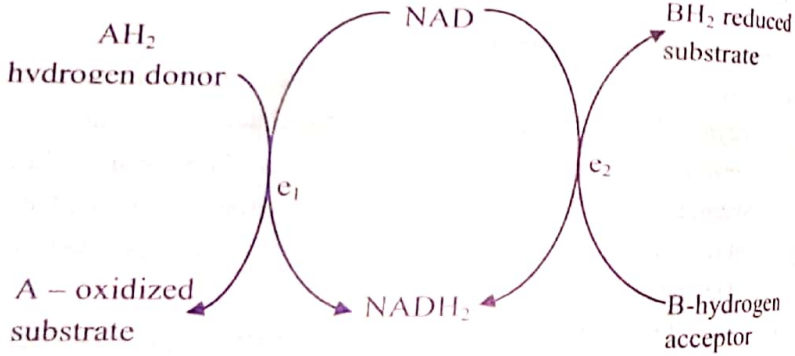
These molecules are coenzymes that are capable of accepting hydrogen ions. Thus, when they accept hydrogen they will be **reduced** and it forms molecules such as **NADH** and **FADH**.

Note: Reduction reaction is a reaction which involves addition of hydrogen and removal of oxygen.

The hydrogen ions found in NADH and FADH are released again when they are oxidized during the electron transport chain. The hydrogen ions are used to turn the rotor of ATP synthase.

Note: Oxidation reaction – is a reaction which involves the addition of oxygen and removal of hydrogen.

For instance, NAD exists in both reduced and oxidized form. In an oxidized state it functions as hydrogen **acceptor** and in a reduced state it acts as a **hydrogen donor**.



- Where e_1 and e_2 are two different dehydrogenate enzymes.
- 2H is transferred from A to B.
- The coenzyme NAD acts as a link between two different enzymes e_1 and e_2.

The stages of aerobic respiration in glucose

There are four stages in the aerobic respiration of glucose.

They are:

1) Glycolysis
2) The link reaction
3) Krebs cycle
4) Electron transport

1. Glycolysis (The first stage of aerobic respiration)

- Glycolysis literally means "**splitting sugar**". Therefore, glycolysis is the sequence of reactions that convert a **glucose** molecule (a six carbon compound) in to two molecules of **pyruvate** (a three carbon compound) with the production of ATP.
- Each reaction of glycolysis is regulated by specific types of enzymes and the whole reactions takes place in the **cytoplasm** of a cell.
- It is the **anaerobic stage of cellular respiration**, since it do not require utilization of oxygen. During this stage of cellular respiration **2 Net ATP** molecules are produced from one glucose molecule.
- The major reasons why glycolysis takes place in the cytoplasm takes place in the mitochondria of the cell are:
  1) The glucose molecule is too large, thus it cannot diffuse through mitochondrial membrane.
  2) There are no carrier proteins used to transport the glucose molecule across the mitochondrial membranes.

After glycolysis the molecule of pyruvate that contains 3 – carbon atoms is relatively smaller and is capable of entering the mitochondria. Accordingly, the other stages of cellular respiration other than glycolysis takes place in the **mitochondoria.**

The major steps of reactions that take place during glycolysis can be discussed as:

1) Two molecules of are **ATP** used to phosphorylate the glucose molecule. This initiates the oxidation of glucose.

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